

Performance Analysis of Real-Time PPP with Ambiguity Resolution in Simulated Scenarios

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Anhang I

Erklärung

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Abstract

High-precision positioning using Global Navigation Satellite System (GNSS) has gained increasing interest due to the growing demand in applications such as autonomous vehicle navigation and precision agriculture. Precise Point Positioning (PPP) offers a distinct advantage over differential techniques by enabling precise position determination of a GNSS rover through externally provided corrections. However, conventional PPP suffers from long convergence times and typically achieves only decimeter-level accuracy.

This thesis investigates the feasibility of applying Ambiguity Resolution (AR) in a dynamic scenario using simulated data, with the goal of reducing convergence time and improving positioning accuracy. A detailed description of the implemented ambiguity resolution approach is provided, together with the achieved results in the simulated scenario. The results demonstrate substantial improvements in positioning accuracy.

Furthermore, the structure of the BKG Ntrip Client (BNC) software is briefly discussed, along with the integration of the AR functionality into BNC. Insights gained during the development process are presented and compared with existing open-source methods. The implementation choices made in this thesis are explained and justified in detail.

Kurzfassung

Die hochpräzise Positionierung mit GNSS hat aufgrund des wachsenden Bedarfs in Anwendungen wie autonomen Fahrzeugsystemen und der Präzisionslandwirtschaft zunehmend an Bedeutung gewonnen. PPP bietet dabei einen wesentlichen Vorteil gegenüber differentiellen Verfahren, da es eine präzise Positionsbestimmung eines GNSS-Rovers mithilfe externer Korrekturen ermöglicht. Allerdings weist das konventionelle PPP eine lange Konvergenzzeit auf und erreicht typischerweise nur eine Genauigkeit im Dezimeterbereich.

Diese Masterarbeit untersucht die Realisierbarkeit der Anwendung von AR in einem dynamischen Szenario unter Verwendung simulierter Daten, mit dem Ziel, die Konvergenzzeit zu verkürzen und die Positionierungsgenauigkeit zu verbessern. Eine detaillierte Beschreibung des implementierten Ambiguitätsauflösungsverfahrens wird gegeben, ebenso wie die erzielten Ergebnisse im simulierten Szenario. Die Resultate zeigen deutliche Verbesserungen in der Positionierungsgenauigkeit.

Darüber hinaus wird die Struktur der BNC-Software kurz erläutert sowie die Integration der AR-Funktionalität in BNC. Erkenntnisse, die im Verlauf der Entwicklung gewonnen wurden, werden dargestellt und mit bestehenden Open-Source-Methoden verglichen. Die im Rahmen dieser Arbeit getroffenen Implementierungsentscheidungen werden ausführlich begründet.

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List of Abbreviations

ANTEX	ANTenna Exchange
AR	Ambiguity Resolution
ARP	antenna reference point
BKG	Bundesamt für Kartographie und Geodäsie
BNC	BKG Ntrip Client
BOC	Binary Offset Carrier
BPSK	Binary Phase Shift Keys
CBOC	Composite Binary Offset Carrier
CDMA	Code Division Multiple Access
CNES	Centre national d'études spatiales, Toulouse, France
DCB	Differential Code Biases
DLL	delay lock loop
ECEF	Earth Centered Earth Fixed
EKF	Extended Kalman Filter
ESKF	Error State Kalman Filter
Galileo HAS	Galileo High Accuracy Service
GNSS	Global Navigation Satellite System
GOOSE	GNSS-Receiver with Open Software Interface
GPS	Global Positioning System
GUI	Graphical User Interface
IAR	Integer Ambiguity Resolution
ICD	Interface Control Document
IDD	Internet Data Distribution
IF	ionosphere-free
IGS	International GNSS Service
LAMBDA	Least squares AMBIGUITY Decorrelation Adjustment

LS	least-squares
MBOC	Multiplexed Binary Offset Carrier
MEO	Medium Earth Orbit
NL	Narrow-Lane
Ntrip	Networked Transport of Radio Technical Commission for Maritime Services (RTCM) via Internet Protocol
OSB	Observable-specific Biases
OSR	Observation Space Representation
PCO	Phase Center Offset
PCV	Phase Center Variation
PNT	Position, Navigation and Timing
PPP	Precise Point Positioning
PPP-AR	Precise Point Positioning with Ambiguity Resolution
PRN	Pseudo Random Noise
RHCP	Right-Hand Circularly Polarized
RTCM	Radio Technical Commission for Maritime Services
RTK	real-time kinematic
RTS	Real-Time Service
SIS	Signal In Space
SSR	State Space Representation
WL	Wide-Lane
ZTD	zenith tropospheric delay

Chapter 1 Introduction

Global Positioning System (GPS) is the world's first satellite navigation system providing continuous, global coverage. It consists of a constellation of Medium Earth Orbit (MEO) satellites that enable users to determine their position, velocity, and time with high accuracy. Since its introduction, GPS has been widely employed in diverse applications such as navigation, autonomous driving, time synchronization, and geodesy.

In recent years, several other global navigation satellite systems have become operational, including GLONASS from Russia, Galileo from the European Union, and BeiDou from China. Each of these systems maintains its own constellation of satellites, ground control infrastructure, and signal specifications. Together, they form the so-called GNSS, which provides users worldwide with more precise, robust, and reliable Position, Navigation and Timing (PNT) services through interoperability and multi-constellation integration.

Measurements obtained from GNSS consist primarily of pseudorange and carrier-phase observations. Pseudorange measurements, derived from code-based correlations, typically provide meter-level positioning accuracy in single point positioning (SPP). Consequently, applications that require high-precision positioning rely mainly on carrier-phase observations due to their much shorter effective wavelength.

Over the past three decades, real-time kinematic (RTK) positioning has been the predominant technique for achieving centimeter- or even millimeter-level accuracy. RTK is a relative positioning method in which the coordinates of an unknown receiver are determined with respect to a reference station of known position. By using simultaneous carrier-phase measurements from both receivers, RTK effectively mitigates or eliminates spatially correlated error sources such as satellite orbit, clock, and atmospheric delays.

However, the main limitation of RTK lies in its reliance on a dense network of reference stations and the restriction of baseline length, as accuracy degrades with increasing distance between the rover and the reference station.

The concept of PPP was first introduced in the early 1970s by R. J. Anderle, who described it as a single-station positioning technique utilizing precise satellite orbits and Doppler-based observations (Anderle, 1977; Kouba and Héroux, 2001). After decades of research and technological advancement, PPP now refers to a positioning method that employs a single GNSS receiver together with carrier-phase and pseudorange measurements, while applying precise satellite orbit and clock products to correct signal path errors. This enables high-precision positioning without the need for a local reference station network.

Today, PPP can typically achieve positioning accuracies at the decimeter level, offering significant advantages such as reduced operational costs and independence from baseline constraints. However, a major limitation of standard PPP remains its relatively long convergence time before reaching decimeter-level accuracy, which restricts its suitability for real-time high-precision applications.

To shorten convergence time and further improve accuracy to a level comparable with RTK, ad-

vanced methods such as Precise Point Positioning with Ambiguity Resolution (PPP-AR) and PPP-RTK have been developed. In conventional PPP, carrier-phase ambiguities are estimated as real-valued (float) parameters. In contrast, PPP-AR resolves these ambiguities as integers, restoring the integer nature of the phase measurements. When additional atmospheric corrections—such as ionospheric and tropospheric delay information—are provided to support the user-side ambiguity resolution, the approach is referred to as PPP-RTK. These techniques significantly enhance the precision and convergence performance of the PPP solution.

An overview of the main differences between PPP, PPP-AR, PPP-RTK, and RTK is shown in Figure 1.1. The figure summarizes the types of corrections and external information required for each method, together with their corresponding positioning accuracy and convergence characteristics. It illustrates the progressive improvement from global, standalone PPP to enhanced PPP-RTK and finally to local RTK, mainly achieved through the inclusion of satellite- and atmosphere-related corrections such as State Space Representation (SSR) products.

It should be noted that, in contrast to the absolute positioning approach using SSR corrections, the differential technique (RTK) relies on Observation Space Representation (OSR) corrections, which provide relative information between reference stations and the rover receiver.

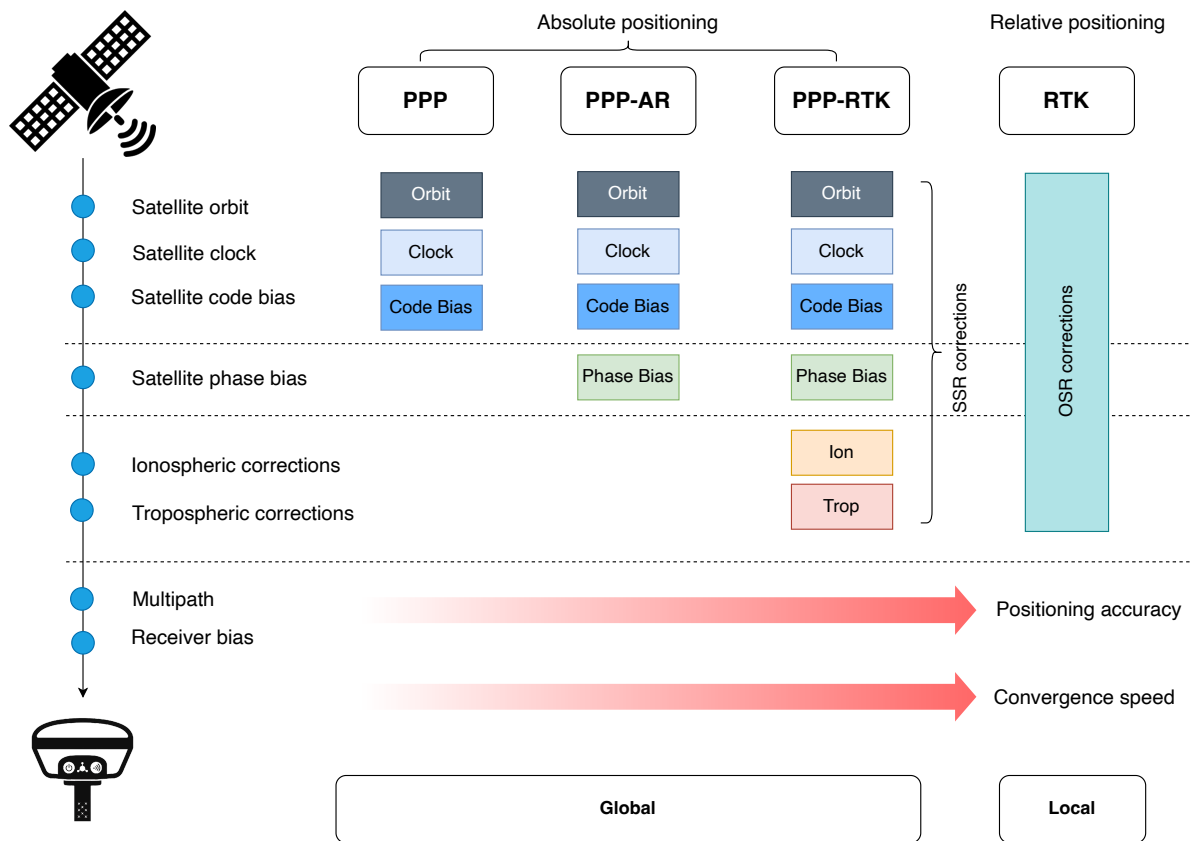


Figure 1.1: Overview of high-precision GNSS positioning techniques, including PPP, PPP-AR, PPP-RTK, and RTK. The diagram illustrates the relationship between these methods in terms of required external corrections, reference network dependency, and achievable accuracy. The progression from standard PPP to PPP-RTK and RTK demonstrates how incorporating integer ambiguity resolution and atmospheric corrections leads to faster convergence and higher positioning precision.

1.1 State-of-the-art Research

Various institutions and developers have created numerous software packages for PPP processing. These can generally be categorized into four groups: research software, commercial software, online processing services, and open-source tools. An overview of representative examples from each category is provided in Table 1.1.

Table 1.1: Software and Services Supporting PPP

Type	Tool Name	Developer or Institution
Research Software	Bernese	University of Bern
	GIPSY-OASIS, GipsyX	JPL (Jet Propulsion Laboratory)
	EPOS	GFZ (GeoForschungsZentrum Potsdam)
	P3	University of Calgary
	PANDA	Wuhan University
Commercial Software	Trimble RTX	Trimble Company
	Grafnav	Waypoint (Hexagon NovAtel) Company
Online Services	APPS	JPL
	GAPS	University of New Brunswick
	CSRS-PPP	NRCan (Natural Resources Canada)
	magicGNSS	GMV (Spain)
Open Source Software	RTKLIB	Tomoji Takasu
	gLAB	Universitat Politècnica de Catalunya (UPC)
	GPSTk	University of Texas at Austin
	PPPH	Hacettepe University (Turkey)
	G-Nut	Geodetic Observatory Pecný (GOP)
	BNC	BKG (Federal Agency for Cartography and Geodesy, Germany)
	PPP-Wizard	CNES (National Centre for Space Studies, France)
	GAMP	Zhou Feng et al.
	PRIDE PPP-AR	Wuhan University
	raPPPid	Vienna University of Technology
	GREAT-PVT	Wuhan University

Scientific PPP software packages are typically complex in structure and have a steep learning curve, as PPP processing often represents only one component of their broader functionality. Commercial software, on the other hand, is usually closed-source and requires expensive licensing fees. Online processing services provide a simple and user-friendly interface but do not allow users to modify or customize the underlying processing algorithms. In contrast, open-source software offers greater

flexibility for learning, research, and further development; however, such tools often suffer from limited maintenance, incomplete documentation, and reduced functionality compared to commercial or institutional software.

The requirements for the software used in this thesis are as follows:

- **Real-time processing:** Capability to ingest RTCM correction streams and provide position estimates in real time.
- **Open source:** The source code must be publicly available for inspection and modification.
- **Kinematic processing:** Support for processing trajectories of moving receivers.
- **PPP-AR:** Ability to resolve carrier-phase ambiguities as integer values.

Most available open-source software packages, such as PRIDE PPP-AR, raPPPid, or GREAT-PVT, are unsuitable for this work because they are limited to post-processing or static-mode applications. Others, such as RTKLIB, no longer support Integer Ambiguity Resolution (IAR) as this functionality was removed in version 2.4.3.

A key reason for selecting the BNC software in this thesis is its strong emphasis on real-time processing capability, which aligns directly with the primary research objective of this work. This focus, combined with its active development and open-source nature, makes BNC the most suitable candidate among the available software packages for implementing and testing real-time PPP-AR solutions.

1.2 BKG Ntrip Client (BNC)

BNC is an open-source software tool developed by the Bundesamt für Kartographie und Geodäsie (BKG). The BNC is designed for the simultaneous retrieval, decoding, conversion, and processing of real-time GNSS data streams using the Networked Transport of Radio Technical Commission for Maritime Services (RTCM) via Internet Protocol (Ntrip) standard. It has been developed within the framework of the IAG Sub-Commission for Europe (EUREF) and the International GNSS Service (IGS), with a strong emphasis on real-time capability and processing. (Weber et al., 2016)

Among its many functions, this thesis focuses on the following key features:

- Performing real-time Precise Point Positioning (PPP) to determine GNSS rover positions.
- Retrieving real-time GNSS data streams distributed via the Ntrip transport protocol.
- Retrieving real-time GNSS data streams directly from an IP address using TCP, without employing Ntrip.

The BNC software is under active development, with frequent updates to its source code. At the start of this thesis (June 1, 2025), the latest available version was 2.13.2, which forms the basis for all developments and analyses presented here. Since then, two additional updates have been released: version 2.13.3 on July 18, 2025, and version 2.13.4 on October 13, 2025 (accessed on October 14, 2025). According to the changelog, these updates do not include modifications related to the PPP processing module.

The BNC software currently supports only real-time float PPP solutions and does not provide functionality for PPP-AR. Consequently, a major part of the work in this thesis involves extending and

enhancing the existing float PPP module within BNC to enable real-time ambiguity resolution capabilities.

1.3 Research Objectives, Scope, and Methodology

The main objective of this research is to investigate the feasibility of implementing a real-time PPP-AR algorithm within the open-source BNC software and to evaluate its performance in a dynamic scenario. The goal is to determine whether the developed approach can achieve improvements over the conventional float PPP solution in terms of positioning accuracy and convergence time under real-time conditions.

To achieve this goal, the specific research objectives are as follows:

1. Analyze the structure of the BNC software and its PPP processing module.
2. Research and implement state-of-the-art AR methods suitable for real-time applications.
3. Process and evaluate the results in a simulated dynamic scenario to assess performance improvements in real-time conditions.

Figure 1.2 illustrates the overall research workflow and technical route followed in this thesis. The process begins with analyzing the BNC source code and reviewing the theoretical foundations of PPP. Based on this understanding, the internal structure of the BNC software and its existing PPP processing module are examined to identify suitable integration points for algorithmic enhancement. Subsequently, the PPP-AR algorithm is developed and implemented by incorporating theoretical concepts from ambiguity resolution methods into the BNC framework. The enhanced system is then tested in a dynamic scenario using simulated real-time data streams to evaluate its positioning performance. Finally, the obtained results are analyzed in terms of accuracy and convergence behavior to assess the improvements over the conventional float PPP solution.

Thesis Outline

The structure of this thesis is organized as follows:

- **Chapter 1** introduces the research background, provides an overview of the state-of-the-art PPP processing software, and outlines the research methods and objectives.
- **Chapter 2** explains the fundamental terminology and methodology, including the basic theory of GNSS, time and coordinate systems and their transformations, GNSS observation models, linear combinations, and PPP error sources.
- **Chapter 3** presents the PPP model, its processing schemes, parameter estimation theories, and the model structure as implemented in the BNC software.
- **Chapter 4** describes the theory, processing scheme, and methodology of PPP-AR, as well as details of its implementation within the BNC framework.
- **Chapter 5** provides the results and analysis of the implemented methods, including performance evaluation under dynamic conditions.

- **Chapter 6** concludes the thesis with a summary of findings and outlines recommendations for future work.

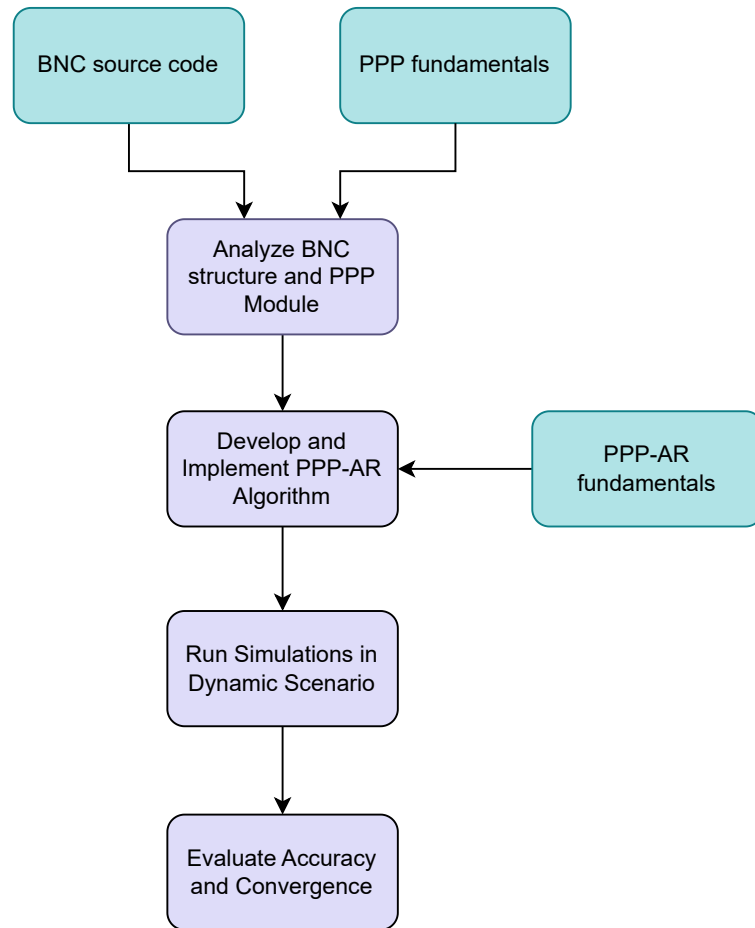


Figure 1.2: Research workflow illustrating the technical route for developing and evaluating the real-time PPP-AR algorithm based on BNC.

Chapter 2 Fundamentals

Conventions

- Scalar variables are in italics, e.g., a .
- Vectors are represented by bold lowercase letters, e.g., $\mathbf{v}$.
- Matrices are represented by bold uppercase letters, e.g., $\mathbf{H}$.

2.1 Global Navigation Satellite System

Global Navigation Satellite System (GNSS) refers to a collection of satellite navigation constellations that provide global positioning, navigation, and timing information. It comprises systems such as GPS, GLONASS, Galileo, and BeiDou. Most GNSS satellites operate in Medium Earth Orbit (MEO), while some function in Geostationary Orbit (GEO) or Inclined Geosynchronous Orbit (IGSO). These satellites transmit modulated signals primarily in the L-band frequency range. In this thesis, the focus is on the GPS and Galileo systems.

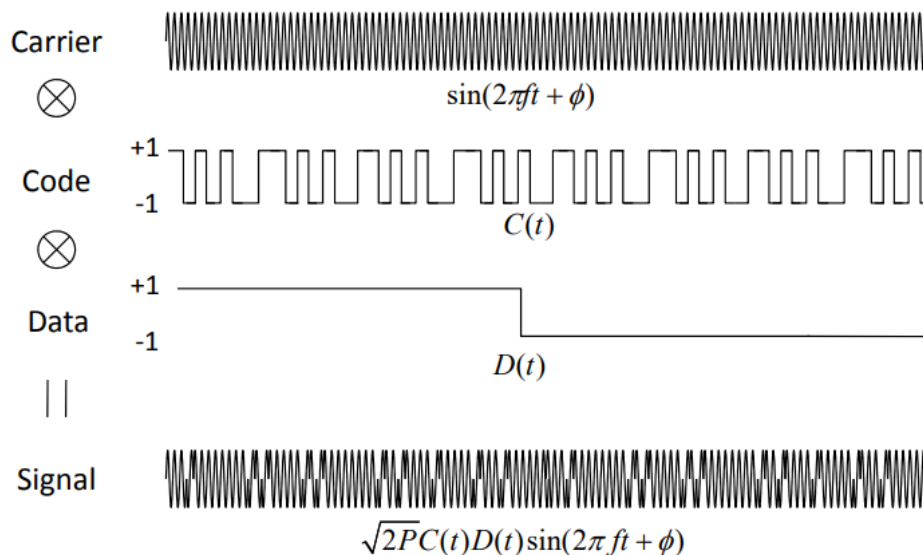


Figure 2.1: Structure of a typical GNSS signal (Takasu and Kubo, 2013, p. 139).

Each GNSS signal consists of three main components: a carrier wave, a ranging code, and a navigation data message. An exception to this are the so-called pilot or data-less signals, which do not

include the data message component. The carrier wave is a sinusoidal electromagnetic signal transmitted in the L-band, selected because this frequency range lies within an atmospheric transmission window where attenuation is minimal, ensuring reliable signal propagation for GNSS applications.

The spreading code is used to distinguish signals transmitted by different satellite vehicles. This allows multiple satellites to transmit simultaneously on the same frequency band without interfering with each other, through a technique known as Code Division Multiple Access (CDMA). These codes are referred to as Pseudo Random Noise (PRN) codes because they appear random but are, in fact, deterministic and defined in the Interface Control Document (ICD). This enables the receiver to generate a local replica of the PRN code and perform correlation with the incoming signal for identification and ranging.

In addition to the spreading code, each GNSS signal also carries navigation data, which provides essential information for positioning. This includes the Keplerian orbital elements describing the satellite's position, almanac data containing coarse orbital parameters of all satellites in the constellation, ionospheric and tropospheric model parameters, as well as satellite clock corrections. The navigation data has a much lower bit rate compared to the spreading code and is modulated together with the PRN code onto the carrier wave using Binary Phase Shift Keys (BPSK). In this modulation scheme, the phase of the carrier is shifted by π radians whenever the chip sequence changes its state from +1 to -1, or vice versa. This phase reversal encodes the binary information of the signal while maintaining constant amplitude, which makes BPSK a robust and widely used modulation method in GNSS signal transmission.

Modern GNSS signals have evolved to incorporate two distinct components to enhance performance and robustness, particularly in challenging environments: the data component and the pilot component.

- **Data component:** This component carries the navigation message, which includes essential information such as satellite ephemerides, clock corrections, and system parameters.
- **Pilot component:** The pilot component is primarily used for signal acquisition and tracking. It does not contain any navigation data but instead transmits a repeating modulation pattern, often a replica of the spreading code itself.

The introduction of the pilot component marks a significant advancement in GNSS signal design. Unlike the data component, which is modulated with navigation bits that can cause abrupt phase transitions, the pilot signal remains free of such discontinuities. This allows for longer coherent integration times, leading to more robust signal acquisition and tracking under weak-signal or high-interference conditions. Consequently, modern GNSS receivers can maintain lock on weaker signals, thereby improving positioning performance in environments such as urban canyons, dense vegetation, or partially obstructed areas.

While the fundamental principles of GNSS signal structure remain consistent across constellations, each system has adopted distinct signal designs and modulation schemes to meet specific performance objectives. For example, the modernized GPS L1C signal employs a Multiplexed Binary Offset Carrier (MBOC) modulation, which enhances tracking accuracy and robustness against interference compared to earlier GPS signal generations. Similarly, Galileo signals make extensive use of Binary Offset Carrier (BOC) and Composite Binary Offset Carrier (CBOC) modulations to achieve improved spectral separation and better multipath mitigation performance.

Table 2.1: GPS and Galileo satellite types and corresponding PRN numbers (operational SVs as of 2025-10-13).

System	Satellite Type	PRN
GPS	IIR	G02, G13, G16, G20, G22
	IIR-M	G05, G07, G12, G15, G17, G29, G31
	IIF	G03, G06, G08, G09, G10, G24, G26, G27, G30, G32
	III	G11, G14, G18, G21, G23, G28
Galileo	IOV	E11, E12, E19
	FOC	E02, E03, E04, E05, E06, E07, E08, E09, E10, E13, E15, E16, E21, E23, E24, E25, E26, E27, E29, E30, E31, E33, E34, E36

2.1.1 GPS System

The GPS was the first satellite system to provide global navigation services. It offers highly stable and reliable positioning performance, forming the foundation for numerous industrial and scientific applications. The system transmits multiple signals and codes that distinguish between the open civil service and the encrypted military service.

The two main frequencies are L1 (1575.42 MHz) and L2 (1227.60 MHz). The newer GPS satellites also transmit on the L5 frequency (1176.45 MHz). On each frequency, one or more codes are sent. The two main legacy codes are the C/A and P(Y) codes, while newer satellites transmit codes such as M-code, L1C, etc.

Table 2.2: Current and new GPS navigation signals. All civil signals are provided free of charge to all users worldwide (Open Services).

Link	Carrier freq. (MHz)	PRN code	Modulation	Code characteristics		Service
				Rate [Mcps]	Length [chips]	
L1	1575.420	C/A	BPSK(1)	1.023	1,023	Civil
		P(Y)	BPSK(10)	10.23	10,230,000	Military
		L1C	MBOC(6,1,1/11)	1.023	10,230	Civil
L2	1227.600	P(Y)	BPSK(10)	10.23	10,230,000	Military
		L2C	BPSK(1)	1.023	10,230	Civil
L5	1176.450	L5-I/Q	BPSK(10)	10.23	10,230	Civil

Table 2.2 summarizes the current and new GPS navigation signals, including their carrier frequencies, modulation types, and code characteristics. All civil signals are freely available under the Open Service policy.

2.1.2 Galileo System

The Galileo system, developed by the European Union, is a relatively new global navigation satellite system that is still under active development. Compared to GPS, Galileo demonstrates several modernization features, such as advanced modulation schemes, a higher number of signal channels, and multiple service types. Most of these signals are available for open access, supporting a wide range of civilian applications.

Table 2.3 summarizes the main Galileo signals, including their carrier frequencies, modulation types, and code characteristics.

Table 2.3: Galileo navigation signals. The E5a and E5b bands are jointly modulated onto a single E5 carrier of 1191.795 MHz using AltBOC(15,10).

Band	Freq.(MHz)	Channel / comp.	Modulation	Code characteristics		Services
				Rate [Mcps]	Length [chips]	
E1	1575.42	A (data)	BOC(15,2.5)	2.5575	4092	PRS
		B (data)	MBOC(6,1,1/11)	1.023	4092	OS, CS, SoL
		C (pilot)	MBOC(6,1,1/11)	1.023	4092	OS, CS, SoL
E6	1278.75	A (data)	BOC(10,5)	5.115	5115	PRS
		B (data)	BPSK(5)	5.115	10,230	CS
		C (pilot)	BPSK(5)	5.115	10,230	CS
E5a	1176.45	I (data)	BPSK(10)	10.23	10,230	OS
		Q (pilot)	BPSK(10)	10.23	10,230	OS
E5b	1207.14	I (data)	BPSK(10)	10.23	10,230	OS, CS, SoL
		Q (pilot)	BPSK(10)	10.23	10,230	OS, CS, SoL

2.1.3 Time Systems

The core principle of GNSS is to determine a user's position by measuring the travel time of a signal from a satellite. This transforms a timing problem into a distance measurement, making a precise, common time reference absolutely fundamental. However, "time" is not a single entity. There is astronomical time, tied to the Earth's physical rotation, and atomic time, based on the highly stable oscillations of atoms. This distinction has led to the development of several distinct time systems, which are summarized in Table 2.4.

Table 2.4: Different time systems based on their underlying physical process (from Hofmann-Wellenhof et al., 2011).

Periodic process	Time
Earth's rotation	Universal Time (UT0, UT1, UT2) Greenwich Sidereal Time (Θ)
Atomic oscillators	International Atomic Time (TAI) Coordinated Universal Time (UTC) GNSS Reference Time

UTC

Coordinated Universal Time (UTC) is the globally recognized standard for civil time. It functions as a hybrid system, blending atomic precision with astronomical reality. The "tick" of a UTC second is identical to that of International Atomic Time (TAI), a composite scale derived from over 400 of the world's most precise atomic clocks.

However, to ensure our daily time remains synchronized with the Earth's day-night cycle, UTC is periodically adjusted to track the planet's irregular rotation, measured by UT1. When the divergence between the uniform TAI and the variable UT1 approaches 0.9 seconds, a **leap second** is added to UTC. This correction is managed by the International Earth Rotation and Reference System Service (IERS), under the authority of the International Bureau of Weights and Measures (BIPM), which maintains UTC.

GPST

The GPS constellation operates on its own distinct timescale, GPS Time (GPST), which is managed by the U.S. Naval Observatory (USNO). GPST is a continuous atomic timescale; it is uniform and is not adjusted with leap seconds.

The starting point, or epoch, for GPST was 00:00:00 UTC on January 6, 1980. At that instant, GPST and UTC were perfectly aligned. Since then, GPST has run continuously, while UTC has been periodically paused for leap seconds. Consequently, a gap has formed between them, which widens with each new leap second (Teunissen, Montenbruck, et al., 2017, p. 159). The conversion is:

$$t_{\text{UTC}} = t_{\text{GPST}} - \Delta t_{LS} \quad (2.1)$$

where Δt_{LS} represents the total integer-second difference between the two scales. In contrast, the relationship between GPST and TAI is fixed:

$$t_{\text{GPST}} = t_{\text{TAI}} - 19\text{ s} \quad (2.2)$$

Galileo System Time (GST)

The European Galileo system uses Galileo System Time (GST), another continuous atomic timescale that is free of leap seconds. Its epoch began on August 22, 1999.

GST is steered to maintain very high fidelity to TAI, remaining within 50 nanoseconds. Although designed for interoperability with GPS, a small, monitored offset known as the Galileo-GPS Time Offset (GGTO) exists between GST and GPST (Teunissen, Montenbruck, et al., 2017, p. 159). For end-users, the relationship between GST and civil time mirrors that of GPS:

$$t_{\text{UTC}} = t_{\text{GST}} - \Delta t_{LS} \quad (2.3)$$

The Galileo navigation message provides users with the necessary data to determine the current integer-second offset (Δt_{LS}) and the GGTO, enabling precise conversion and positioning.

2.2 Coordinate Systems

To describe a position on Earth, an appropriate coordinate system must be defined. Several coordinate systems can be used depending on the application, as some are more suitable than others. The **Cartesian coordinate system** represents positions using three orthogonal axes (x, y, z) in a three-dimensional space. Alternatively, considering the approximately spherical shape of the Earth, a **geocentric coordinate system** can be defined, where a point is described by its geodetic latitude, longitude, and height above the reference ellipsoid.

In BNC, both receiver and satellite positions are internally represented as X , Y , and Z components in the Earth-Centered Earth-Fixed (ECEF) coordinate system.

2.2.1 Transformation from Geodetic to ECEF Coordinates

A geodetic position is defined on a reference ellipsoid by its latitude φ , longitude λ , and ellipsoidal height h . The corresponding position in the ECEF system can be expressed as

$$\mathbf{r}_r = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (N + h) \cos \varphi \cos \lambda \\ (N + h) \cos \varphi \sin \lambda \\ [N(1 - e^2) + h] \sin \varphi \end{bmatrix}, \quad (2.4)$$

where N is the prime vertical radius of curvature, defined as

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad (2.5)$$

with a denoting the semi-major axis of the reference ellipsoid and e its eccentricity.

2.2.2 Transformation from ECEF to Geodetic Coordinates

The inverse transformation from ECEF coordinates to geodetic coordinates is given by

$$\mathbf{r}_{\text{ell}} = \begin{bmatrix} \lambda \\ \varphi \\ h \end{bmatrix} = \begin{bmatrix} \arctan\left(\frac{y}{x}\right) \\ \arctan\left(\frac{z + e^2 N \sin \varphi}{p}\right) \\ \frac{p}{\cos \varphi} - N \end{bmatrix}, \quad \text{with } p = \sqrt{x^2 + y^2}. \quad (2.6)$$

Since the latitude φ appears implicitly in its own definition through both $\sin \varphi$ and N , it cannot be solved in closed form. Therefore, φ is obtained iteratively, starting from an initial approximation and updating until the result converges within a specified tolerance.

2.2.3 Transformation from ECEF to Local Coordinates

The local topocentric coordinate system (East–North–Up, ENU) can be derived from the ECEF frame using the following rotation matrix:

$$\mathbf{E}_r = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix}. \quad (2.7)$$

Here, φ and λ denote the geodetic latitude and longitude of the receiver. Using $\mathbf{E}_r$ and the receiver position $\mathbf{r}_r$ in ECEF coordinates, any position $\mathbf{r}_{\text{ECEF}}$ can be transformed into the local ENU frame as

$$\mathbf{r}_{\text{local}} = \mathbf{E}_r (\mathbf{r}_{\text{ECEF}} - \mathbf{r}_r). \quad (2.8)$$

2.3 GNSS Observations

2.3.1 Pseudorange

The fundamental principle of GNSS is to determine the apparent signal travel time from the satellite to the user. The receiver clock measures the difference between the signal transmission time and its reception time. To achieve this, the receiver's delay lock loop (DLL) generates a local replica of the known code sequence and aligns it with the incoming signal, with the required time shift providing the travel time estimate. Since the receiver and satellite clocks are not perfectly synchronized and operate independently, the measurement is biased. The resulting biased range, known as the pseudorange, is defined as the estimated transit time multiplied by the speed of light in vacuum.

Code-based measurements are affected by noise and multipath at the meter level, which typically limits the achievable positioning accuracy to the meter range.

2.3.2 Carrier Phase

In addition to measuring pseudoranges, a GNSS receiver also tracks the carrier signal using a phase-locked loop (PLL). The receiver generates a replica of the carrier wave, aligns it with the incoming signal, and measures the fractional number of cycles corresponding to the observed phase shift. Furthermore, it keeps count of the complete carrier cycles, thereby providing a continuous carrier phase measurement. Carrier phase measurements are significantly more precise than pseudorange measurements, as the carrier wavelength is on the order of 19–25 cm, depending on the carrier frequency.

This high precision comes at the cost that the integer number of full carrier cycles accumulated along the signal path remains unknown. In contrast to pseudorange, the carrier-phase measurement is ambiguous because the integer number of cycles between the satellite and the receiver at the start of phase tracking is unknown. This unknown is referred to as the ambiguity, and the process of resolving it is known as ambiguity resolution.

2.3.3 Doppler

The relative motion between a satellite and the user causes a change in the observed frequency of the satellite signal. In addition, the receiver clock bias contributes to a frequency offset or drift. This Doppler shift is continuously measured by the receiver's carrier-tracking loop. The frequency adjustment required to compensate for the relative motion, as well as frequency deviations in the receiver or satellite clock, is reported by the receiver as the Doppler measurement.

Although the RTCM Multiple Signal Messages (MSM) include phase-rate information, which is essentially a Doppler-type observable. This information is not used by the BNC processing engine. The reason is that BNC focuses solely on position determination and does not estimate receiver velocity. Consequently, Doppler measurements are not incorporated into the estimation process and are ignored in the final PNT solution.

2.3.4 GNSS Observation Equations

GNSS positioning primarily relies on two types of measurements: pseudorange and carrier phase. Pseudorange generation is based on measuring the apparent travel time of the signal transmitted

by the navigation satellite, as received by the GNSS receiver. By multiplying the signal travel time with the speed of light, the pseudorange can be obtained.

The GNSS observation equations between a satellite s and a receiver r can be written for the pseudorange and phase measurements in units of meters as follows (Teunissen, Montenbruck, et al., 2017, p. 583, 584):

$$\begin{aligned} p_{r,j}^s(t) &= \rho_r^s(t) + \xi_{r,j}^s(t) + c \left(dt_r(t) - dt^s(t) + \delta t^{\text{rel}}(t) \right) \\ &\quad + I_{r,j}^s(t) + T_r^s(t) + B_j^s + B_{r,j} + e_{r,j}^s(t) \\ \varphi_{r,j}^s(t) &= \rho_r^s(t) + \zeta_{r,j}^s(t) + c \left(dt_r(t) - dt^s(t) + \delta t^{\text{rel}}(t) \right) \\ &\quad - I_{r,j}^s(t) + T_r^s(t) + b_j^s + b_{r,j} + \lambda_j \left(\omega_r^s(t) + N_{r,j}^s \right) + \epsilon_{r,j}^s(t) \end{aligned} \quad (2.9)$$

In the above equations, $p_{r,j}^s(t)$ and $\varphi_{r,j}^s(t)$ represent the pseudorange and carrier-phase observations, respectively, between receiver r and satellite s on frequency j , both expressed in meters. The term $\rho_r^s(t)$ denotes the geometric distance between the satellite and receiver antenna phase centers. The parameters $\xi_{r,j}^s(t)$ and $\zeta_{r,j}^s(t)$ represent antenna-related corrections, including phase center offsets and variations.

The constant c is the speed of light in vacuum, while $dt_r(t)$ and $dt^s(t)$ are the receiver and satellite clock offsets relative to GNSS system time. The relativistic clock correction due to satellite motion is expressed as $\delta t^{\text{rel}}(t)$. The ionospheric and tropospheric propagation delays are denoted by $I_{r,j}^s(t)$ and $T_r^s(t)$, respectively.

The hardware-related delays B_j^s and $B_{r,j}$ correspond to code biases from the satellite and receiver, while b_j^s and $b_{r,j}$ denote the corresponding carrier-phase hardware biases. The carrier wavelength on frequency j is λ_j , and $N_{r,j}^s$ is the integer ambiguity term, representing the unknown number of carrier cycles between the satellite and receiver. The variable $\omega_r^s(t)$ accounts for the phase wind-up effect, and the terms $e_{r,j}^s(t)$ and $\epsilon_{r,j}^s(t)$ represent measurement noise and residual unmodeled errors for the code and phase observations, respectively.

2.4 Single Difference

The single-difference observations between satellites k and l observed by the same receiver r can be formed for both pseudorange and carrier-phase measurements as:

$$\begin{aligned} p_r^{kl} &= p_r^k - p_r^l \\ \varphi_r^{kl} &= \varphi_r^k - \varphi_r^l \end{aligned} \quad (2.10)$$

Figure 2.2 illustrates the concept of between-satellite single differencing, where observations from two satellites are differenced to remove receiver-dependent terms.

The observation equations for the single-differenced pseudorange p_r^{kl} and carrier-phase φ_r^{kl} can then be written as (Teunissen, Montenbruck, et al., 2017, p. 596):

$$p_r^{kl} = \rho_r^{kl} + c \left(dt^{kl} + \delta t^{\text{rel},kl} \right) + B_r^{kl} + \xi_r^{kl} + T_r^{kl} + I_r^{kl} + e_r^{kl}, \quad (2.11)$$

$$\begin{aligned} \varphi_r^{kl} &= \rho_r^{kl} + c \left(dt^{kl} + \delta t^{\text{rel},kl} \right) + b_r^{kl} + \zeta_r^{kl} + T_r^{kl} - I_r^{kl} + \epsilon_r^{kl} \\ &\quad + \lambda \left(\omega_r^{kl} + N_r^{kl} \right), \end{aligned} \quad (2.12)$$

where ρ_r^{kl} represents the geometric range difference between satellites k and l as observed by receiver r . The terms d^{kl} and $\delta t^{\text{rel},kl}$ denote the satellite clock offset difference and the relativistic correction difference, respectively. T_r^{kl} and I_r^{kl} represent the tropospheric and ionospheric delay differences, while b_r^{kl} and d_r^{kl} denote the satellite-dependent carrier and code hardware bias differences.

In the carrier-phase observation, $\omega_r^{kl} = \omega_r^k - \omega_r^l$ and $N_r^{kl} = N_r^k - N_r^l$ denote the differential phase wind-up and differential integer ambiguity, respectively. It is assumed that both phase measurements are made on the same frequency, and thus share a common wavelength λ , which can be factored out of the difference in Eq. (2.12).

The time variable (t) has been omitted from the notation for simplicity. Since all quantities in Eqs. (2.11)–(2.12) refer to the same observation epoch, explicitly writing (t) for each term would unnecessarily complicate the expressions. It is therefore common practice in GNSS modeling to drop the time argument unless temporal variation or time derivatives are specifically analyzed.

In this differencing process, the receiver clock offset and its associated relativistic effects are eliminated, as they are common to both satellite observations. Additionally, the receiver-dependent hardware delays cancel out, leaving only satellite-dependent and propagation-related terms in the single-differenced observations.

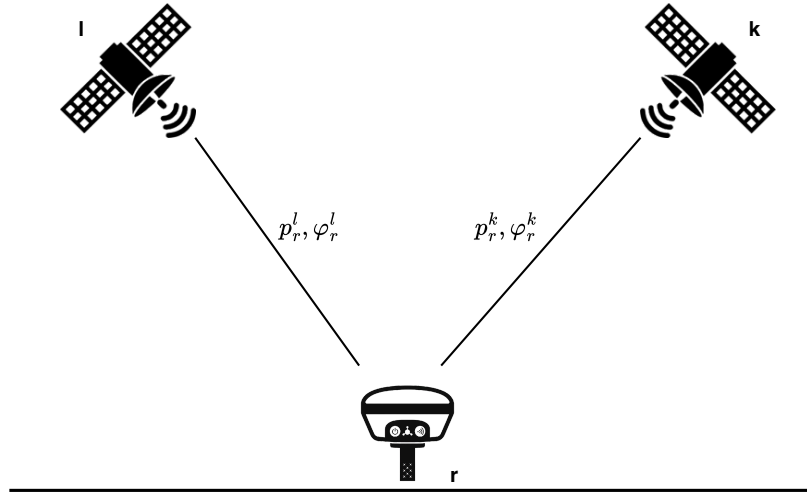


Figure 2.2: Single difference of observations from receiver r between satellites k and l

2.5 Linear Combination

The concept of linear combinations in GNSS involves forming new observables by combining multiple measurements of the same or different types. Such combinations can yield pseudo-observations that offer advantages over the raw measurements, for example by eliminating nuisance parameters, reducing common errors, or emphasizing specific parameters of interest.

2.5.1 Ionosphere-free Linear Combination

When two different frequencies are used, an ionosphere-free combination can be formed. This exploits the dispersive nature of the ionosphere, whereby signals at different frequencies are affected differently. By forming such a combination, the first-order ionospheric delay can be effectively eliminated. The equations for the ionosphere-free pseudorange and carrier-phase combinations can be expressed as (Teunissen, Montenbruck, et al., 2017, p. 590):

$$p_{r,IF}^s = \frac{f_1^2}{f_1^2 - f_2^2} p_{r,1}^s - \frac{f_2^2}{f_1^2 - f_2^2} p_{r,2}^s \quad (2.13)$$

$$\varphi_{r,IF}^s = \frac{f_1^2}{f_1^2 - f_2^2} \varphi_{r,1}^s - \frac{f_2^2}{f_1^2 - f_2^2} \varphi_{r,2}^s \quad (2.14)$$

However, forming the ionosphere-free combination increases the measurement noise and reduces the number of independent observations by one compared to the original uncombined dual-frequency measurements.

Forming the ionosphere-free combination increases the observation noise. If the individual measurement noises on f_1 and f_2 are uncorrelated with standard deviations σ_1 and σ_2 , the IF noise standard deviation is

$$\sigma_{IF} = \sqrt{\left(\frac{f_1^2}{f_1^2 - f_2^2}\right)^2 \sigma_1^2 + \left(\frac{f_2^2}{f_1^2 - f_2^2}\right)^2 \sigma_2^2}.$$

For equal noise levels ($\sigma_1 = \sigma_2 = \sigma$),

$$\sigma_{IF} = \sigma \sqrt{\frac{f_1^4 + f_2^4}{(f_1^2 - f_2^2)^2}}.$$

Example (GPS L1/L2): with $f_1 = 1575.42$ MHz and $f_2 = 1227.60$ MHz,

$$\frac{\sigma_{IF}}{\sigma} \approx 2.98 \ (\approx 3).$$

Thus, the IF combination roughly triples the measurement noise (for both code and carrier-phase when expressed in meters) and reduces the number of independent observations by one.

2.5.2 Narrow- and Wide-Lane Linear Combinations

The idea of forming narrow-lane and wide-lane combinations is to generate artificial wavelengths that differ from the individual carrier wavelengths. These synthetic wavelengths are particularly useful for integer ambiguity resolution. In the case of the *wide-lane* (WL) combination, the resulting wavelength is longer than the longest individual carrier wavelength, which improves the reliability of ambiguity fixing. Conversely, the *narrow-lane* (NL) combination produces a wavelength shorter than the shortest carrier wavelength, thereby increasing the sensitivity of the observable but also amplifying the noise.

For illustration purposes, let us first consider the wide-lane combination of carrier-phase measurements on two frequencies. By ignoring additional error sources (e.g., atmospheric delays, clock

offsets, and hardware biases) and expressing the carrier phases in units of cycles, the wide-lane observable can be written as (Misra and Enge, 2011, p. 254):

$$\begin{aligned}\Phi_{r,WL}^s &= \Phi_{r,1}^s - \Phi_{r,2}^s \\ &= \rho_r^s \left(\frac{1}{\lambda_1 - \lambda_2} \right) + (N_1 - N_2) + \varepsilon_{\Phi_{WL}} \\ &= \frac{\rho_r^s}{\lambda_{WL}} + N_{WL} + \varepsilon_{\Phi_{WL}},\end{aligned}\tag{2.15}$$

with the corresponding WL wavelength

$$\lambda_{WL} = \frac{c}{f_1 - f_2},\tag{2.16}$$

where c is the speed of light and f_1, f_2 are the carrier frequencies.

Similarly, a geometry-preserving narrow-lane carrier-phase combination can be formed as

$$\Phi_{r,NL}^s = \Phi_{r,1}^s + \Phi_{r,2}^s,\tag{2.17}$$

with the corresponding NL wavelength

$$\lambda_{NL} = \frac{c}{f_1 + f_2}.\tag{2.18}$$

2.5.3 Melbourne–Wübbena Linear Combination

The Melbourne–Wübbena (MW) combination is a linear combination of dual-frequency carrier-phase and pseudorange measurements. It is constructed in such a way that both the first-order ionospheric delay and the geometric range terms are eliminated, leaving an observable that primarily depends on the wide-lane ambiguity, hardware biases, and measurement noise. The MW combination can be expressed either in cycles or in meters and is particularly useful for ambiguity resolution, since it provides a direct estimator of the wide-lane integer ambiguity. However, because pseudorange measurements are involved, the MW combination is considerably noisier than pure carrier-phase combinations and therefore requires smoothing or averaging before it can be reliably applied.

The MW combination can be formed using dual-frequency pseudorange and carrier-phase measurements as follows:

$$\begin{aligned}MW^s &= \left(\frac{f_1}{f_1 - f_2} \varphi_{r,1}^s - \frac{f_2}{f_1 - f_2} \varphi_{r,2}^s \right) - \left(\frac{f_1}{f_1 + f_2} P_{r,1}^s + \frac{f_2}{f_1 + f_2} P_{r,2}^s \right) \\ &= \lambda_{WL} \underbrace{(N_{r,1}^s - N_{r,2}^s)}_{N_{WL}^s} + \lambda_{WL} \underbrace{\left(\frac{b_{r,1}}{\lambda_1} - \frac{b_{r,2}}{\lambda_2} \right)}_{b_{WL}} - \lambda_{NL} \underbrace{\left(\frac{B_{r,1}}{\lambda_1} + \frac{B_{r,2}}{\lambda_2} \right)}_{b_{WL}^s} + \lambda_{WL} \underbrace{\left(\frac{b_1^s}{\lambda_1} - \frac{b_2^s}{\lambda_2} \right)}_{b_{WL}^s} - \lambda_{NL} \underbrace{\left(\frac{B_1^s}{\lambda_1} + \frac{B_2^s}{\lambda_2} \right)}_{b_{WL}^s} + \varepsilon_{MW} \\ &= \lambda_{WL} N_{WL}^s + b_{WL} + b_{WL}^s + \varepsilon_{MW}.\end{aligned}\tag{2.19}$$

where $\lambda_{WL} = \frac{c}{f_1 - f_2}$ and $\lambda_{NL} = \frac{c}{f_1 + f_2}$ denote the wide-lane and narrow-lane wavelengths, respectively, with c the speed of light. The corresponding wide-lane ambiguity is defined as $N_{WL} = N_1 - N_2$, where N_1 and N_2 are the carrier-phase ambiguities on frequencies f_1 and f_2 . The values of λ_{WL} and λ_{NL} depend on the specific signal frequencies used in the linear combination. Since GPS and

Table 2.5: Wide- and narrow-lane combinations of signals for different frequencies of GPS and Galileo.

Sys.	Sig.	f_i (MHz)	λ_i (m)	Comb. ij	λ_w (m)	λ_N (m)
GPS	L1	1575.420	0.190	L1,L2	0.862	0.107
	L2	1227.600	0.244	L1,L5	0.751	0.109
	L5	1176.450	0.255	L2,L5	5.861	0.125
Galileo	E1	1575.420	0.190	E1,E5b	0.814	0.108
	E5b	1207.140	0.248	E1,E5a	0.751	0.109
	E5a	1176.450	0.255	E5b,E5a	9.768	0.126

Galileo operate on multiple carrier frequencies, these wavelengths differ across signal pairs. Table 2.5 summarizes the wide- and narrow-lane combinations for the most commonly used GPS and Galileo signals.

Since the MW combination includes pseudorange measurements, its noise level is dominated by code noise rather than carrier-phase noise. For GPS L1/L2, this corresponds to a noise amplification factor of approximately three compared to the individual pseudorange observables. Consequently, the MW observable is very noisy and must be averaged or smoothed before being used for reliable ambiguity resolution.

2.6 Error Sources in PPP

Precise Point Positioning concept

The fundamental idea of PPP is to incorporate precise carrier-phase observations, alongside pseudoranges, while replacing broadcast satellite orbits and clocks with high-precision products. In contrast to traditional double-difference relative positioning, PPP enables absolute positioning without relying on a nearby reference station or baseline geometry. This, however, necessitates the introduction of additional initial phase ambiguity unknowns, causing a fairly long upto 15mins or longer initial convergence time (Teunissen, Montenbruck, et al., 2017, p. 723). PPP also requires significantly more rigorous modeling of all relevant error sources, as these effects cannot be eliminated through double differencing and must instead be accounted for explicitly in the estimation process.

According to their origin, the errors in PPP can be classified into satellite-related errors, atmospheric delays, site displacements, and receiver-related errors. An overview of the pseudorange-related errors is illustrated in Figure 2.3. In addition to the aforementioned errors, several further effects must be considered in order to fully exploit precise carrier-phase measurements. These include site displacements, antenna phase center offsets, antenna phase center variations, and the phase wind-up effect.

To come to the PPP's goal of centimeter precision, there are mainly two strategies to be considered:

- **Error modelling:** Depending on the underlying mechanism of the error, a correction model can be derived, or an empirical model can be established based on a large amount of historical observations. Corrections for ionospheric delay, tropospheric delay, and site displacements fall into this category.
- **Unknown parameter estimation:** For error sources that are difficult to model accurately, the standard approach is to estimate them as unknown parameters within the adjustment process. Typical examples include the receiver clock offset and the wet component of the tropospheric delay.

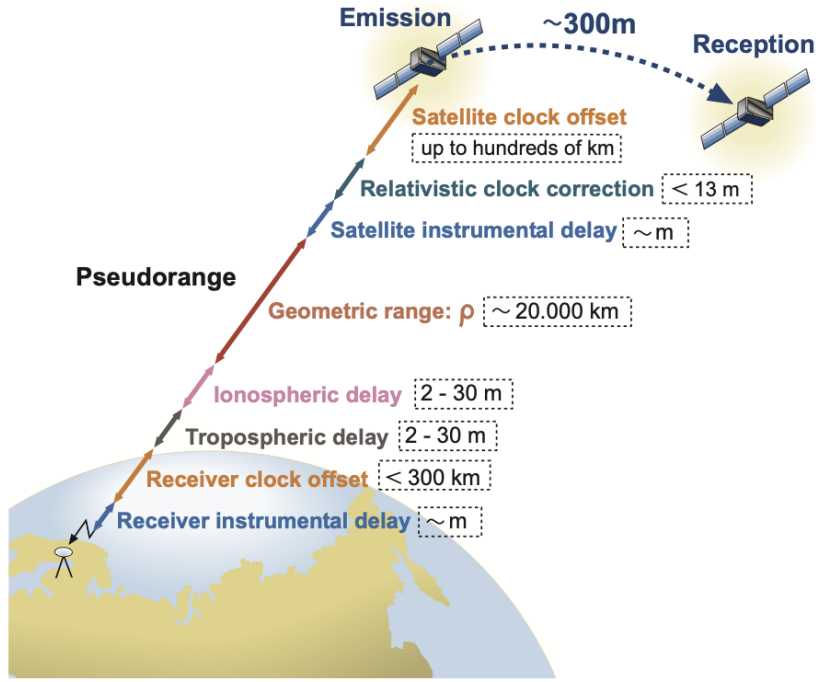


Figure 2.3: Pseudorange measurement error sources (Subirana et al., 2013)

A summary of the modeled corrections and their corresponding magnitudes is provided in Table 2.6.

2.6.1 Relativistic Effects

The relativistic corrections for GNSS observations account for both the Shapiro effect and the relativistic clock correction.

The Shapiro effect is a propagation delay of the satellite signal caused by the Earth's gravitational field, which introduces an additional delay due to the curvature of space-time. The corresponding correction can be computed as

$$\delta t_{\text{stc}}^{\text{rel}} = \frac{2\mu}{c^3} \ln \left(\frac{\|\mathbf{r}^s\| + \|\mathbf{r}_r\| + \rho_r^s}{\|\mathbf{r}^s\| + \|\mathbf{r}_r\| - \rho_r^s} \right), \quad (2.20)$$

where μ is the Earth's gravitational constant. The maximum Shapiro effect is approximately 60 ps, corresponding to about 2 cm, for MEO satellites of GNSS (p. 564, Teunissen, Montenbruck, et al., 2017).

The motion of the satellite, as well as variations in the gravitational potential, introduce a relativistic clock error. This effect is largely mitigated by applying a frequency offset to the satellite clock. However, deviations from a perfectly circular orbit cause residual errors with respect to this mean offset. The additional effect due to orbital eccentricity can be calculated as

$$\delta t_{\text{clk}}^{\text{rel}} = -\frac{2}{c^2} \sqrt{a\mu} e \sin E = -\frac{2}{c^2} (\mathbf{r}^s \cdot \mathbf{v}^s), \quad (2.21)$$

where a is the semimajor axis, e is the orbital eccentricity, E is the eccentric anomaly of the satellite orbit, $\mathbf{r}^s$ and $\mathbf{v}^s$ denote the satellite position and velocity vectors in ECEF, respectively. The typical

Table 2.6: Selective PPP a priori correction models. (Teunissen, Montenbruck, et al., 2017, p. 727)

Model component	Magnitude	Uncertainty	Notes
Satellite			
Antenna phase center offset	0.5–3 m	10 cm	IGS ANTEX model
Phase center variations	5–15 mm (GPS)	0.2–1 mm	IGS ANTEX model
Clock offset	< 1 ms	75 ps (GPS)	From precise clock product
Relativistic clock effects	10–20 m	–	Eccentricity-dependent effect
Differential code biases	up to 15 ns, 5 m	0.1–1 ns	From biases produces
Fractional phase biases	up to 0.5 cy	0.01 cy	From biases produces for ambiguity resolution
Atmosphere			
Troposphere (dry)	2.3 m	5 mm	Models available
Troposphere (wet)	up to 0.3 m	up to 100%	Estimated due to insufficient a priori models
Ionosphere (1st-order)	up to 30 m	– / 1 m	Corrected through ionosphere-free combination
Ionosphere (higher-order)	0–2 cm	1–2 mm	Higher-order ionospheric effects
Site displacement			
Plate motion	up to 0.1 m/y	0.3 mm/y	Corrections for expressing measured positions in a conventional terrestrial reference frame
Solid Earth tide	up to 0.4 m	1 mm	Solid Earth tide models available
Ocean loading (tidal)	1–10 cm	1–2 mm	Tidal ocean loading effects
Ocean loading (nontidal)	up to 15 mm	1 mm	Nonconventional correction
Pole tide	25 mm	–	Conventional pole tide correction
Atmospheric loading (tidal)	up to 1.5 mm	–	Conventional atmospheric tide correction
Atm. loading (nontidal)	up to 20 mm	15%	Nonconventional correction
Receiver			
Phase center offset	5–15 cm	–	IGS ANTEX model
Phase center variations	up to 3 cm	1–2 mm	IGS ANTEX model
Others			
Phase wind-up	10 cm	see 2.6.4	Wavelength dependent; correction subject to knowledge of satellite/receiver antenna orientation

magnitude of the relativistic clock correction is on the order of nanoseconds. For example, assuming the maximum tolerated eccentricity of 0.02 for GPS, the maximum value of $\delta t_{\text{clk}}^{\text{rel}}$ is about 45.0 ns, corresponding to a range error of approximately 13.5 m (p. 564, Teunissen, Montenbruck, et al., 2017).

2.6.2 Atmospheric Signal Delays

GNSS satellites transmit signals from space through the Earth's atmosphere to the user. At an altitude of approximately 1000 km, the signals encounter the ionosphere, a layer of charged particles that extends down to about 50 km. Below the ionosphere lies the troposphere, which is composed primarily of neutral gases and constitutes the lowest layer of the atmosphere.

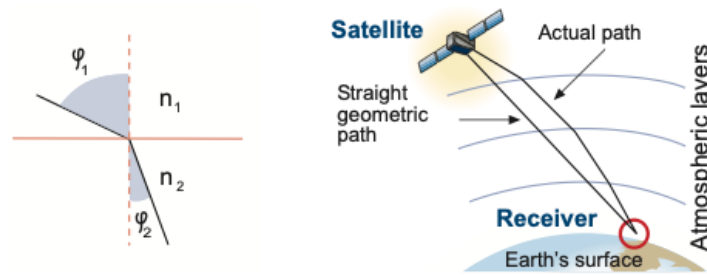


Figure 2.4: Snell's law and its effects over the ray path. (Subirana et al., 2013)

The density of atmospheric gases and plasma is not homogeneous. This leads to spatial and temporal variations in the refractive index, causing GNSS signals to follow an effective path that is longer than the geometric free-space path when traveling through the atmosphere (Subirana et al., 2013, p. 109). Snell's law describes this bending of the signal, as illustrated in Fig. 2.4. This effect introduces delays in GNSS observations, which must be modeled and corrected.

Ionospheric delay

The ionosphere is a layer of the Earth's atmosphere that contains free electrons, which can cause delays in the propagation of GNSS signals. This delay is dispersive, meaning it affects different frequencies differently. In the context of dual-frequency PPP, the ionosphere-free (IF) combination is employed to eliminate the first-order ionospheric error. The IF combination can effectively eliminate up to 99.9% of the slant delay.

Another possibility is to compute the total ionospheric delay from the Total Electron Content (TEC), since the propagation speed of GNSS electromagnetic signals in the ionosphere depends on the electron density. If a TEC-based correction is available, the ionospheric delay can be calculated as

$$I_r^s = \frac{40.3 \cdot 10^6}{f^2} \cdot \text{STEC} \quad (2.22)$$

Tropospheric delay

The troposphere is a non-dispersive layer of the neutral atmosphere for L-band signals. Consequently, both pseudorange and carrier-phase measurements experience the same delay in this re-

gion of the atmosphere, independent of their carrier frequencies. Thus, the only way to mitigate the tropospheric effect is to use models and/or estimate it from the observational data.

Most of the tropospheric delay (about 90%) comes from the predictable hydrostatic component. The total delay is decomposed to a dry component and a wet contribution.

$$T_r^s = \text{ZTD}_d m_d(E) + \text{ZTD}_w m_w(E) \quad (2.23)$$

where ZTD_d and ZTD_w are the dry and wet ZTDs, and m_d and m_w are the corresponding mapping functions.

Most of the mapping functions currently used to scale the zenith delay to a slant delay at the satellite elevation angle E are based on a continued fraction expansion of $1/\sin(E)$. This approach was first suggested by *Marini* (Marini, 1972) and later formalized by *Herring* (Herring, 1992).

In the BNC, a simple mapping function is implemented, which depends solely on the satellite elevation angle and does not include any additional coefficients:

$$m(E) = \frac{1}{\sin(E)}. \quad (2.24)$$

Other empirical models have also been developed to improve the mapping of zenith delays. The Niell Mapping Function (NMF) provides globally valid coefficients based on latitude, station height, and day of year, but its accuracy is limited by its reliance on climatological averages (Niell, 1996). The Vienna Mapping Function 1 (VMF1), derived from numerical weather models, adapts to actual atmospheric conditions and therefore offers higher accuracy, especially for precise GNSS applications (Böhm et al., 2006). Additionally, the Global Pressure and Temperature models (GPT, GPT2, and GPT3) provide site-dependent atmospheric parameters from climatological data and are frequently used to estimate the zenith hydrostatic delay or to supply coefficients for empirical mapping functions when numerical weather model data are unavailable (Böhm et al., 2007).

The BNC employs a very general standard-atmosphere model to estimate pressure and temperature using only the station height. The standard atmosphere can be expressed as

$$p(h) = 1013.25 \text{ hPa} \left(1 - 2.25 \cdot 10^{-5} \text{ m}^{-1} \cdot h\right)^{5.225}, \quad (2.25)$$

$$T(h) = T_0 - 0.0065 \cdot h, \quad (2.26)$$

$$e(T) = \frac{h}{100} \cdot \exp(-37.2465 + 0.213166 \cdot T - 0.000256908 \cdot T^2). \quad (2.27)$$

Here, $p(h)$ denotes the barometric pressure as a function of altitude, which follows the empirical model of (Berg, 1948). The function $T(h)$ represents the temperature profile of the U.S. Standard Atmosphere, with $T_0 = 288.15 \text{ K}$ as the standard sea-level temperature. In this model, the temperature decreases at a lapse rate of approximately 6.5 K per kilometer of altitude. Similarly, $e(T)$ corresponds to the partial pressure of water vapor as a function of temperature, which is also derived from the standard atmosphere representation.

Based on these atmospheric parameters, the total hydrostatic delay of the troposphere can be computed using the Saastamoinen model:

$$T_r^s = \frac{0.002277}{\cos z} \left[p + \left(\frac{1255}{T} + 0.05 \right) e - B \tan^2 z \right], \quad (2.28)$$

where z is the zenith angle, defined as $z = \frac{\pi}{2} - E_r^s$. The parameter B is a correction factor that accounts for variations in gravity with height, and thus depends on the station altitude above sea level.

In the BNC, a simplified implementation is used in which B is taken from a reduced version of the numerical coefficient table originally provided by Saastamoinen (Saastamoinen, 1973). Only a subset of the tabulated values is included, as shown in Table 2.7.

Station Height Above Sea Level	B, mb
0 km	1.156
1 km	1.006
2 km	0.874
3 km	0.757
4 km	0.654
5 km	0.563

Table 2.7: Saastamoinen B correction factor as a function of station height above sea level (Saastamoinen, 1973).

This simplified lookup-table approach reduces computational complexity while still retaining the main dependency of the hydrostatic delay on station altitude. However, it neglects finer variations in gravity, pressure, and temperature that can be captured by more advanced tropospheric models or by numerical weather models. As a result, while sufficient for standard positioning applications, it may introduce small residual errors in precise GNSS processing.

2.6.3 Site Displacement Effects

Solid Earth tides

Solid Earth tides are caused by the gravitational attraction of the Moon and the Sun, which deform the Earth's crust by up to several decimeters. This deformation is commonly described using the Love numbers (h_2, l_2), which scale the radial and horizontal displacements relative to the applied tidal forces. Considering only the degree-2 terms of the tidal potential is adequate to account for nearly all of the solid Earth tide correction. Following this approximation, the vector displacement of the station due to the degree 2 tides is given by (McCarthy, 1996):

$$\Delta \mathbf{r}_{\text{solid}} = \sum_{j=\odot, \oplus} \frac{GM_j}{GM} \frac{\|\mathbf{r}_r\|^4}{\|\mathbf{R}_j\|^3} \left\{ 3l_2(\hat{\mathbf{R}}_j \cdot \hat{\mathbf{r}}_r) \hat{\mathbf{R}}_j + \left[3\left(\frac{h_2}{2} - l_2\right)(\hat{\mathbf{R}}_j \cdot \hat{\mathbf{r}}_r)^2 - \frac{h_2}{2} \right] \hat{\mathbf{r}}_r \right\}$$

where:

- GM_j : gravitational parameter for the Moon or the Sun
- GM : gravitational parameter for the Earth
- $\hat{\mathbf{R}}_j, \|\mathbf{R}_j\|$: unit vector from the geocenter to the Moon or Sun and the magnitude of that vector
- $\hat{\mathbf{r}}_r, \|\mathbf{r}_r\|$: unit vector from the geocenter to the station and the magnitude of that vector
- l_2 : nominal degree 2 Shida number
- h_2 : nominal degree 2 Love number

Other effects

For the site displacement effects other than the solid Earth tides, no corrections are applied in the BNC, as their magnitudes are relatively small and their impact on the final positioning results is considered negligible. Atmospheric loading effects are typically very small, while ocean tidal loading can reach up to about 10 cm for coastal stations. In BNC, ocean loading is corrected only for static station processing; however, it is not applied in this study, as the receiver is moving and the effect is negligible for kinematic PPP processing.

Rotational deformation due to polar motion (polar tides) is also not corrected, as this effect is relatively small (typically below 2 cm) and therefore has a negligible influence on the positioning performance.

2.6.4 Phase Wind-up

GNSS satellites primarily transmit Right-Hand Circularly Polarized (RHCP) electromagnetic waves. This polarization helps mitigate multipath effects and avoids constraints related to the relative orientation between satellite and receiver antennas. (Teunissen, Montenbruck, et al., 2017, p. 569, 570) However, the change in the orientation of the two antennas relative to the line of sight will cause a variation in the carrier phase measurements. This is known as *phase wind-up*. (Wu et al., 1993)

The phase wind-up effect can be modeled using the effective dipole method (Wu et al., 1993). In this approach, both the satellite and the receiver antenna are represented as crossed dipole antennas. Let $\hat{\mathbf{k}} = -\mathbf{e}_r^s$ denote the unit vector from the satellite to the receiver antenna. Then, the effective dipole vectors of the satellite, $\mathbf{D}_s$, and the receiver, $\mathbf{D}_r$, can be expressed as follows:

$$\mathbf{D}_s = \hat{\mathbf{x}}_s - \hat{\mathbf{k}}(\hat{\mathbf{k}} \odot \hat{\mathbf{x}}_s) + \hat{\mathbf{k}} \times \hat{\mathbf{y}}_s \quad (2.29)$$

$$\mathbf{D}_r = \hat{\mathbf{x}}_r - \hat{\mathbf{k}}(\hat{\mathbf{k}} \odot \hat{\mathbf{x}}_r) + \hat{\mathbf{k}} \times \hat{\mathbf{y}}_r \quad (2.30)$$

where $\hat{\mathbf{x}}_s, \hat{\mathbf{x}}_r$ and $\hat{\mathbf{y}}_s, \hat{\mathbf{y}}_r$ are the satellite and receiver dipole unit vectors.

The fractional component of the phase wind-up correction in radians can be written as

$$\Delta\varphi_w = \text{sign}(\zeta) \arccos\left(\frac{\mathbf{D}_s \odot \mathbf{D}_r}{\|\mathbf{D}_s\| \|\mathbf{D}_r\|}\right) \quad (2.31)$$

$$\zeta = \hat{\mathbf{k}} \odot (\mathbf{D}_s \times \mathbf{D}_r) \quad (2.32)$$

To maintain the continuity of the phase wind-up correction, the integer part N is determined using the preceding correction value $\varphi_{w_{prev}}$ as follows:

$$N = \text{round}\left(\frac{\varphi_{w_{prev}} - \Delta\varphi_w}{2\pi}\right) \quad (2.33)$$

The value of N can be initialized with zero. The total phase wind-up correction becomes

$$\varphi_w = 2\pi N + \Delta\varphi_w \quad (2.34)$$

The final carrier-phase observation is to be subtracted by the phase wind-up correction.

2.6.5 Antenna Phase-Center Offset and Variations

Receiver antenna phase center

The distance measurements of GNSS are referenced to the phase centers of the satellite and receiver antennas. For receiver antennas, the antenna phase center is usually specified relative to the antenna reference point (ARP), which serves as a common physical reference point for all signals. According to the IGS, the ARP is defined for most antennas as the intersection of the vertical symmetry axis of the antenna with the bottom of the ground plane or choke ring. The mean electrical phase center differs from the ARP by the phase center offset vector $\mathbf{r}_{PCO,j}$, which is frequency dependent. (Teunissen, Montenbruck, et al., 2017, p. 573)

In addition, the residual deviations of the mean antenna phase center should be considered in the modeling. These phase center variations (PCV) of the transmitting antenna, denoted as $\zeta_{PCV,r,j}$, are frequency-, elevation-, and azimuth-dependent. The corrections for the PCO and PCV are obtained from calibration procedures carried out by institutions such as NGS, Geo++ , Bonn University, and Hannover University (Kallio et al., 2018; Kouba, 2015; Schmid et al., 2016) and are distributed to users in the ANTEX format (*ANTEX: The Antenna Exchange Format, Version 1.4*, 2015).

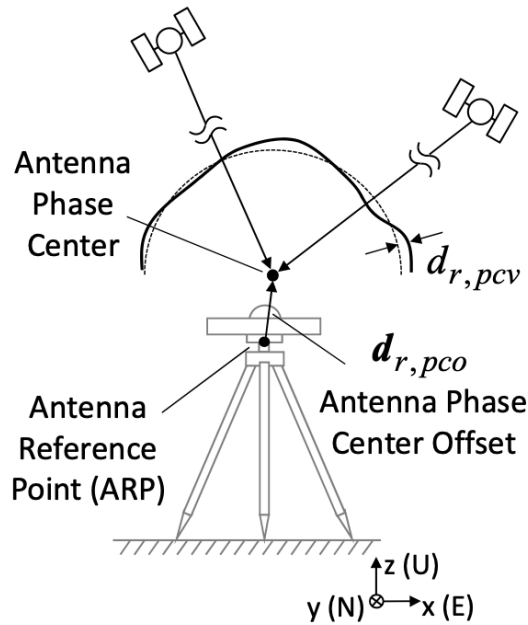


Figure 2.5: Receiver Antenna Phase Center (Takasu and Kubo, 2013)

Satellite antenna phase center

In high-accuracy orbit products used for PPP applications, the satellite position is typically referenced to its center of mass (CoM), which serves as the primary reference point for orbit determination. However, GNSS signals are transmitted from the phase center of the satellite's antenna, not from the CoM (Teunissen, Montenbruck, et al., 2017, p. 730). Therefore, it is essential to account for the CoM offset between these two reference points, as well as the orientation of the offset vector in space.

In addition to the phase center offset (PCO) defined in the satellite body-fixed frame, knowledge of the satellite's attitude, its orientation relative to the terrestrial reference frame, is required to

correctly project this offset and determine the actual phase center position in Earth-centered coordinates.

Furthermore, phase center variations (PCV) must be considered in high-precision carrier-phase modeling to account for small, direction-dependent deviations of the phase center. Incorporating both PCO and PCV corrections enables consistent modeling of all signals and lines of sight with respect to a common effective phase center.

2.6.6 Signal Biases

Each ranging code generated onboard a satellite experiences a distinct delay between its generation and transmission from the antenna phase center. This delay arises from the different analog and digital signal paths associated with each signal (Kaplan and Hegarty, 2017). These hardware-induced biases affect not only the pseudorange but also the carrier-phase observations. Since most of these biases are not directly observable or measurable, they must be estimated or calibrated through ground-based processing (Teunissen, Montenbruck, et al., 2017, p.576).

Code biases originate from time delays introduced by the satellite hardware, which affect the pseudorange observations. For GPS, the published clock offsets, both in the broadcast ephemerides and in the precise products, are derived from the ionosphere-free combination of P(Y)-code observations on the L1 and L2 frequencies (Teunissen, Montenbruck, et al., 2017, p. 577).

Differential Code Biases

It is important to note that the absolute delay of an individual signal is unobservable; only Differential Code Biases (DCB) between pairs or combinations of signals can be determined. These biases are obtained by selecting one signal (or a specific combination) as a reference and expressing the delay of other signals relative to it (Teunissen, Montenbruck, et al., 2017, p. 577).

When dual-frequency signals different from those used in the satellite clock computation are employed, satellite-specific DCBs must be applied. These corrections account for the group delay differences between the signals tracked by the receiver and those used as the clock reference signals (Montenbruck and Hauschild, 2013).

Observable-Specific Biases (OSB)

A more consistent and user-friendly approach to applying bias and clock corrections in precise point positioning is through Observable-Specific Biases (OSBs). The OSB concept allows analysis centers to provide signal-specific corrections that maintain internal consistency between satellite clock and bias products (Villiger et al., 2019).

Unlike traditional Differential Code Biases (DCBs), which represent relative delays between pairs of signals, OSBs are defined per observable (i.e., per signal). This enables a consistent application of corrections across different constellations and signal combinations. OSBs are typically distributed in the Bias-SINEX format for post-processing applications, where each correction corresponds to a specific satellite and signal.

Importantly, the real-time RTCM-SSR format also adopts the OSB concept for distributing code and phase bias corrections. This ensures compatibility between real-time and post-processed PPP and PPP-AR applications, allowing users to apply the same bias model seamlessly in both domains.

By providing signal-specific corrections rather than differential ones, OSBs eliminate the need for users to know the reference signals or processing strategies used by the analysis centers. This greatly simplifies user-side implementation while ensuring internal consistency between clocks and biases, which is essential for reliable ambiguity resolution in PPP-AR.

Phase biases

Similar to the pseudorange measurements, carrier-phase observations are also affected by instrumental delays originating from both the satellite and the receiver hardware. These carrier-phase biases are particularly critical because their fractional components prevent the integer resolution of the phase ambiguities. In precise positioning, such biases can be mitigated either by forming double-differenced observations with respect to a reference station or by applying externally provided correction values derived from global or regional reference networks. Although the exact definition of these corrections depends on the parameterization of the estimation model, extensive efforts have been made to characterize and separate the satellite- and receiver-dependent components, thereby enabling integer ambiguity resolution through the use of phase bias corrections (Teunissen, Montenbruck, et al., 2017, p.578).

2.7 Real-Time Corrections

In real-time applications, GNSS orbit and clock corrections must be delivered instantaneously to the point of data collection. Transmission of real-time corrections for PPP is typically performed via the Internet using dedicated transport protocols, such as the Ntrip standard.

As one of the services offered by Galileo, the Galileo High Accuracy Service (Galileo HAS) is a free and open-access service that provides precise corrections to users. These corrections are transmitted via the Galileo E6 signal (E6-B, data component), allowing improved positioning performance even without Internet connectivity. This is referred to as the Galileo HAS Signal In Space (SIS). In addition, the same corrections are distributed over the Internet through the Galileo HAS Internet Data Distribution (IDD) interface, also based on the Ntrip protocol.

Similar to the dissemination of SSR corrections via satellite links in Galileo HAS, other regional satellite navigation systems have also implemented comparable services. The Japanese Quasi-Zenith Satellite System (QZSS) provides the Centimeter Level Augmentation Service (CLAS) through its L6D signal, delivering precise orbit, clock, and bias corrections over Japan and the surrounding regions (Hao et al., 2020). Likewise, China's BeiDou Global Navigation Satellite System (BDS-3) officially introduced its satellite-based PPP service in 2021, broadcasting precise corrections via the B2b signal (PPP-B2b) from three geostationary satellites (Gao et al., 2024).

2.7.1 State Space Representation

The term SSR refers to the concept of modeling individual GNSS error sources to provide users with improved positioning accuracy. The SSR framework typically consists of the following parameters ("IGS State Space Representation (SSR) Format, Version 1.00", 2020):

- Satellite orbit errors
- Satellite clock errors

- Satellite signal biases (code and phase)
- Ionospheric and tropospheric delays

Satellite orbit and clock errors are estimated to refine the broadcast ephemerides, which describe the satellite's position and clock offset. Additionally, satellite code and phase biases account for hardware-induced delays specific to each signal.

SSR corrections are derived from the processing of GNSS observations collected within a reference station network. The resulting SSR parameters are transmitted to users, who then apply them as corrections to their own observations.

RTCM-SSR Messages

The RTCM SC-104 SSR Working Group was established in 2007 to define standardized message formats for the dissemination of real-time GNSS corrections. These messages provide orbit, clock, and bias corrections to be applied to the broadcast ephemerides, enabling high-precision positioning techniques such as PPP.

In the RTCM-SSR specification, standardized message types have been defined for the GPS and GLONASS constellations. Each constellation includes separate message types for orbit corrections, clock corrections, combined orbit–clock corrections, high-rate clock updates, code bias corrections, and user range accuracy parameters (RTCM Special Committee No. 104, 2016).

While additional parameters, such as phase bias corrections and support for other GNSS constellations, have been proposed within the RTCM-SSR framework, these extensions have not yet been formally standardized.

IGS RTS

Although the RTCM-SSR specification laid the foundation for standardized real-time GNSS corrections, it has not seen major updates in recent years, leading to limitations in message completeness and update frequency. To overcome these constraints, the International GNSS Service (IGS) developed its own SSR convention, which closely follows the RTCM structure but extends it with additional metadata, message identifiers, and bias-handling conventions. These enhancements ensure global consistency, interoperability, and improved temporal resolution in real-time correction products (“IGS State Space Representation (SSR) Format, Version 1.00”, 2020).

To meet the growing demand for real-time PPP, the IGS officially introduced the Real-Time Service (RTS) in 2013 (Elsobeiey and Al-Harbi, 2016). Today, most real-time correction services, such as the IGS-RTS, adopt the IGS SSR format. This standardization ensures compatibility with widely used software tools, including the BNC, which applies these SSR corrections directly to raw GNSS observations for real-time PPP processing.

Currently, real-time SSR corrections are available from several IGS analysis centers, including Wuhan University, the Centre National d'Études Spatiales (CNES), and the BKG. A complete list of SSR stream providers can be found in (“List of SSR Providers”, 2025).

The application of SSR corrections enables users to refine the broadcast satellite orbit and clock information in real-time. These corrections are transmitted as incremental updates relative to the broadcast ephemerides and must be applied to obtain precise satellite positions and clock offsets that are consistent with the reference frame of the correction provider.

In practice, SSR corrections include both orbit and clock parameters, expressed in a compact form suitable for real-time transmission. The orbit corrections are defined in a local satellite reference frame, consisting of the radial, along-track, and cross-track directions, while the clock corrections are modeled as a polynomial function of time. Together, these parameters describe the temporal evolution of the corrections during the validity period of the message.

The complete set of SSR correction parameters can be expressed as:

$$\Delta_{\text{SSR}}(t_0, \text{IOD}) = (\delta O_{\text{radial}}, \delta O_{\text{along}}, \delta O_{\text{cross}}, \delta \dot{O}_{\text{radial}}, \delta \dot{O}_{\text{along}}, \delta \dot{O}_{\text{cross}}, C_0, C_1, C_2) \quad (2.35)$$

where t_0 is the reference epoch and IOD denotes the Issue of Data identifier that ensures temporal consistency between orbit and clock information.

2.7.2 Orbit Corrections

The RTCM-SSR orbit corrections describe the deviation between the broadcast and precise satellite positions. These are defined in the satellite's local orbital frame and expressed as both position and velocity terms:

$$\delta \mathbf{O} = [\delta O_{\text{radial}} \ \delta O_{\text{along}} \ \delta O_{\text{cross}}] + [\delta \dot{O}_{\text{radial}} \ \delta \dot{O}_{\text{along}} \ \delta \dot{O}_{\text{cross}}](t - t_0) \quad (2.36)$$

The orthogonal unit vectors defining the satellite reference frame are computed as:

$$\mathbf{e}_{\text{along}} = \frac{\mathbf{v}_{\text{broadcast}}^s(t)}{|\mathbf{v}_{\text{broadcast}}^s(t)|} \quad \mathbf{e}_{\text{cross}} = \frac{\mathbf{r}_{\text{broadcast}}^s(t) \times \mathbf{v}_{\text{broadcast}}^s(t)}{|\mathbf{r}_{\text{broadcast}}^s(t) \times \mathbf{v}_{\text{broadcast}}^s(t)|} \quad \mathbf{e}_{\text{radial}} = \mathbf{e}_{\text{along}} \times \mathbf{e}_{\text{cross}} \quad (2.37)$$

with

- $\mathbf{r}_{\text{broadcast}}^s$: Earth Centered Earth Fixed (ECEF) satellite broadcast position vector
- $\mathbf{v}_{\text{broadcast}}^s$: ECEF satellite broadcast velocity vector
- $\mathbf{e}_i$: direction unit vector, $i = \{\text{radial}, \text{along}, \text{cross}\}$

The corrected satellite position is then obtained as:

$$\mathbf{r}^s(t) = \mathbf{r}_{\text{broadcast}}^s(t) + [\mathbf{e}_{\text{radial}}, \mathbf{e}_{\text{along}}, \mathbf{e}_{\text{cross}}] \delta \mathbf{O} \quad (2.38)$$

2.7.3 Clock Corrections

The SSR clock correction provides the deviation of the satellite clock from the broadcast value and is expressed as a low-degree polynomial with an optional high-rate term:

$$\delta C = C_0 + C_1(t - t_0) + C_2(t - t_0)^2 + \delta C_{\text{High Rate}} \quad (2.39)$$

where:

- t : current epoch
- t_0 : reference epoch from the SSR Clock Correction message

- C_i : polynomial coefficients, $i = 0, 1, 2$
- $\delta C_{\text{High Rate}}$: high-rate clock correction term

The corrected satellite clock offset is then computed as:

$$dt^s(t) = dt_{\text{broadcast}}^s(t) - \frac{\delta C}{c} \quad (2.40)$$

with:

- t_{sat} : satellite time
- c : speed of light

2.7.4 Bias corrections

After applying the orbit and clock corrections, signal-specific bias corrections must also be added to the raw measurements. These biases, provided in the form of Observable-specific Biases (OSB), ensure consistency between satellite and receiver signals. Since OSB are defined relative to an internally established datum or reference, users can directly apply them without concern for reference signal definitions.

The sign convention used in the SSR satellite bias messages specifies that the reported code and phase biases must be added to the corresponding pseudorange or carrier-phase measurements in order to obtain the corrected observables (“IGS State Space Representation (SSR) Format, Version 1.00”, 2020).

Chapter 3 PPP Models

3.1 Precise Point Positioning

Precise Point Positioning (PPP) is a GNSS-based method that enhances navigation accuracy by combining pseudorange measurements with more precise carrier phase observations. Relative positioning techniques require measurements from nearby reference stations to calculate precise corrections. PPP enables absolute positioning by utilizing precise satellite orbit and clock products, along with satellite signal bias corrections. By replacing the satellite broadcast ephemerides with precise data sets, PPP can achieve centimeter-level accuracy without relying on local infrastructure.

3.2 Conventional Model

The most commonly used observation model in the context of PPP is the **ionosphere-free (IF) combination**. This combination exploits the dispersive nature of the ionospheric delay to effectively eliminate its first-order effect, reducing approximately 99% of the total ionospheric error. The ionosphere-free pseudorange and carrier-phase combinations are expressed as

$$p_{r,IF}^s = \rho_r^s + c(dt_r - dt^s) + T_r^s + e_{IF}, \quad (3.1)$$

$$\varphi_{r,IF}^s = \rho_r^s + c(dt_r - dt^s) + T_r^s + \lambda_{IF} A_{IF} + \varepsilon_{IF}, \quad (3.2)$$

where ρ_r^s is the geometric range between the receiver and satellite, $c(dt_r - dt^s)$ represents the receiver and satellite clock offsets, and T_r^s denotes the tropospheric delay. The terms e_{IF} and ε_{IF} represent the measurement noise and unmodeled errors of the pseudorange and carrier-phase observables, respectively.

The ambiguity term A_{IF} and the corresponding IF wavelength and ambiguity are defined as follows:

$$A_{IF} = N_{IF} + \frac{(b_{r,\varphi_{IF}} - b_{r,p_{IF}}) - (b_{\varphi_{IF}}^s - b_{p_{IF}}^s)}{\lambda_{IF}}, \quad (3.3)$$

$$\lambda_{IF} = \frac{f_1^2}{f_1^2 - f_2^2} \lambda_1 - \frac{f_2^2}{f_1^2 - f_2^2} \lambda_2, \quad (3.4)$$

$$N_{IF} = \frac{f_1^2}{f_1^2 - f_2^2} N_1 - \frac{f_2^2}{f_1^2 - f_2^2} N_2. \quad (3.5)$$

Here, $b_{r,\varphi_{IF}}$ and $b_{r,p_{IF}}$ denote the receiver hardware biases of the carrier-phase and pseudorange observables, respectively, and $b_{\varphi_{IF}}^s$ and $b_{p_{IF}}^s$ represent the corresponding satellite biases. λ_{IF} and N_{IF} are the effective wavelength and ambiguity of the ionosphere-free combination.

3.3 SSR Corrections

For PPP, it is required to use the most precise satellite orbits and clock corrections available. The satellites' ephemerides are treated as known quantities. This means that the PPP model does not consider the errors in these precise ephemerides, therefore the accuracy level of the satellite orbits and clocks are the limiting factor of the final solution. The products are divided into two different categories, post-processed and real-time. The post-processed product can be accessed from ftp-servers of the IGS and contributing analysis centers with different latencies. The another possibility is to receive real-time correction streams, which are transmitted via a new RTCM standard - the SSR messages.

Table 3.1: Relevant RTCM-SSR messages for GPS and Galileo corrections (as used by IGS and CNES real-time services).

RTCM Message	IGS Message	Description	Occurrence (s)
1057, 1240	–	GPS, Galileo Orbit Corrections	5
1058, 1241	–	GPS, Galileo Clock Corrections	5
1060, 1243	23, 63	Combined GPS, Galileo Orbit and Clock Corrections	5
1059, 1242	25, 65	GPS, Galileo Code Biases	5
1265, 1267	26, 66	GPS, Galileo Phase Biases (for PPP-AR)	5

Note: Messages 1060 and 1243 provide a combined orbit and clock correction and are the standard in IGS real-time products (e.g., IGS Message 23,63). Messages 1057/1058 are alternative formats used by some providers (e.g., CNES, GFZ) transmitting orbit and clock corrections separately.

3.4 Adjustment in PPP

3.4.1 Bancroft Algorithm

The observation equations in GNSS are inherently non-linear and are typically solved using iterative methods that linearize the equations by constructing a matrix of partial derivatives. In contrast, the Bancroft algorithm (Bancroft, 2007) offers a direct, non-iterative solution for estimating the receiver's position and clock offset.

The first step of the Bancroft method involves constructing the B matrix. When four or more pseudorange measurements are available, this matrix can be formed to include the coordinates of all satellites along with the corresponding pre-fit residuals, denoted by x^j, y^j, z^j, PR^j for satellite j .

$$B = \begin{bmatrix} x^1 & y^1 & z^1 & PR^1 \\ x^2 & y^2 & z^2 & PR^2 \\ \vdots & \vdots & \vdots & \vdots \\ x^n & y^n & z^n & PR^n \end{bmatrix} \quad (3.6)$$

Further terms for the computation are being formed using the inner product of Lorentz.

$$\Lambda = \frac{1}{2} \left\langle \begin{bmatrix} \mathbf{r} \\ cdt_r \end{bmatrix}, \begin{bmatrix} \mathbf{r} \\ cdt_r \end{bmatrix} \right\rangle, \quad \mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \quad \text{with } a_j = \frac{1}{2} \left\langle \begin{bmatrix} \mathbf{r}^j \\ PR^j \end{bmatrix}, \begin{bmatrix} \mathbf{r}^j \\ PR^j \end{bmatrix} \right\rangle \quad (3.7)$$

with $\mathbf{r} = [x, y, z]^T$ being the unknown receiver position and cdt_r being the receiver clock offset. Using these definitions, Bancroft showed that the receiver position can be obtained by solving a single quadratic equation in the unknown parameter Λ :

$$\langle \mathbf{B}^{-1} \mathbf{1}, \mathbf{B}^{-1} \mathbf{1} \rangle \Lambda^2 + 2 [\langle \mathbf{B}^{-1} \mathbf{1}, \mathbf{B}^{-1} \mathbf{a} \rangle - 1] \Lambda + \langle \mathbf{B}^{-1} \mathbf{a}, \mathbf{B}^{-1} \mathbf{a} \rangle = 0 \quad (3.8)$$

The unknown state vector in Equation (3.8) can be obtained as the direct solution of a quadratic equation. The correct root can be verified by substituting it back into the original pseudorange equations and checking for consistency.

If more than four observations are available, the matrix $\mathbf{B}$ is no longer square and therefore cannot be inverted directly. In this case, $\mathbf{B}^{-1}$ is replaced by the generalized inverse,

$$\mathbf{B}^+ = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T, \quad (3.9)$$

which allows the algorithm to incorporate all available satellites while maintaining the same algebraic solution structure.

3.4.2 Kalman Filter

The Kalman filter is a recursive algorithm for estimating the random states of a dynamic system such that the mean squared prediction error is minimized. Following its initialization, the filter proceeds in a recursive two-step cycle consisting of a state prediction and a state update.

Due to the non-linearity of the observation model, an Extended Kalman Filter (EKF) is employed, which linearizes the measurement model. A state vector $\mathbf{x}$ for unknown model parameters and its covariance matrix $\mathbf{P}$ can be estimated with a measurement vector $\mathbf{y}_k$ at epoch t_k by:

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{x}}_k^-)) \quad (3.10)$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_k^- \quad (3.11)$$

Time update of the state vector and its covariance matrix for EKF is expressed as:

$$\hat{\mathbf{x}}_k^- = \Phi_{k-1} \hat{\mathbf{x}}_{k-1}^+ \quad (3.12)$$

$$\mathbf{P}_k^- = \Phi_{k-1} \mathbf{P}_{k-1}^+ \Phi_{k-1}^T + \mathbf{Q}_{k-1}^k \quad (3.13)$$

where Φ_{k-1} is the state transition matrix describing the system dynamics between epochs $k-1$ and k , and $\mathbf{Q}_{k-1}^k$ denotes the process noise covariance, accounting for model uncertainties during the propagation step.

The so-called *Kalman gain* $\mathbf{K}_k$, which minimizes the mean-square estimation error, is computed as:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1}, \quad (3.14)$$

where $\mathbf{P}_k^-$ is the predicted state covariance matrix, $\mathbf{H}_k$ the measurement design (sensitivity) matrix, and $\mathbf{R}_k$ the measurement noise covariance matrix.

3.4.3 Parameters and Observables

The state vector $\mathbf{x}$ contains all the estimated parameters in the IF PPP model, including receiver coordinates, receiver clock offsets, tropospheric delay, and the float ambiguities for each tracked satellite in unit of cycles. The corresponding observation vector $\mathbf{y}$ consists of dual-frequency pseudorange and carrier-phase measurements, both expressed in their ionosphere-free combinations.

$$\mathbf{x} = \begin{bmatrix} X \\ Y \\ Z \\ cdt_r^{\text{GPS}} \\ cdt_r^{\text{Galileo}} \\ T_{\text{ZWD}} \\ A_{\text{IF}}^1 \\ \vdots \\ A_{\text{IF}}^n \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} p_{\text{IF}}^1 \\ \varphi_{\text{IF}}^1 \\ \vdots \\ p_{\text{IF}}^n \\ \varphi_{\text{IF}}^n \end{bmatrix} \quad (3.15)$$

The number of estimated parameters in the Kalman filter depends on the number of tracked satellites, as summarized in Table 3.2. Each satellite contributes one ambiguity term in the IF model, while the other parameters are common to all satellites.

Table 3.2: Parameter amount for estimated filter states for IF model

Estimated parameter	Amount
Receiver Coordinates	3
Receiver Clock GPS	1
Receiver Clock Galileo	1
Tropospheric Delay	1
Float Ambiguities	n
Combined	$6 + n$

The functional relationship between the observations and estimated parameters is expressed through the design matrix $\mathbf{H}$, which contains the partial derivatives of each observation with respect to the estimated parameters. This matrix defines how small changes in the state vector affect the modeled

observations and forms the core of the least-squares adjustment or Kalman filtering process.

$$H = \begin{bmatrix} \frac{\partial p_{IF}^1}{\partial X} & \frac{\partial p_{IF}^1}{\partial Y} & \frac{\partial p_{IF}^1}{\partial Z} & \frac{\partial p_{IF}^1}{\partial cdt_r^{GPS}} & \frac{\partial p_{IF}^1}{\partial cdt_r^{Galileo}} & \frac{\partial p_{IF}^1}{\partial T_{ZWD}} & \frac{\partial p_{IF}^1}{\partial A_{IF}^1} & \cdots & \frac{\partial p_{IF}^1}{\partial A_{IF}^n} \\ \frac{\partial \varphi_{IF}^1}{\partial X} & \frac{\partial \varphi_{IF}^1}{\partial Y} & \frac{\partial \varphi_{IF}^1}{\partial Z} & \frac{\partial \varphi_{IF}^1}{\partial cdt_r^{GPS}} & \frac{\partial \varphi_{IF}^1}{\partial cdt_r^{Galileo}} & \frac{\partial \varphi_{IF}^1}{\partial T_{ZWD}} & \frac{\partial \varphi_{IF}^1}{\partial A_{IF}^1} & \cdots & \frac{\partial \varphi_{IF}^1}{\partial A_{IF}^n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial p_{IF}^n}{\partial X} & \frac{\partial p_{IF}^n}{\partial Y} & \frac{\partial p_{IF}^n}{\partial Z} & \frac{\partial p_{IF}^n}{\partial cdt_r^{GPS}} & \frac{\partial p_{IF}^n}{\partial cdt_r^{Galileo}} & \frac{\partial p_{IF}^n}{\partial T_{ZWD}} & \frac{\partial p_{IF}^n}{\partial A_{IF}^1} & \cdots & \frac{\partial p_{IF}^n}{\partial A_{IF}^n} \\ \frac{\partial \varphi_{IF}^n}{\partial X} & \frac{\partial \varphi_{IF}^n}{\partial Y} & \frac{\partial \varphi_{IF}^n}{\partial Z} & \frac{\partial \varphi_{IF}^n}{\partial cdt_r^{GPS}} & \frac{\partial \varphi_{IF}^n}{\partial cdt_r^{Galileo}} & \frac{\partial \varphi_{IF}^n}{\partial T_{ZWD}} & \frac{\partial \varphi_{IF}^n}{\partial A_{IF}^1} & \cdots & \frac{\partial \varphi_{IF}^n}{\partial A_{IF}^n} \end{bmatrix} \quad (3.16)$$

$$= \begin{bmatrix} \mathbf{e}_r^1 & 1 & 0 & \frac{1}{\sin(E_r^1)} & 0 & \cdots & 0 \\ \mathbf{e}_r^1 & 1 & 0 & \frac{1}{\sin(E_r^1)} & \lambda_{IF} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{e}_r^n & 0 & 1 & \frac{1}{\sin(E_r^n)} & 0 & \cdots & 0 \\ \mathbf{e}_r^n & 0 & 1 & \frac{1}{\sin(E_r^n)} & 0 & \cdots & \lambda_{IF} \end{bmatrix} \quad (3.17)$$

$\mathbf{e}_r^j$ denotes the line-of-sight unit vector pointing from satellite j toward the receiver. The term E_r^j represents the elevation angle of satellite j as seen from the receiver, which is used in the tropospheric mapping function.

It should be noted that the ionosphere-free wavelength λ_{IF} used internally in the BNC differs from the theoretical definition given in 3.4. The wavelength derived from 3.4 is generally non-integer (i.e., a fractional wavelength), whereas the ambiguities in the BNC are handled as parameters expressed in cycles. To preserve the integer nature of the ambiguity states, BNC introduces an *artificial ionosphere-free wavelength*, defined as

$$\lambda_{IF} = \frac{c}{f_1 + f_2}. \quad (3.18)$$

This wavelength is frequency-dependent and therefore differs between GNSS constellations. For example:

- GPS L1/L2: $\lambda_{IF} \approx 10.7$ cm
- Galileo E1/E5a: $\lambda_{IF} \approx 10.9$ cm

This artificial wavelength ensures that the ionosphere-free ambiguities remain compatible with the internal cycle-based state representation used by the BNC.

3.5 Implementation in BNC

The requirement for PPP is to take the raw GNSS observations together with the necessary correction terms as input, and to output a PNT solution. Depending on the processing structure and the selected models, the detailed implementation can differ between software packages. However, the overall processing concept remains analogous: the observations are corrected and modeled, to estimate the receiver state and the associated parameters.

In Figure 3.1, an overview of the processing scheme, from the input RTCM data streams to the output of the float PPP solution, is presented. The complete internal structure and implementation details of BNC can be found in its publicly available source code. Here, only the conceptual processing steps and key components relevant to this thesis are outlined.

The primary components include:

1. **RTCM Decoder:** Responsible for decoding both raw GNSS observations from the receiver and real-time correction data streams (e.g., IGS, Galileo HAS).
2. **Bancroft Model:** Provides an initial estimate of the receiver position and clock offset.
3. **PPP Model:** Implements the core PPP algorithms, including the application of satellite, receiver, and atmospheric corrections, as well as other relevant models.
4. **Error-State Kalman Filter:** Estimates and predicts the receiver state errors with respect to a priori initial values, thereby refining the position solution over time.

The details of binary stream decoding are not discussed further, as they do not contribute directly to the evaluation of PPP performance.

3.5.1 Bancroft Model

In the BNC software, the Bancroft algorithm is executed in two iterations. The first iteration provides a coarse estimate of the receiver position, which is necessary for computing various correction terms using the models implemented in the PPP module. Once these error sources are modeled, the updated measurements are used to form refined pre-fit residuals. In the second iteration, the Bancroft algorithm is applied again using these corrected observations to obtain a more accurate estimate of the receiver position and receiver clock offset.

The two iterations differ in how the pre-fit residuals of IF combination are computed, depending on the corrections available:

$$PR^s = \begin{cases} p_{IF}^s + c dt_{IF}^s, & \text{if } i = 1, \\ p_{IF}^s + c dt_{IF}^s - a_{ecc}^s - T_r^s - T_{earth} - c \delta t^{rel} - \xi_{r, IF}^s - B_{IF}^s, & \text{if } i = 2, \end{cases} \quad (3.19)$$

The additional correction terms applied in the second iteration, including the satellite antenna eccentricity correction, tropospheric delay, Earth tidal correction, relativistic effect, antenna phase center variation, and satellite code bias, are summarized in Table 3.3. Afterward, the solution, namely the coarse receiver position and clock offset, is obtained according to Section 3.4.1.

3.5.2 Error State Kalman Filter

In the BNC, an Error State Kalman Filter (ESKF) is employed. The main distinction from a conventional Kalman filter is that the ESKF estimates the errors of the individual state parameters rather than their absolute values. The resulting error-state vector, denoted as $\Delta \hat{x}$, represents the deviations of the estimated parameters from their a priori or predicted values.

An important advantage of using the ESKF formulation lies in its improved numerical stability. In a conventional Kalman filter, the state vector directly contains the absolute receiver coordinates expressed in the ECEF frame, whose values are typically on the order of 10^6 m. Such large magnitudes can cause poor conditioning of the covariance matrix and increase susceptibility to numerical rounding errors during matrix inversion and update operations.

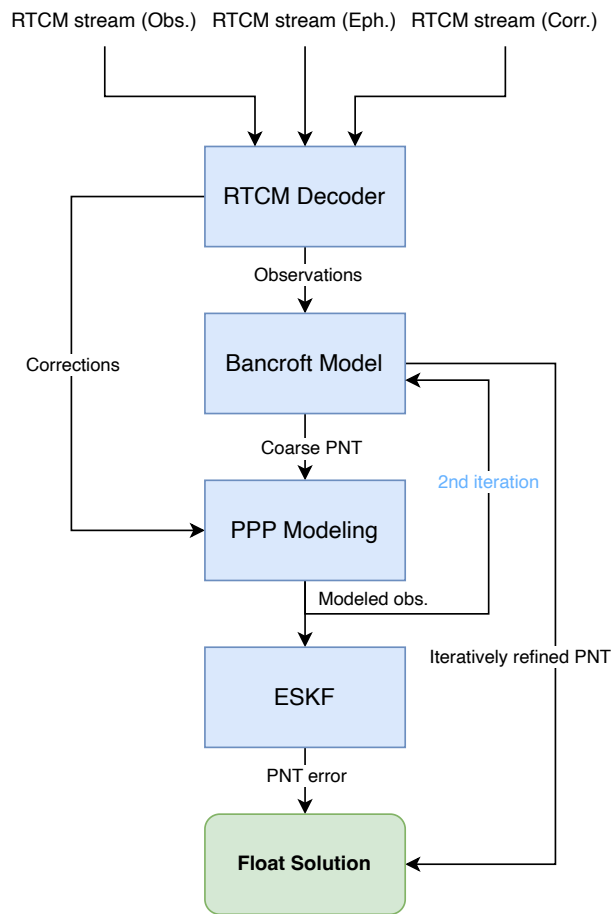
Epoch-wise processing:

Figure 3.1: Processing flow of the real-time Precise Point Positioning (PPP) module implemented in BNC. The system receives multiple RTCM input streams containing raw observations, broadcast ephemerides, and SSR correction data. The RTCM decoder extracts and synchronizes these data streams, which are processed epoch by epoch. A coarse position, navigation, and timing (PNT) solution is first obtained using a Bancroft model and iteratively refined through modeled observations and error corrections. The final stage applies an Extended State Kalman Filter (ESKF) to estimate the float PPP solution, producing precise receiver coordinates and associated quality metrics.

By contrast, the ESKF expresses the states as small deviations rather than absolute coordinates. Since these position errors are typically only at the centimeter or millimeter level, the associated covariance matrices involve much smaller numerical values and are thus better conditioned. This significantly enhances the numerical stability and computational efficiency of the filter, particularly in high-dynamic or long-duration processing scenarios where accumulated numerical errors could otherwise degrade estimation performance.

Table 3.3: PPP Modeled Corrections

Model Component	Symbol	Calculation	Description
Geometric range	ρ_r^s	$\ \rho_r^s\ $	$\rho_r^s = r^s - r_r$, r_r : receiver position in ECEF, r^s : satellite position in ECEF
Receiver Clock offset	t_r	t_r	from Bancroft
Satellite Clock offset	t^s	t^s	from correction stream
Satellite Code Bias	B_j^s	B_j^s	from correction stream
Satellite Phase Bias	b_j^s	b_j^s	from correction stream
Sagnac Effect	S_g	$\omega_E \cdot (r^s \times r_r)$	$\omega_E = [0 \ 0 \ \omega_E/c]^T$, ω_E : Earth's angular velocity, $\times$: Cross product
Antenna Eccentricity	a_{ecc}^s	$-(e_{ecc} \cdot \rho_r^s)/\rho_r^s$	e_{ecc} : Antenna eccentricity in ECEF, $\cdot$: Dot product
Antenna Phase Center Offset and Variation for Code	$\xi_{r,j}^s$	$\xi_{r,j} + \xi_j^s$	$\xi_{r,j}$: receiver antenna phase center offset and variation on j frequency; ξ_j^s : satellite antenna phase center offset and variation on j frequency
Antenna Phase Center Offset and Variation for Phase	$\zeta_{r,j}^s$	$\zeta_{r,j} + \zeta_j^s$	analogously as code
Phase windup	φ_w	$\text{windUp}(t, s, r^s, \psi, v^s)$	See section 2.6.4, correction due to RHCP
Tropospheric Delay	T_r^s	$m(E_r^s) \cdot T_{ZTD}$	E_r^s : Elevation, $m(E)$: elevation dependent mapping function, T_{ZTD} : Zenith total delay from Saastmoinen model section 2.6.2
Earth Tidal Correction	T_{earth}	$-(o_{tide} \cdot \rho_v)/\rho$	o_{tide} : $earth(k, r_r)$, Earth tidal loading via model in section 2.6.3
Relativistic effect due to Earth gravity	$c \delta t^{rel}$	$\frac{2\mu}{c^2} \cdot \ln \left(\frac{\ r^s\ + \ r_r\ + \rho_r^s}{\ r^s\ + \ r_r\ - \rho_r^s} \right)$	$\mu = 3.986004418 \times 10^{14} \text{ m}^3/\text{s}^2$

State prediction

No system model has been used in the BNC. The transition matrix of the state prediction will simply be defined as an identity matrix.

$$\Delta \hat{x}_k^- = \Delta \hat{x}_{k-1}^+ \quad (3.20)$$

The covariance matrix is added by the process noise:

$$P_k^- = P_{k-1}^+ + Q_k^{k+1} \quad (3.21)$$

Square root method

BNC applies the square root method to compute both the Kalman gain and the updated covariance matrix. This approach enhances the numerical stability of the filter, guarantees a positive definite covariance matrix, and offers improved computational efficiency. The basic idea is to propagate the standard deviation ($\sqrt{P}$) rather than variance (P). (Brown and Hwang, 2012)

The square root method is based on a Cholesky decomposition of the predicted covariance matrix and the covariance matrix of the measurement errors, P^- , R , into lower and upper triangular square root factors, here denoted as:

$$P_k^- = \left(\sqrt{P_k^-} \right)^T \sqrt{P_k^-}, \quad R_k = \left(\sqrt{R_k} \right)^T \sqrt{R_k} \quad (3.22)$$

$$\sqrt{P_k^-} = chol(P_k^-)^T, \quad \sqrt{R_k} = chol(R_k)^T = (R_k)^{1/2} \quad (3.23)$$

where:

- $\sqrt{*}$: upper triangular matrix of their square root counterparts from Cholesky decomposition
- $chol(*)$: Cholesky decomposition

The measurement noise covariance matrix R_k is diagonal, in which case the Cholesky decomposition reduces to taking the square root of each diagonal element. In a subsequent step, a QR decomposition is applied to the matrix A , which is constructed from the corresponding square root factors.

$$U = qr(A) = \begin{bmatrix} \left(\sqrt{B_k} \right)^T & W_k^T \\ \mathbf{0} & \left(\sqrt{P_k^+} \right)^T \end{bmatrix} \quad \text{with} \quad A = \begin{bmatrix} \left(\sqrt{R_k} \right)^T & \mathbf{0} \\ \left(\sqrt{P_k^-} \right)^T H_k^T & \left(\sqrt{P_k^-} \right)^T \end{bmatrix} \quad (3.24)$$

The elements of the resulting submatrices are then combined to compute the Kalman gain and to update the covariance matrix:

$$K_k = W_k \left(\sqrt{B_k} \right)^{-1} \quad (3.25)$$

$$P_k^+ = \left(\sqrt{P_k^+} \right)^T \sqrt{P_k^+} \quad (3.26)$$

State update

After obtaining the Kalman gain, the error states of the parameters are updated in an analogous manner:

$$\Delta \hat{\mathbf{x}}_k^+ = \Delta \hat{\mathbf{x}}_k^- + K_k (y_k - h(\hat{\mathbf{x}}_k^-)) \quad (3.27)$$

The processing flow of the square-root formulation of the Kalman filter is illustrated in Figure 3.2.

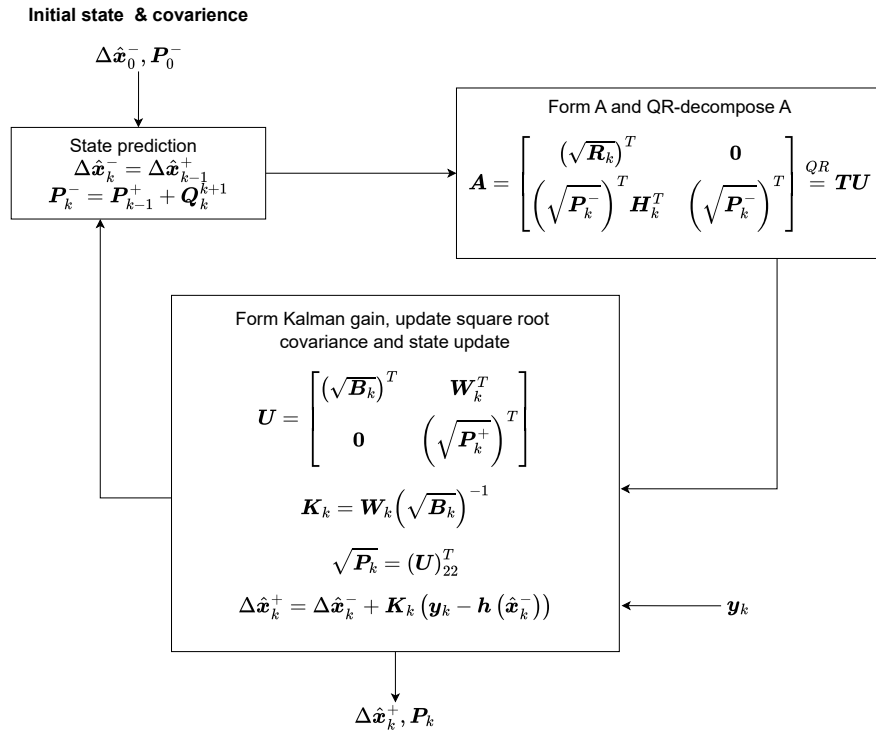


Figure 3.2: Flow chart for square-root Kalman filter

Chapter 4 PPP with Ambiguity Resolution

In the previous chapter, the concept of PPP and its advantages over traditional differential positioning methods were discussed. A major drawback of PPP, however, is its long convergence time to achieve high-accuracy positioning. This is primarily due to the carrier phase ambiguities being affected by receiver-related phase biases, which are typically estimated as float values. To overcome this limitation, the concept of IAR can be applied to PPP, enabling the ambiguities to be fixed to their correct integer values. This enhanced technique is commonly referred to as PPP-AR.

The idea of ambiguity resolution is to exploit the integer constraints, $N \in \mathbb{Z}^n$. In this context, the problem of PNT becomes a mixed-integer estimation problem, where the integer-valued unknown ambiguities must be estimated alongside real-valued parameters such as receiver position coordinates, atmospheric delay parameters, and clock offsets. The integer ambiguities in the mixed-integer GNSS model can be resolved in the following steps:

1. *Float Solution*: In the first step, the integer nature of the ambiguities is ignored, and a standard estimation procedure is applied. This yields the so-called *float solution* for the ambiguities, together with its associated variance–covariance matrix. The parameter estimation can be carried out using standard least-squares (LS) techniques or by applying a Kalman filter.
2. *Integer Solution*: In this step, integer constraints are imposed on the float ambiguities. For this purpose, a mapping $\mathcal{I} : \mathbb{R}^n \mapsto \mathbb{Z}^n$ is introduced, which maps the float ambiguities to their corresponding integer values:

$$\check{N} = \mathcal{I}(\hat{N}).$$

Popular choices of the mapping are integer rounding (IR), integer bootstrapping (IB) and integer least-squares (ILS).

3. *Fixed Solution*: Once the integer ambiguity vector $\check{N}$ is validated, it is used to readjust the float solution of the remaining parameters, yielding the so-called *fixed solution*.

Fixing ambiguities to integer values in the context of PPP offers two main benefits: enhanced positioning accuracy, particularly in the east component, and, in the case of filter-based PPP implementations, a significant reduction or even elimination of the initial convergence period. The latter advantage is especially important for real-time PPP services, which are the focus of this thesis, as it enables faster attainment of high-accuracy positioning. (p 739, Teunissen, Montenbruck, et al., 2017)

In the context of this thesis, the focus is on fixing the ambiguity term N_{IF} in Equation (2.14). The ionosphere-free (IF) ambiguity is a linear combination of the ambiguities of the dual-frequency

carrier-phase measurements and is often decomposed into two components as follows:

$$\begin{aligned}
\lambda_{IF} N_{IF} &= \frac{1}{f_1^2 - f_2^2} (f_1^2 \lambda_1 N_1 - f_2^2 \lambda_2 N_2) = \frac{1}{f_1^2 - f_2^2} \left(f_1^2 \lambda_1 N_1 - f_2^2 \lambda_2 \left(N_1 - \underbrace{(N_1 - N_2)}_{N_{WL}} \right) \right) \\
&= \frac{c(f_1 - f_2)}{f_1^2 - f_2^2} N_1 + \frac{cf_2}{f_1^2 - f_2^2} N_{WL} = \underbrace{\frac{c}{f_1 + f_2} N_1}_{\lambda_{NL} N_{NL}} + \frac{cf_2}{f_1^2 - f_2^2} N_{WL} \\
&= \lambda_{NL} N_{NL} + \frac{cf_2}{f_1^2 - f_2^2} N_{WL}
\end{aligned} \tag{4.1}$$

As an example of the phase observations of GPS L1, L2 in cycles:

$$N_{IF} = 17N_1 + 60(N_1 - N_2) = 17N_{NL} + 60N_{WL} \tag{4.2}$$

where N_1 and N_2 represent the ambiguity of the phase observations at the L1 and L2 GPS frequencies, respectively.

The difficulty arises when comparing this to the raw phase measurement in Equation (3.2), where the ionosphere-free ambiguity is estimated as A_{IF} together with the hardware delays originating from both the satellite and the receiver. Therefore, appropriate pre-processing is essential in order to recover the integer nature of the ambiguities before they can be reliably fixed.

4.1 Existing approaches for PPP-AR

The potential to preserve carrier-phase ambiguities as integer values has spurred a significant effort by the GNSS community to better understand these biases, and techniques for PPP-AR. Analysis centers of the IGS have developed different approaches to achieve PPP-AR.

This is important to note that depending on the source of the correction data as well as the method being used in the server side. The AR approach would be different.

4.1.1 Uncalibrated Phase Delay

In the uncalibrated phase delay (UPD) approach, receiver-dependent biases are eliminated through single differencing, and the resulting single-differenced satellite UPDs are estimated modulo one cycle for both wide-lane (WL) and narrow-lane (NL) combinations. The WL UPDs are typically modeled as daily constants, while the NL UPDs are represented as piecewise linear functions over shorter intervals. These network-derived products are then applied in PPP processing to constrain the corresponding single-differenced ambiguities. For more details, see Geng et al., 2012.

4.1.2 Decoupled Clock Model

The Decoupled Clock Model (DCM) separates the pseudorange and carrier-phase clock components by forming two ionosphere-free (IF) combinations and the Melbourne–Wübbena (MW) combination. Each IF combination is associated with distinct satellite and receiver clocks that include the

respective biases. The MW combination is parameterized using wide-lane (WL) and narrow-lane (NL) ambiguities along with corresponding satellite and receiver WL biases. To preserve the integer nature of ambiguities, a minimum set of ambiguities is fixed to arbitrary integers, serving as an ambiguity datum. The DCM therefore requires estimating two receiver clocks, one for the code and one for the carrier phase, as well as satellite- and receiver-specific WL biases. This model provides a consistent framework for integer ambiguity resolution in PPP. For more details, see Collins et al., 2010.

4.1.3 Integer Clock Model

The Integer Recovery Clock (IRC) approach employs the same observation combinations as the UPD method, two ionosphere-free (IF) and one wide-lane (WL) combination, but with a different parameterization. In addition to satellite and receiver WL biases and ambiguous phase clocks, code-phase biases are introduced to represent the difference between pseudorange-specific and carrier-phase-specific clocks. The satellite WL biases are modeled as daily constants, while the receiver WL biases are estimated epoch by epoch under a mean constraint. As in the DCM model, a subset of integer ambiguities is arbitrarily defined, and the remaining WL and NL ambiguities are resolved through a bootstrapping integer search. The resulting phase clocks are therefore consistent with the pseudorange-specific clocks. For more details, see Laurichesse et al., 2009.

4.1.4 Observable-Specific Biases

In the three previously described methods, the ambiguity resolution performance depends strongly on the chosen parameter estimation strategy. Since each analysis center may adopt a different approach, users face difficulties when applying corrections from multiple providers in a consistent manner.

To address this issue and improve interoperability, the International GNSS Service (IGS) established a dedicated working group during the 2018 IGS Workshop in Wuhan, China. One of its main objectives is to define a unified standard for PPP-AR products. In this framework, analysis centers are encouraged to provide OSB corrections together with their satellite orbit and clock products. The inclusion of OSB ensures consistency between clock and bias products, simplifying their application by end users.

In this thesis, the OSB-based correction approach is adopted. The biases are provided as absolute values for each individual signal, ensuring a consistent and reference-independent correction scheme suitable for real-time and post-processed PPP applications. A detailed analysis of the interoperability among different PPP-AR correction strategies can be found in Banville et al., 2020.

4.2 Ambiguity Fixing Approach

In this thesis, a PPP-AR strategy is implemented, in which single-differenced (SD) ambiguities between satellites are fixed. This approach has been widely used in previous studies (e.g., Ge et al., 2008, Huber, 2015, Glaner, 2022).

The main processing steps applied in this work are as follows:

1. Apply SSR corrections
2. Perform wide-lane (WL) ambiguity fixing
3. Perform narrow-lane (NL) ambiguity fixing
4. Compute the final fixed solution

4.2.1 Applying SSR Corrections to GNSS Measurements

Before forming the ambiguity combinations, the raw GNSS measurements must be corrected using the SSR products introduced earlier in Table 3.1. These corrections account for satellite orbit, clock, and signal biases, ensuring that the subsequent ambiguity estimation is based on bias-reduced observables.

The corrected pseudorange and carrier-phase observations on frequency j can be expressed as:

$$\begin{aligned}\tilde{p}_{r,j}^s &= p_{r,j}^s + c dt_{IF}^s + B_{IF,j}^s = \rho_r^s + c dt_{r,IF} + B_{r,j} + I_{r,j}^s + T_r^s, \\ \tilde{\varphi}_{r,j}^s &= \varphi_{r,j}^s + c dt_{IF}^s + b_{IF,j}^s = \rho_r^s + c dt_{r,IF} + b_{r,j} - I_{r,j}^s + T_r^s + \lambda_j N_{r,j}^s,\end{aligned}\quad (4.3)$$

where the tilde symbol ($\tilde{\cdot}$) denotes observations that have been corrected using the SSR products.

It should be noted that in the original PPP implementation of the BNC software, the SSR corrections are applied in the computation of the modeled observations, meaning the correction terms are subtracted internally to form residuals in an “observed minus computed” sense.

However, when forming wide-lane ambiguities in the Melbourne–Wübbena combination without including satellite-specific bias corrections, the corrections must instead be added to the raw measurements to recover the true satellite-referenced observables. This logic is implemented separately in this work to ensure consistency with the ambiguity resolution process.

4.2.2 Wide-lane Fixing

The wide-lane ambiguities are first obtained by forming the Melbourne–Wübbena combination using the corrected pseudorange and carrier-phase measurements:

$$\widetilde{MW}^s(\tilde{p}_i^s, \tilde{p}_j^s, \tilde{\varphi}_i^s, \tilde{\varphi}_j^s) = \lambda_{WL} N_{WL,ij}^s + b_{MW,ij} + \epsilon_{MW} \quad (4.4)$$

$$\text{with } \lambda_{WL} = \frac{c}{f_i - f_j}, \quad N_{WL,ij}^s = N_i^s - N_j^s, \quad b_{WM,ij} = \frac{b_{r,IF,i}}{\lambda_i} - \frac{b_{r,IF,j}}{\lambda_j} + \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \left(\frac{B_{r,IF,i}}{\lambda_i} + \frac{B_{r,j}}{\lambda_j} \right) \quad (4.5)$$

The MW combination effectively generates a signal with a significantly longer wavelength. However, this comes at the cost of increased measurement noise, which for the GPS L1/L2 combination is amplified by a factor of about six. To counteract this, smoothing with a sliding window is typically applied during processing.

$$\langle \widetilde{MW}^s \rangle_n \approx \lambda_{WL} N_{WL,ij}^s + b_{MW,ij} \quad (4.6)$$

The operator $\langle \cdot \rangle$ denotes a smoothing or averaging operation, typically implemented as a sliding-window mean over a certain number of epochs. This reduces the measurement noise of the wide-lane observable while preserving its integer nature.

To eliminate the receiver-dependent biases, a single differencing between a satellite s and a chosen reference satellite ref is performed:

$$\begin{aligned} \langle \widetilde{\text{MW}}^{\text{ref}} \rangle_n - \langle \widetilde{\text{MW}}^s \rangle_n &= \lambda_{\text{WL}} (N_{\text{WL}}^{\text{ref}} - N_{\text{WL}}^s) \\ \widetilde{\text{MW}}_{\text{SD}}^s &= \lambda_{\text{WL}} N_{\text{WL,SD}}^s \end{aligned} \quad (4.7)$$

where $N_{\text{WL,SD}}^s$ is the single-differenced wide-lane ambiguity.

The reliability of integer ambiguity estimation depends on the carrier wavelength: the longer the wavelength, the more robust the estimation. Owing to their significantly larger effective wavelength, wide-lane ambiguities can often be fixed reliably by a simple rounding operation.

$$\check{N}_{\text{WL,SD}}^s = \text{round} \left(\widetilde{\text{MW}}_{\text{SD}}^s / \lambda_{\text{WL}} \right) \quad (4.8)$$

where $\widetilde{\text{MW}}_{\text{SD}}^s$ denotes the averaged single-differenced Melbourne–Wübbena combination, computed according to (4.7).

The check mark $\check{N}$ is used to denote the fixed integer ambiguity, i.e., the ambiguity after applying the integer constraint. As an example, Figure 4.1 illustrates the wide-lane ambiguity fixing process based on single differencing. The result is obtained using real-time SSR correction streams provided by CNES. The figure demonstrates how smoothing of the MW combination enables reliable identification and rounding of the wide-lane ambiguity to the nearest integer.

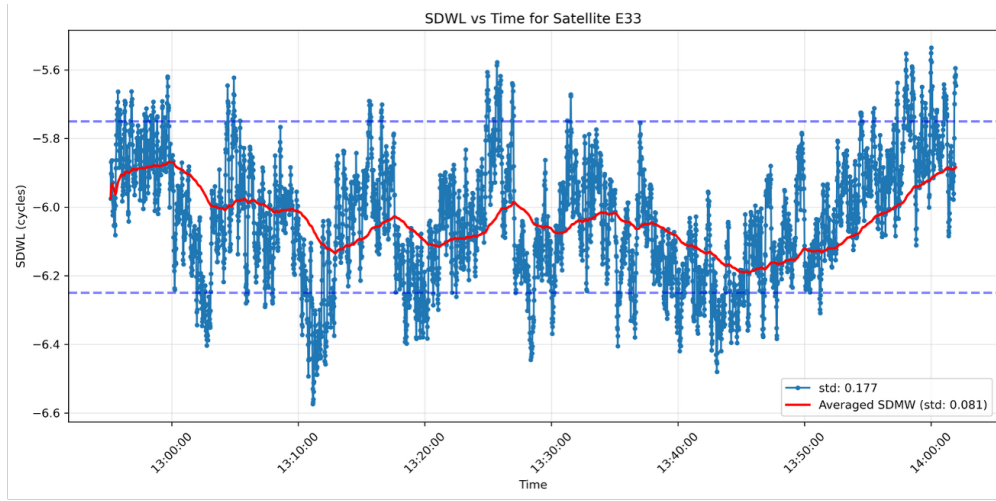


Figure 4.1: Wide-lane ambiguity fixing using single differencing between satellites. The plot shows the single-differenced wide-lane (SDWL) observable for satellite E33 with respect to the reference satellite E26. A moving average with a 300-epoch window size (red line) is applied to smooth the measurements. The ambiguity is fixed by rounding to the closest integer value within a threshold of ± 0.25 cycles.

Selection of the Reference Satellite

To form the single-differenced (SD) observations, a reference satellite must be selected for each constellation. The reference satellite is chosen according to the following criteria:

1. **Ascending satellite:** to ensure the longest possible continuous tracking period.
2. **Highest elevation angle:** to minimize measurement noise and multipath effects.

Since two constellations are used in this thesis, a total of two reference satellites are chosen. Once the reference satellite descends below the predefined elevation mask, a new reference satellite is automatically selected according to the established criteria.

4.2.3 Narrow-lane Fixing

To eliminate receiver-dependent phase biases, the ionosphere-free (IF) ambiguities are first single-differenced with respect to a chosen reference satellite. For a non-reference satellite s and a reference satellite ref , this gives

$$\begin{aligned} A_{\text{IF}}^{\text{ref}} - A_{\text{IF}}^s &= \lambda_{\text{IF}} (N_{\text{IF}}^{\text{ref}} - N_{\text{IF}}^s) \\ &= \lambda_{\text{IF}} N_{\text{IF,SD}}^s. \end{aligned} \quad (4.9)$$

For all non-reference satellites, the system can be expressed compactly in matrix form:

$$\lambda_{\text{IF}} N_{\text{IF,SD}} = \mathbf{C} \cdot \mathbf{A}_{\text{IF}}, \quad \text{with} \quad \mathbf{C} = \begin{bmatrix} 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & -1 \end{bmatrix}, \quad (4.10)$$

and the corresponding covariance matrix of the single-differenced IF ambiguities is

$$\mathbf{Q}_{N,\text{SD}} = \mathbf{C} \cdot \mathbf{Q}_N \cdot \mathbf{C}^T. \quad (4.11)$$

Here, the first column corresponds to the chosen reference satellite (index 1), while the remaining columns correspond to the non-reference satellites.

Once the IF ambiguities are expressed in single-difference form, the narrow-lane ambiguities can be obtained by combining them with the already fixed wide-lane ambiguities:

$$\lambda_{\text{NL}} N_{\text{NL,SD}}^s = \lambda_{\text{IF}} N_{\text{IF,SD}}^s - \frac{cf_2}{f_1^2 - f_2^2} \tilde{N}_{\text{WL,SD}}^s, \quad (4.12)$$

$$N_{\text{NL,SD}}^s = \frac{\lambda_{\text{IF}}}{\lambda_{\text{NL}}} N_{\text{IF,SD}}^s - \frac{f_2}{f_1 - f_2} \tilde{N}_{\text{WL,SD}}^s. \quad (4.13)$$

4.2.4 LAMBDA Method

The Least squares AMBiguity Decorrelation Adjustment (LAMBDA) method is one of the most widely used techniques for integer ambiguity resolution in precise GNSS positioning. In this approach, the float ambiguities are first decorrelated using an integer transformation matrix, commonly referred to as the Z-transformation. This decorrelation step reduces the correlations among the ambiguities, thereby improving the efficiency of the subsequent integer least-squares (ILS) search.

The core principle of the method is to exploit the integer nature of the ambiguities by searching for the integer vector $\mathbf{N} \in \mathbb{Z}^n$ that minimizes the squared residuals between the float ambiguity estimates, $\hat{\mathbf{N}}$, and their integer candidates, weighted by the inverse of their cofactor matrix $\mathbf{Q}_{\hat{\mathbf{N}}}$. The objective function can be written as:

$$\tilde{\mathbf{N}} = \arg \min_{\mathbf{N} \in \mathbb{Z}^n} (\hat{\mathbf{N}} - \mathbf{N})^T \mathbf{Q}_{\hat{\mathbf{N}}}^{-1} (\hat{\mathbf{N}} - \mathbf{N}), \quad (4.14)$$

where $\tilde{N}$ represents the fixed integer ambiguity solution. Note that either the cofactor or covariance matrix can be used, since they are proportional by a constant scale factor.

Applied to the narrow-lane ambiguities in this study, the LAMBDA method can be expressed as:

$$\tilde{N}_{NL} = \text{LAMBDA}(N_{NL,SD}, Q_{N,SD}), \quad (4.15)$$

where $N_{NL,SD}$ denotes the float single-differenced narrow-lane ambiguities, and $Q_{N,SD}$ is their associated covariance matrix.

The original LAMBDA toolbox is available in both MATLAB and Python implementations. As part of this thesis, the relevant components of the toolbox were translated into C++ to ensure seamless integration with the BNC software environment. Although the toolbox provides a variety of functions for ambiguity resolution and validation, only the core LAMBDA integer least-squares estimation method is implemented in this work. For a comprehensive description of the theoretical background and algorithmic details, the reader is referred to De Jonge, Tiberius, et al., 1996 and Massarweh et al., 2025, among others.

Full Ambiguity Resolution and Partial Ambiguity Resolution

After decorrelation and performing the integer least-squares (ILS) search, a key decision must be made on whether to fix the ambiguities or not, as incorrect fixing can lead to significant positioning errors.

Full Ambiguity Resolution (FAR) aims to fix all available ambiguities simultaneously. However, this approach is only feasible under favorable conditions, such as short baselines and high-quality, low-noise observations, where the risk of incorrect fixing is minimal.

Due to the limitations in data quality and the higher noise level in dynamic and real-time scenarios, this thesis adopts a **Partial Ambiguity Resolution (PAR)** strategy. The concept of PAR is to fix only a subset of ambiguities that exhibit the highest reliability, thereby improving the overall solution robustness.

In this work, satellite elevation angle is used as the primary indicator of observation quality. A fixed subset consisting of four satellites (two from GPS and two from Galileo) with the highest elevation angles is selected for fixing. This choice ensures that only the most stable and geometrically strong ambiguities are resolved, while the remaining ambiguities are retained as float values. The fixed number of four satellites is chosen as the minimal reliable subset, balancing computational efficiency and ambiguity correlation while allowing systematic evaluation of the fixing performance.

Ambiguity Ratio Test

The ambiguity ratio test is a commonly used quality control procedure to assess the reliability of integer ambiguity resolution. It determines whether there is sufficient statistical confidence to accept the fixed integer solution or whether the float solution should be retained. The ambiguity resolution (AR) ratio is defined as the ratio between the squared weighted residuals of the second-best integer candidate and those of the best integer candidate obtained from the LAMBDA search. A higher ratio indicates greater confidence that the best integer solution is correct:

$$R = \frac{(\tilde{N}_2 - \hat{N})^T Q_N^{-1} (\tilde{N}_2 - \hat{N})}{(\tilde{N} - \hat{N})^T Q_N^{-1} (\tilde{N} - \hat{N})} > R_{\text{thres}}, \quad (4.16)$$

where $\tilde{N}$ and $\tilde{N}_2$ represent the best and second-best integer candidates, respectively, $\hat{N}$ is the float ambiguity vector, and Q_N is its covariance matrix.

A known drawback of this approach is that determining an appropriate threshold value, R_{thres} , is not straightforward. As discussed by Verhagen and Teunissen, 2006, the threshold depends on several factors such as the number of ambiguities, measurement geometry, and noise characteristics. In practice, the threshold is typically chosen empirically through extensive testing. Commonly adopted values for the ratio threshold range between $R_{\text{thres}} = 3.0$ and $R_{\text{thres}} = 5.0$, with higher thresholds providing more conservative and reliable fixing at the expense of longer convergence times. In this thesis, a middle value of $R_{\text{thres}} = 4.0$ is adopted to achieve a balance between reliable fixing and shorter convergence time.

4.2.5 Implemented AR Structure

The ambiguity resolution (AR) module developed in this thesis is built upon the float PPP solution framework of the BNC software. The overall processing sequence of the implemented PPP-AR algorithm is illustrated in Figure 4.2. It outlines the main functional steps from the float solution to the final ambiguity-fixed positioning result, including ambiguity formation, quality control, and validation.

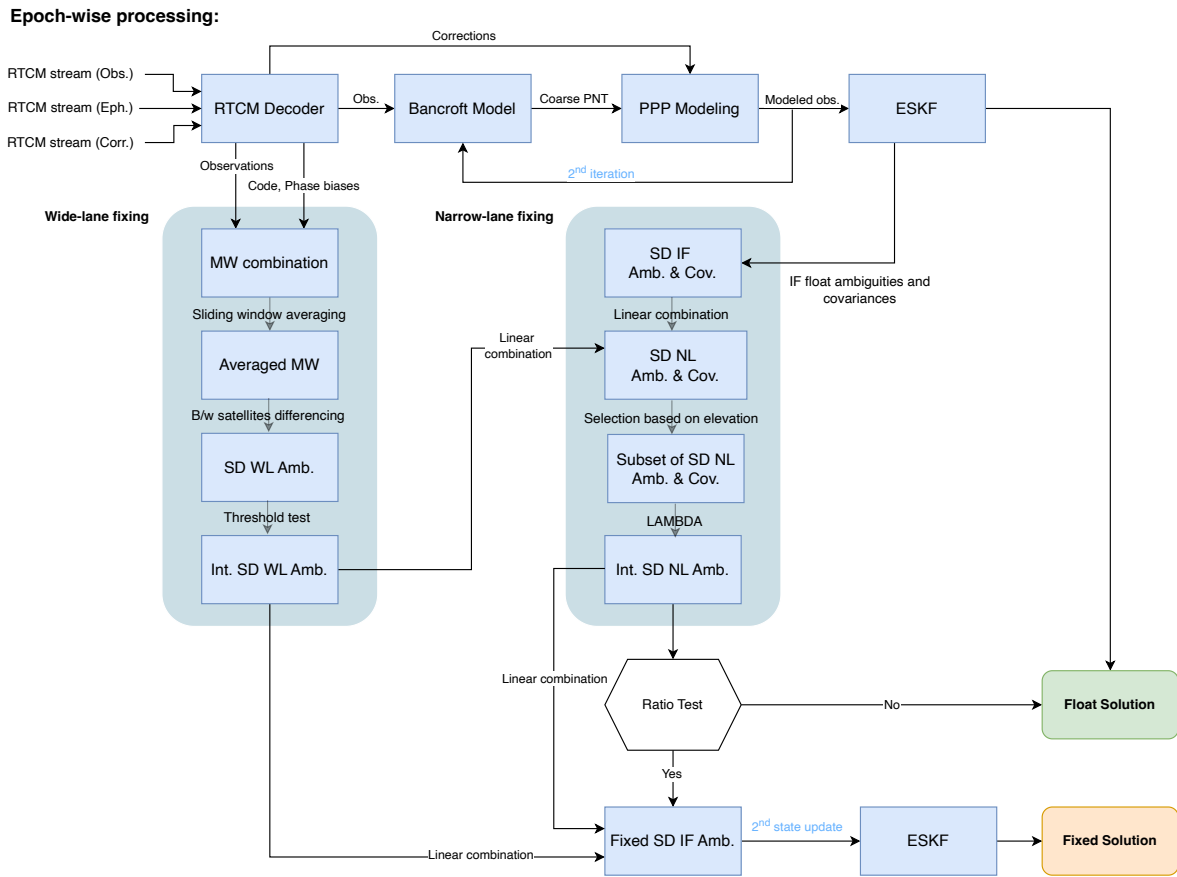


Figure 4.2: Processing flow of the implemented PPP-AR module.

Firstly, the decoded observations, together with the corresponding code and phase biases, are used to generate the corrected measurements according to (4.3). Subsequently, the Melbourne–Wübbena (MW) combination is formed as described in (4.4). Owing to the high noise level inherent in this

combination, a moving average filter with a window size of 600 epochs (corresponding to 10 minutes) is applied to smooth the data. This operation is performed for each satellite above the defined elevation mask for ambiguity fixing.

At this stage, the averaged MW combinations still contain receiver-dependent hardware biases, for which no direct correction terms are available. To mitigate these effects, single differencing is performed between each satellite and a designated reference satellite. This operation restores the integer nature of the wide-lane ambiguities in a single-differenced sense, i.e. it yields the integer number of wide-lane cycles between a satellite and its reference. The resulting values are verified against a predefined threshold (e.g., 0.33 cycles) to ensure ambiguity consistency. Once the criterion is satisfied, the wide-lane ambiguities are rounded to their nearest integer values.

For the narrow-lane ambiguity resolution, the process begins with the computation of single-differenced ionosphere-free (IF) ambiguities and their associated covariance matrix, which is derived through error propagation as shown in (4.10) and (4.11). The single-differenced narrow-lane (NL) ambiguities are then obtained from the fixed single-differenced wide-lane (WL) and ionosphere-free (IF) ambiguities according to (4.12) and (4.13). A subset of ambiguities is subsequently selected, typically two satellites with the highest elevations from each constellation, resulting in a total of four.

The LAMBDA method is applied to this first subset to estimate the most likely integer values of the single-differenced NL ambiguities. Finally, a ratio test is conducted to validate the correctness of the integer solutions before they are accepted.

For this first subset of satellites, several quality control criteria are applied in addition to the ratio test to ensure the reliability of the ambiguity resolution process. The following three checks are performed before accepting the fixed integer solution:

1. **Ratio test:** The ratio between the squared residuals of the best and second-best integer candidates obtained from the LAMBDA search must exceed a predefined threshold to ensure ambiguity reliability:

$$R > 4.0 \quad (4.17)$$

2. **Standard deviation of float ambiguities:** The formal precision of the float ambiguities, represented by their standard deviation from the Kalman filter estimates, is checked. Only ambiguities with sufficiently small uncertainty are considered for fixing:

$$\sigma_{N_{IF}} < 0.1 \text{ cycle} \quad (4.18)$$

3. **Float-fixed consistency check:** The difference between the float and fixed narrow-lane ambiguities is evaluated. Large deviations indicate unreliable fixing and lead to rejection of the solution:

$$|\hat{N}_{NL} - \check{N}_{NL}| < 0.8 \text{ cycle} \quad (4.19)$$

Only when all three validation criteria are satisfied are the fixed ambiguities accepted and used to form the initial fixed subset. Once this first subset of four satellites is successfully resolved, the remaining satellites are evaluated using a simplified deviation-based criterion. In this subsequent stage, the ratio test is no longer applied, this is discussed in Section 5.3.1. Instead, the validation relies solely on assessing the deviation between the fixed and float narrow-lane ambiguities to ensure consistency:

$$|\hat{N}_{NL} - \check{N}_{NL}| < 0.2 \text{ cycles} \quad (4.20)$$

If the deviation remains below this threshold, the ambiguity is accepted as fixed; otherwise, it is retained as a float value. A relatively conservative threshold is selected to minimize the risk of incorrect fixing, since even a single wrongly fixed ambiguity can significantly deteriorate the overall positioning accuracy.

4.2.6 Fix-and-hold

A fix-and-hold strategy, similar to that implemented in RTKLIB, is adopted in this thesis due to its effectiveness in improving ambiguity fixing for moving receivers. The “Fix and Hold” mode was first introduced in RTKLIB version 2.4.0 to enhance the fixing ratio, particularly in kinematic applications involving dynamic receivers (Takasu and Kubo, 2013).

In this approach, integer ambiguities are continuously estimated and monitored. Once the validation criteria are satisfied, the corresponding ambiguities are fixed to their integer values and held as tightly constrained parameters in the filter. This allows subsequent epochs to benefit from the fixed ambiguities, improving both the stability and continuity of the real-time solution.

Finally, the fixed ionosphere-free (IF) ambiguities are reconstructed from the fixed narrow-lane (NL) and wide-lane (WL) ambiguities by rearranging the terms in (4.12).

$$\lambda_{IF} \check{N}_{IF,SD}^s = \lambda_{NL} \check{N}_{NL,SD}^s + \frac{cf_2}{f_1^2 - f_2^2} \check{N}_{WL}^s \quad (4.21)$$

In the BNC, an artificial wavelength for IF is defined as $\lambda_{IF} = \frac{c}{f_1 + f_2}$ and thus the fixed IF ambiguities can be convert to unit cycle, which is the difference between IF ambiguity of one satellite and its reference satellite.

$$\begin{aligned} \check{N}_{IF,SD}^s &= \frac{1}{\lambda_{IF}} \left(\lambda_{NL} \check{N}_{NL,SD}^s + \frac{cf_2}{f_1^2 - f_2^2} \check{N}_{WL}^s \right) = \frac{\lambda_{NL}}{\lambda_{IF}} \check{N}_{NL,SD}^s + \frac{f_2}{f_1 - f_2} \check{N}_{WL,SD}^s \\ &= \check{N}_{IF}^{\text{ref}} - \check{N}_{IF}^s \end{aligned} \quad (4.22)$$

After the fixed ionosphere-free (IF) ambiguities are obtained, they are reintroduced into the Error State Kalman Filter (ESKF) as pseudo-observations with tight constraints, ensuring that the fixed integer values are maintained in subsequent filter updates.

$$\underbrace{\begin{bmatrix} \check{N}_{IF,SD}^1 \\ \vdots \\ \check{N}_{IF,SD}^n \end{bmatrix}}_y = \underbrace{\begin{bmatrix} \mathbf{0} & 1 & -1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & 1 & 0 & \cdots & -1 & 0 & 0 & \cdots & 0 \\ \mathbf{0} & 0 & 0 & \cdots & 0 & 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & -1 \end{bmatrix}}_H \underbrace{\begin{bmatrix} \mathbf{r}_r \\ c dt_r^{\text{GPS}} \\ c dt_r^{\text{Galileo}} \\ T_{ZWD} \\ N_{IF}^1 \\ \vdots \\ N_{IF}^n \end{bmatrix}}_x \quad (4.23)$$

The index “1” indicates the reference satellite, while the non-reference satellites are represented by “-1” in the corresponding columns. This structure ensures that each single-differenced ambiguity is formed between the reference satellite and a non-reference satellite for both constellations.

The tight constraint is implemented by defining small standard deviations in the covariance matrix of the pseudo-observations:

$$\mathbf{R} = \text{diag}(\sigma_c^2, \sigma_c^2, \sigma_c^2, \dots), \quad \text{with } \sigma_c = 0.03 \text{ cycles.} \quad (4.24)$$

This effectively enforces strong confidence in the fixed ambiguities during the update. After defining the covariance matrix, a second update is performed using the Kalman filter to incorporate the pseudo-measurements and refine the state estimates.

Implementation in BNC

It should be noted that the BNC employs an Error-State Kalman Filter (ESKF), where the state vector represents deviations from predefined or computed nominal values. Consequently, the input observations must be expressed as observed-minus-computed (OMC) residuals:

$$\Delta \mathbf{y} = \mathbf{y} - \begin{bmatrix} N_0^{\text{ref}} - N_0^1 \\ \vdots \\ N_0^{\text{ref}} - N_0^n \end{bmatrix} \quad (4.25)$$

Here, N_0^s denotes the internally initialized ambiguities in BNC, which serve as preliminary estimates. These values are derived from the observed-minus-computed carrier-phase differences according to

$$N_0^s = \text{round} \left(\frac{\varphi_{\text{IF}}^s - \varphi_{\text{IF,model}}^s}{\lambda_{\text{IF}}} \right), \quad (4.26)$$

where the artificial wavelength for the ionosphere-free (IF) combination is defined as

$$\lambda_{\text{IF}} = \frac{c}{f_1 + f_2}. \quad (4.27)$$

Subsequently, the Kalman gain is computed according to (3.25), followed by the state update step defined in (3.27).

Chapter 5 Test setup & results

5.1 BNC Configuration

The BNC software offers convenient configuration options, either through its Graphical User Interface (GUI) or by directly modifying the configuration file.

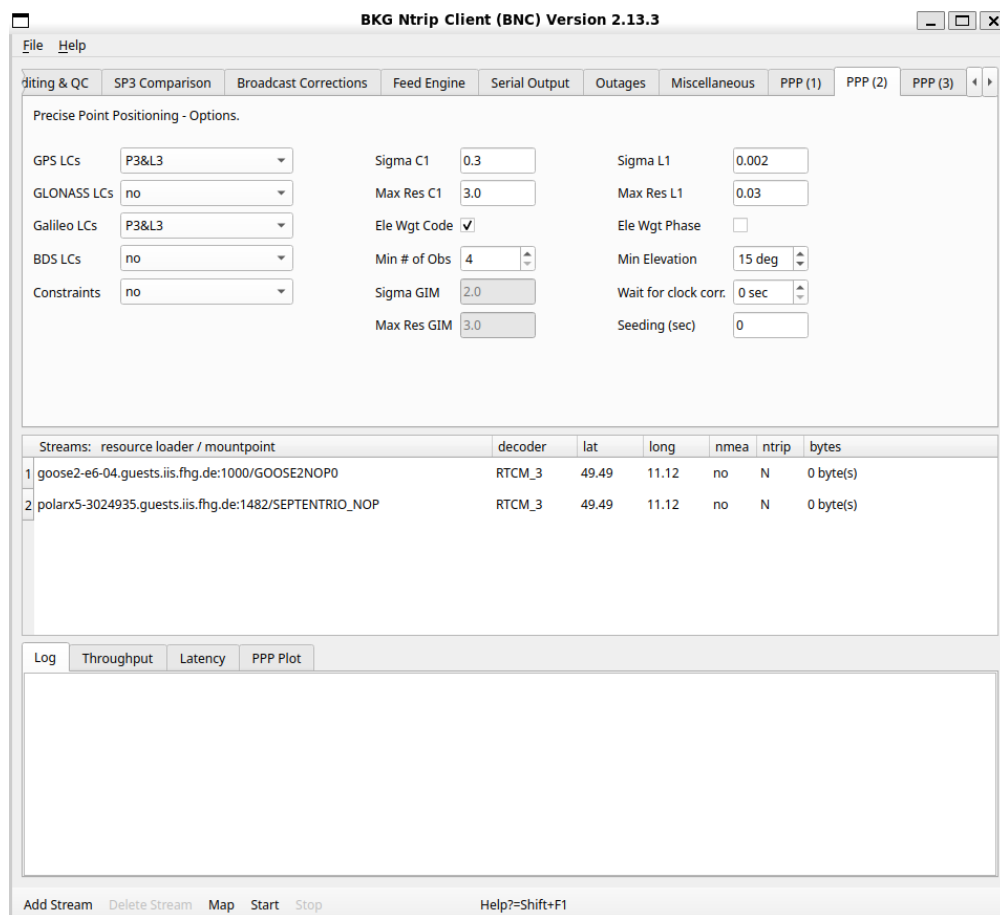


Figure 5.1: BNC GUI PPP Model config

Input streams:

- GOOSE (SSR corrections):
RTCM Stream: `goose2-e6-04.guests.iis.fhg.de:1000/GOOSE2NOP0`
- Septentrio (Observations and Ephemeris):
RTCM Stream: `polarx5-3024935.guests.iis.fhg.de:1482/SEPTENTRIO_NOP`

The ionosphere-free (IF) model (P3 & L3) is selected for both code and carrier-phase measurements. The standard deviations for the code and phase observations can also be configured, along with a maximum residual threshold used to discard outlier measurements. Additionally, an elevation-dependent mask can be applied to exclude satellites below a specified elevation angle. The corresponding parameters are shown in the Figure 5.1.

In the Figure 5.2, the user can set the standard deviation and the process noise for individual parameters in the Kalman filter, as well as the GNSS signals to be used.

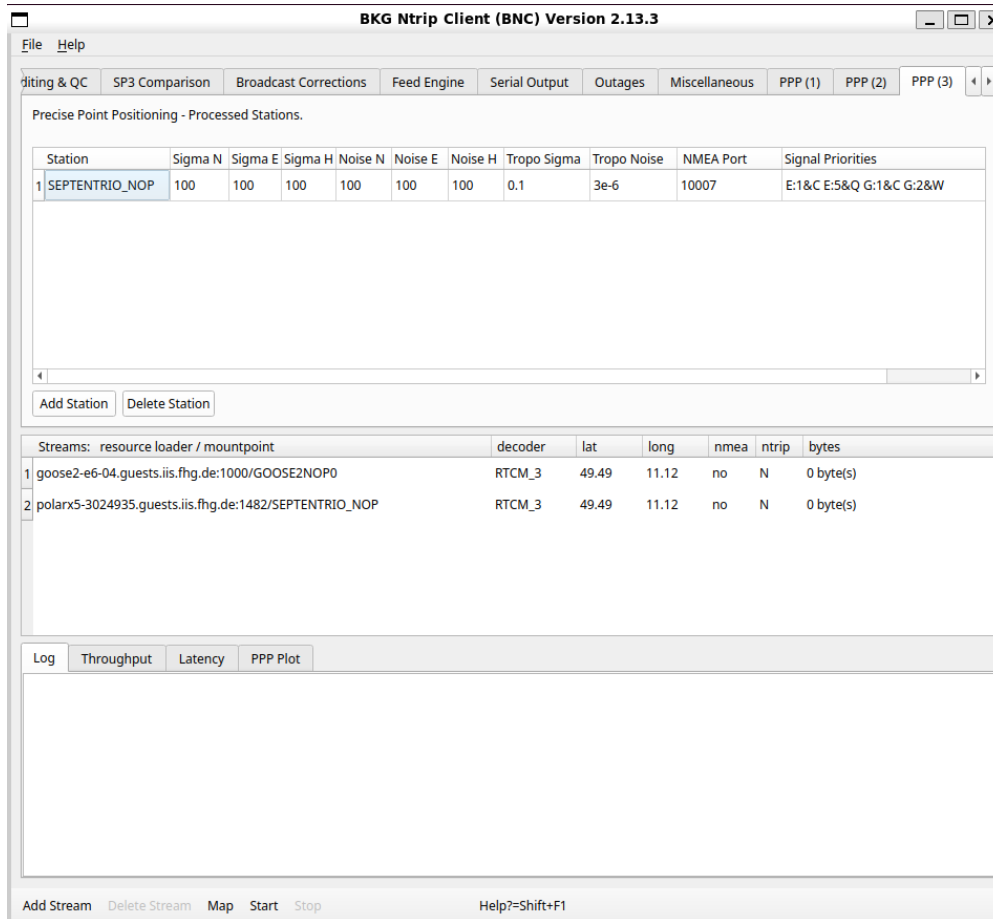


Figure 5.2: Filter parameter settings in the BNC GUI

The standard deviations for all NEU coordinate components, as well as their corresponding process noise levels, are set to 100 m. For the tropospheric delay, a much smaller standard deviation of 0.1 m and a process noise of $3 \times 10^{-6} m/\sqrt{s}$ are used, reflecting the relative temporal stability of the troposphere. The signals included in the processing are Galileo E1 and E5a, and GPS L1 and L2. These parameter settings follow the recommendations provided in the BNC manual. The author also performed several tests with modified values; however, no significant changes in the positioning performance were observed.

5.1.1 Include AR Option into GUI

For this thesis, the GUI of BNC was further developed to enhance support for PPP-AR processing. This modification allows direct configuration of parameters related to the ambiguity resolution

(AR) process. An overview of the newly added interface elements is shown in Figure 5.3.

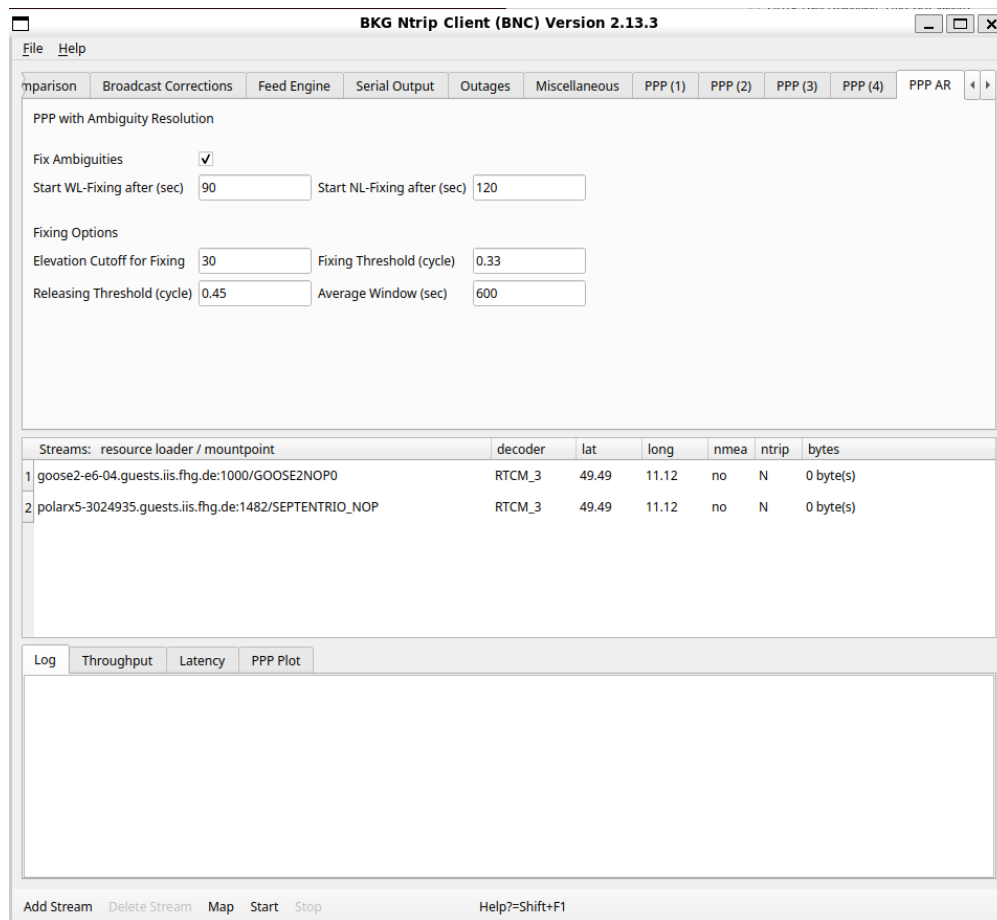


Figure 5.3: BNC GUI extension for ambiguity resolution configuration

- **Fix Ambiguities:** Enables or disables the ambiguity resolution function.
- **Start Fixing Time:** Defines the starting epoch for the wide-lane (WL) and narrow-lane (NL) fixing procedures.
- **Elevation Cutoff:** Specifies the minimum satellite elevation angle for inclusion in the AR process.
- **Fixing Threshold:** Sets the rounding threshold for WL ambiguities (only ambiguities with differences smaller than this threshold from an integer value are fixed).
- **Releasing Threshold:** Determines when a previously fixed satellite should be reverted to float status (e.g., when the deviation exceeds 0.45 cycles).
- **Average Window:** Defines the window length for the moving average filter applied to the Melbourne–Wübbena (MW) combination.

5.2 Spirent GNSS Simulator

The Spirent GSS9000 GNSS simulator is a high-end multi-frequency, multi-constellation testing system designed for precise evaluation of positioning, navigation, and timing (PNT) applications. It



Figure 5.4: Spirent GSS9000

supports the generation of signals from multiple constellations, including GPS, Galileo, GLONASS, BeiDou, QZSS, and NavIC. The simulator can reproduce realistic signal conditions by incorporating modeled error sources such as ionospheric and tropospheric delays, multipath, and signal dynamics, enabling controlled testing of receiver performance.

By accurately reproducing real-world GNSS signal effects, the Spirent GSS9000 enables a controlled and repeatable testing environment for evaluating receiver performance. This capability is particularly valuable for high-precision applications, where even minor errors can lead to significant degradations in positioning accuracy.

5.2.1 Test Setup

The experimental setup used in this thesis is designed to simulate real-time GNSS signal reception and correction processing under controlled laboratory conditions. An overview of the setup is illustrated in Figure 5.5. It consists of the following main components:

- **Spirent GSS9000 GNSS Simulator:** Generates multi-constellation, multi-frequency GNSS signals and emulates realistic signal conditions, including typical GNSS error sources. It also simulates the Galileo HAS signal as if transmitted from space.
- **Septentrio PolaRx5 GNSS Receiver:** Receives the Radio Frequency signals generated by the simulator and outputs observation data and navigation messages as an Ntrip data stream, which can be directly imported into the BNC software for real-time PPP processing.
- **GOOSE Receiver:** Decodes the Galileo Galileo HAS SIS corrections and converts them into an SSR correction stream. The GOOSE receiver is an in-house development of Fraunhofer IIS.

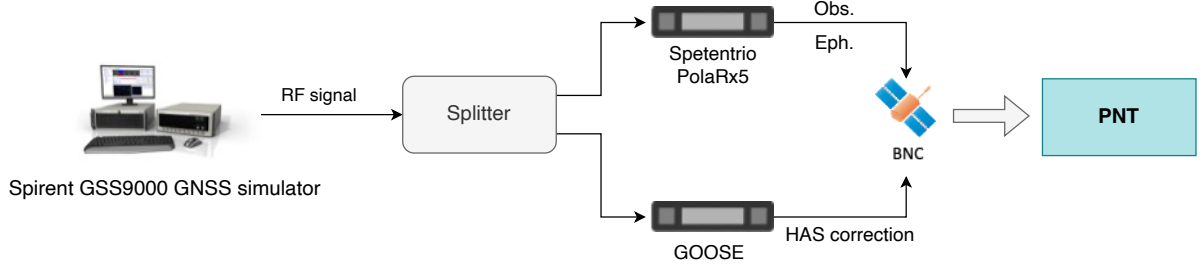


Figure 5.5: Overview of the test setup showing the signal flow from the simulator through the receivers to the processing software.

Simulation scenario

In the simulation setup, GPS and Galileo signals are generated, and the corresponding SSR correction data are transmitted as simulated Galileo HAS corrections on the E6 frequency. To realistically emulate space-to-Earth signal propagation, atmospheric effects are included: tropospheric delays are modeled using the STANAG model, while ionospheric delays are computed using the Klobuchar model.

The total duration of the simulation is 2 hours, with a maximum receiver velocity of approximately 15 m/s. The trajectory used in the simulation scenario is shown in Figure 5.6.

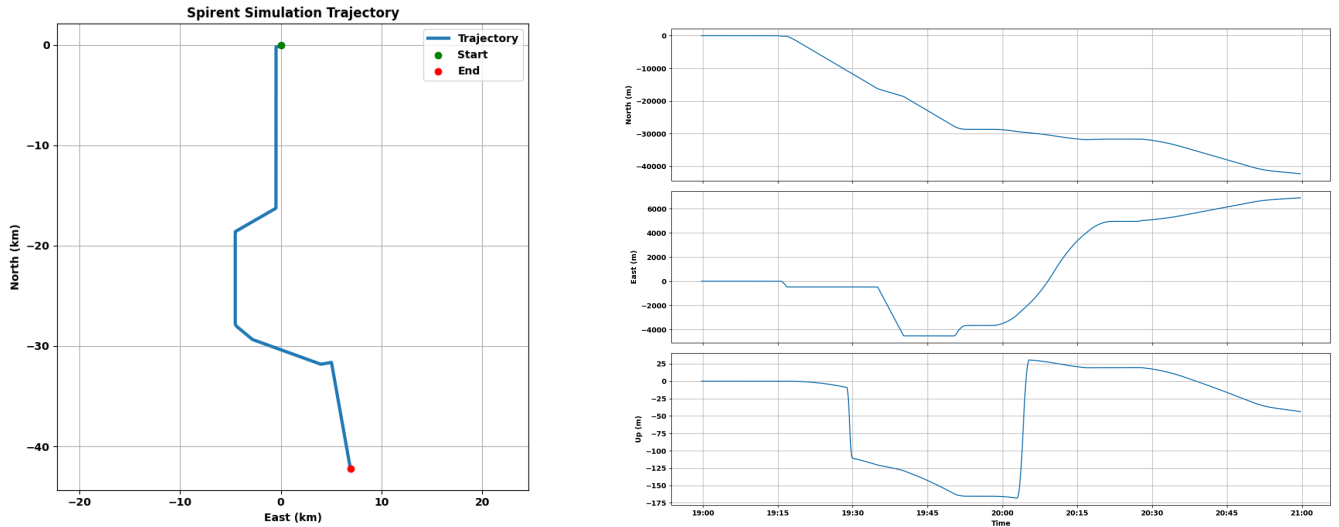


Figure 5.6: Simulated trajectory and corresponding position deviations in the North, East, and Up components.

Galileo HAS correction

The real Galileo High Accuracy Service (HAS) correction is still under development. At the time of this thesis, the service remains in its initial phase, and the phase bias corrections have not yet been officially provided or validated. (European GNSS Service Centre (GSC), 2025)

Therefore, the HAS corrections used in this study are simulated. They are generated using the Spirent GSS9000 GNSS simulator, which emulates the transmission of HAS corrections as if they were broadcast from the Galileo satellites in space.

Limitations of simulation compared to real observations

It is also informative to compare the simulated observations and corrections with those obtained from real GNSS data. For a preliminary analysis, this comparison was performed using the single-differenced wide-lane (WL) ambiguities. Since SDWL ambiguities exhibit the integer property, their values should lie close to integers and can therefore be used as a consistency check.

Table 5.1 presents several example results for selected satellites, showing the single-differenced WL ambiguities computed from both real data and simulated data.

Table 5.1: Comparison of single-differenced wide-lane (SDWL) ambiguities between real GNSS data (left) and simulated data (right).

PRN	SDWL (cycles)	PRN	SDWL (cycles)
G05	-21.9403	G10	6.9773
G07	-16.7544	G12	10.9862
G09	-17.0771	G19	2.9787
G11	-29.8779	G25	8.9657
E05	-6.3573	E02	12.9950
E13	-2.0179	E03	5.9991
E15	-23.6498	E09	25.0001
E16	19.9253	E10	11.0006

(a) SDWL ambiguities computed from real GNSS observations using CNES SSR corrections.

(b) SDWL ambiguities computed from simulated GNSS observations using Spirent-generated Galileo HAS corrections.

A clear difference is observed between the SDWL ambiguities computed from real observations and those obtained from the simulated scenario. In the simulation, the SDWL values lie almost exactly at integer cycle levels, with deviations on the order of a few thousandths of a cycle. This behavior is expected, as the simulator generates idealized measurements with perfectly consistent correction data and without unmodeled effects.

In contrast, the real data show noticeably larger deviations from integer values. Only a subset of satellites produces SDWL values sufficiently close to an integer to allow confident rounding. The remaining satellites exhibit deviations caused by real-world effects such as measurement noise, multipath, atmospheric modelling errors, and residual inconsistencies in the correction products.

This comparison highlights that the simulated environment is “too perfect,” while real observations contain many practical imperfections that make ambiguity resolution more challenging.

5.2.2 Float Solution

One of the objectives of this thesis is to improve upon the standard PPP float solution. Therefore, it is essential to first analyze and benchmark the performance of the float solution itself.

Table 5.2 summarizes the root mean square (RMS) positioning errors in the NEU components. Overall, the float solution achieves only decimeter-level accuracy, with a 3D RMS error of approximately 20 cm.

Divergence in the float solution

Figure 5.7 shows the time series of coordinate errors in the North, East, and Up (NEU) directions for the float solution. After an initial convergence period of approximately one hour, the solution

Table 5.2: Root Mean Square (RMS) errors of the float solution.

RMS_N	RMS_E	RMS_U	RMS_{2D}	RMS_{3D}
5.60 cm	10.98 cm	11.89 cm	12.33 cm	17.13 cm

stabilizes within a decimeter-level accuracy range. However, a gradual drift can still be observed in the East and Up components toward the end of the session. This behavior reflects the inherent limitation of the float PPP solution, which is without integer ambiguity resolution cannot achieve the centimeter-level precision attainable with fixed ambiguities.

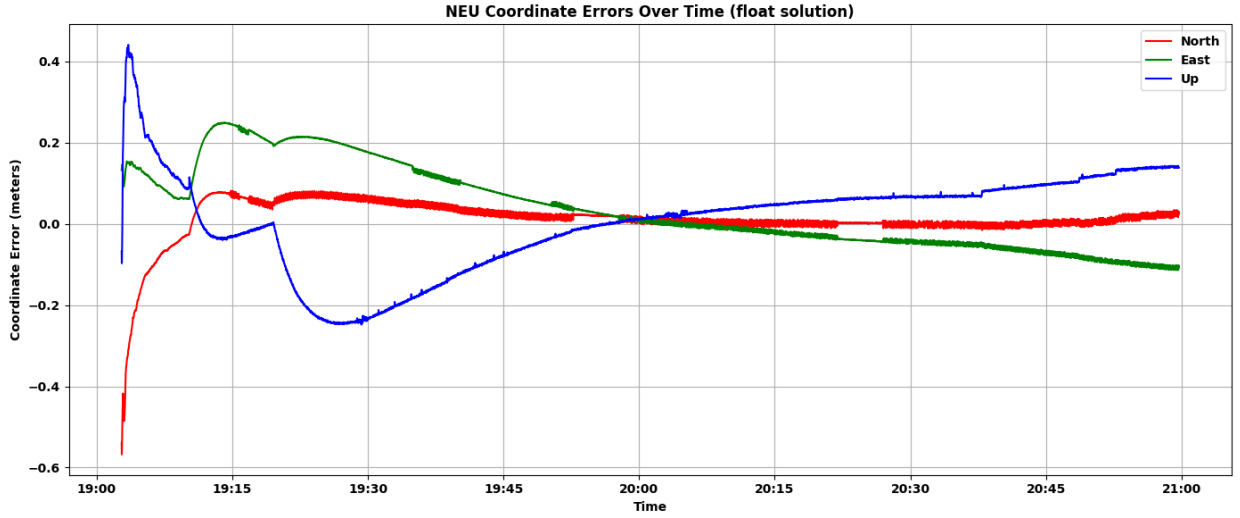


Figure 5.7: NEU coordinate errors over time for the float solution. After convergence, the solution remains stable within decimeter-level accuracy, with small drifts visible in the East and Up components.

5.3 Lessons Learned and Final Implementation Strategy

During the development of the AR approach implemented in this thesis, several modifications and refinements were introduced to address issues encountered at different stages of the process. The overall evolution of the methodology is summarized in the flow diagram shown in Figure 5.8. Each step represents a key stage in the development, where specific challenges were identified, analyzed, and resolved through targeted adjustments to the processing strategy.

Further details of each step are presented in the following subsections.

5.3.1 raPPPid Method

In the early stage of this work, the author followed the procedure described in the raPPPid software (Glaner, 2022), which employs a Kalman filter to compute the float solution while using a standalone least-squares procedure to obtain the fixed solution. A similar strategy is also described in Huber, 2015.

The AR functionality was initially implemented based on the raPPPid source code. However, several issues were identified during preliminary testing with the simulated scenario, which revealed

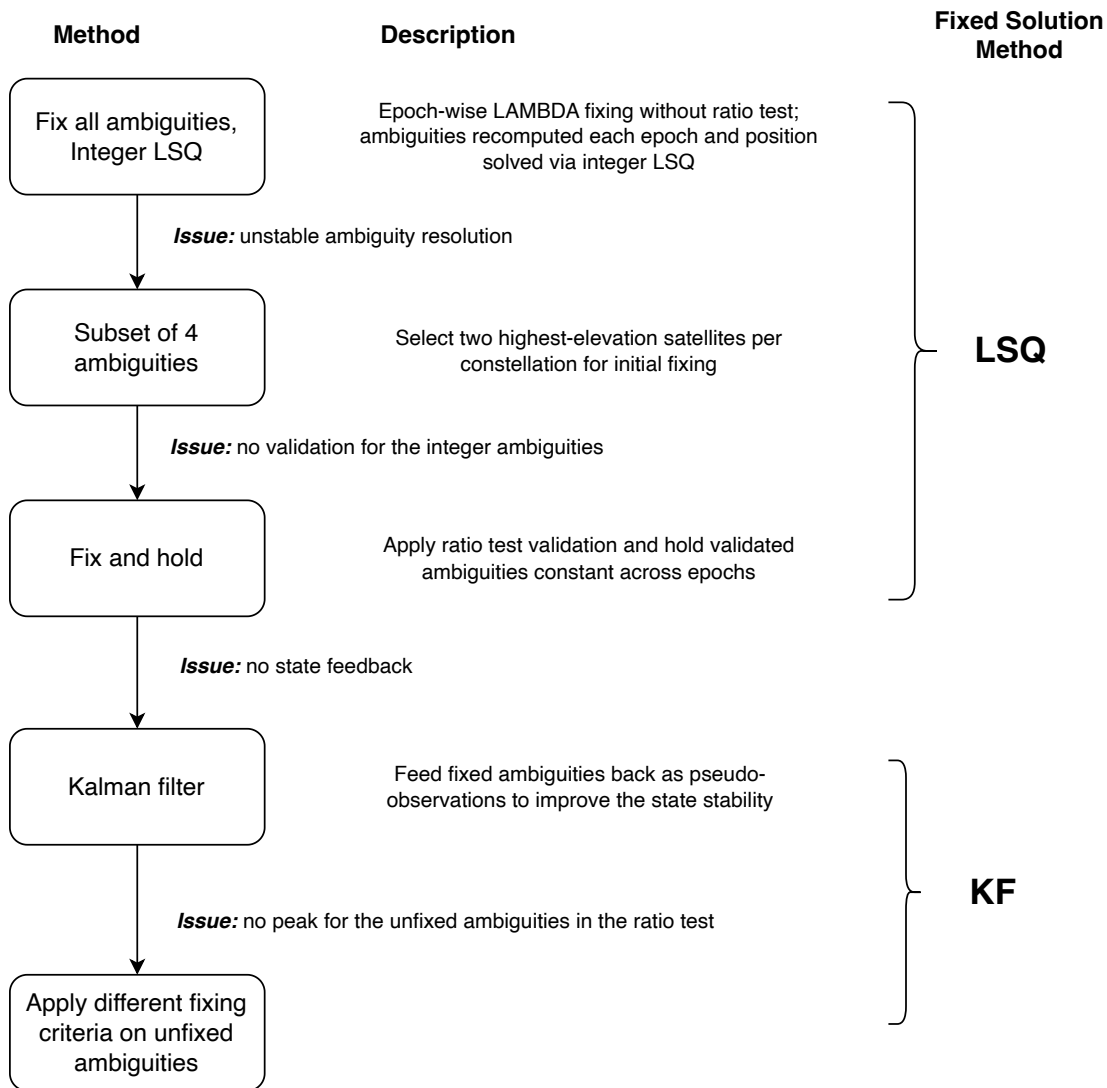


Figure 5.8: Development process of the PPP-AR implementation showing the progression from independent integer least-squares fixing to the final ESKF-integrated approach.

fundamental limitations of this approach when applied to dynamic conditions.

The main issues identified with this implementation are summarized as follows:

1. **Lack of ambiguity validation:** No validation step was implemented to assess the reliability of the integer ambiguity estimates produced by the LAMBDA method.
2. **Full-set fixing strategy:** The algorithm attempted to fix all ambiguities simultaneously, without distinguishing between well-determined and weakly-determined parameters, which often led to incorrect fixing.
3. **No temporal constraint on fixed ambiguities:** The fixed ambiguities were treated purely as integer values, without enforcing temporal consistency. Their constant nature over time was therefore not maintained.
4. **Absence of feedback to the Kalman filter:** The fixed ambiguities were not fed back into the

Kalman filter for state refinement, preventing the filter from benefiting from the improved ambiguity information.

Validation of integer ambiguities

In the original implementation of the raPPPid software, only the best integer candidate obtained from the LAMBDA method was selected and used directly to compute the fixed solution, without applying any validation test. The residuals of the subsequent least-squares adjustment were then checked as a form of quality control. However, testing with the simulated dataset in this thesis revealed that this residual-based check is insufficient. It was observed that incorrect ambiguity fixing can still produce relatively small residuals, particularly because the least-squares adjustment in this approach is based on observations from a single epoch. This limitation became evident in the simulation experiments conducted in this work.

After reviewing various open-source software implementations and research studies related to PPP and PPP-AR, the ratio test method was adopted to validate the integer ambiguity candidates obtained from the LAMBDA method. A detailed description of the ratio test is provided in Section 4.2.4.

Subset of 4 ambiguities

After implementing the ratio test, the author investigated the feasibility of fixing the set of ambiguities corresponding to satellites above the elevation mask (i.e., 30°). The integer estimates of the narrow-lane ambiguities obtained from the LAMBDA method, together with the corresponding ratio test time series, are presented in Figure 5.9. The temporal variations in the estimated ambi-

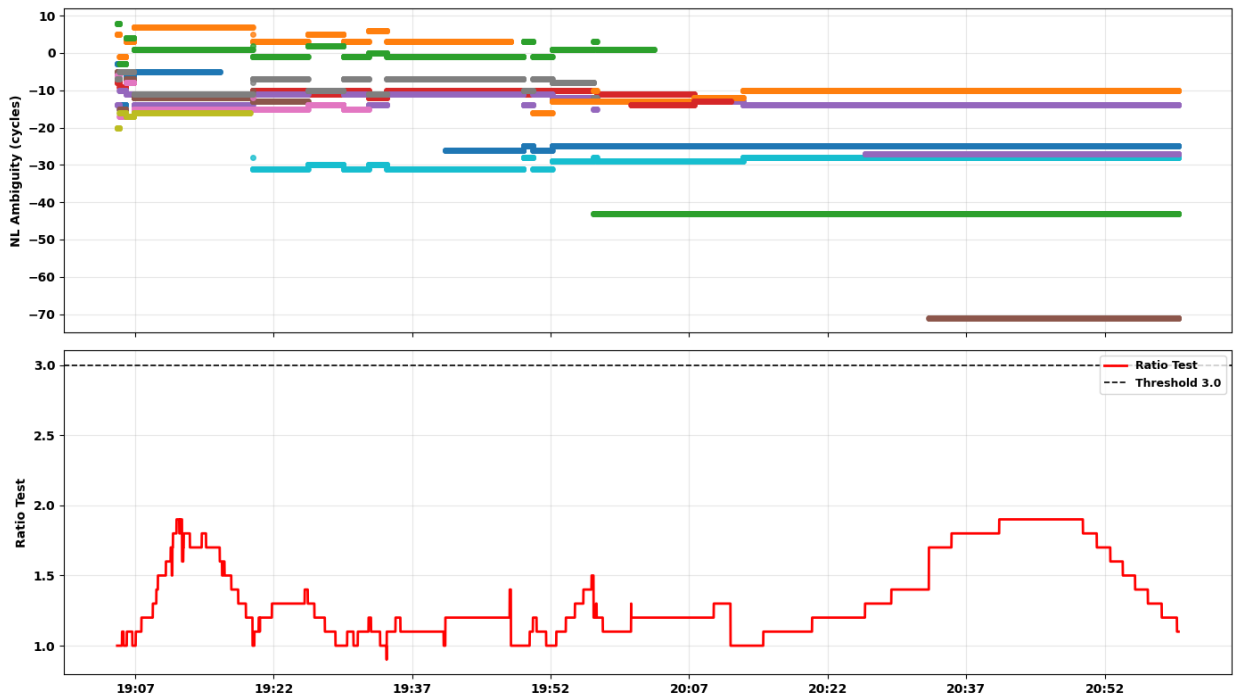


Figure 5.9: Time series of integer single-differenced narrow-lane ambiguities (top) and corresponding ratio test values (bottom).

guities can be clearly observed in the upper plot. Meanwhile, the ratio values remain consistently

low throughout the processing period, staying below the predefined fixing threshold. As a result, it is not possible to reliably conclude whether the set of ambiguities has been correctly fixed.

The initial objective was to fix a complete set of ambiguities simultaneously using the LAMBDA method. However, as the ratio test did not provide sufficient validation for the correctness of the fixed ambiguities, a second investigation was conducted to identify potential limiting factors. One possible source of error was hypothesized to be the tropospheric delay.

One of the advantages of using a simulator is the ability to design controlled scenarios tailored to specific testing needs. Therefore, the tropospheric delay was deactivated in the simulation setup, and the ambiguity estimation was repeated using the LAMBDA method and the ratio test. As shown in Figure 5.10, no significant improvement or difference was observed compared to the previous case.

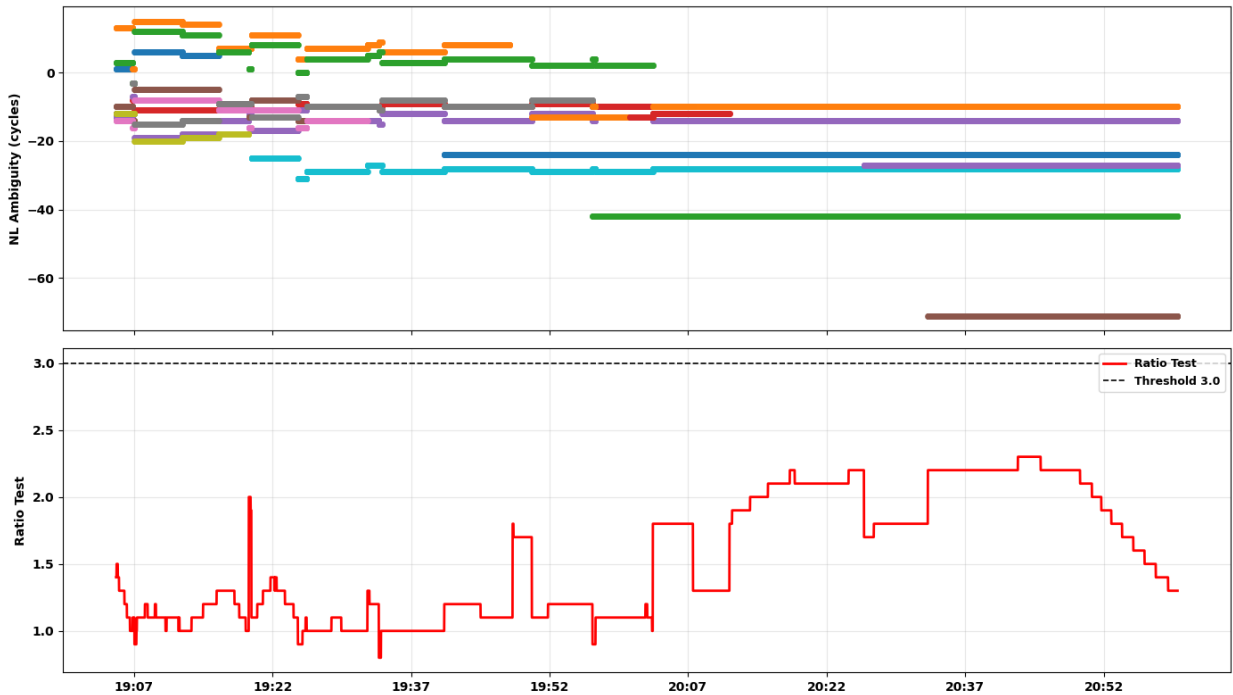


Figure 5.10: Time series of integer ambiguities and ratio test values with the tropospheric delay model deactivated in the simulation.

The next approach focused on reducing the number of unknowns in the LAMBDA method by limiting the set of ambiguities used as input. A minimal subset was selected, two satellites per constellation, to maintain sufficient correlation between ambiguities while still benefiting from the decorrelation process within the LAMBDA algorithm. The results, shown in Figure 5.11, demonstrate that the number of ambiguities included in the LAMBDA estimation has a significant impact on the solution quality, particularly on the ratio values in the ratio test. Compared to the time series where all satellites were included in the fixing process, limiting the subset to only four satellites resulted in a noticeable improvement. A distinct peak appeared in the ratio test, indicating a higher level of confidence in the integer ambiguity search. This confirms that fixing a smaller, well-selected subset of satellites can significantly enhance the reliability of the ambiguity resolution process.

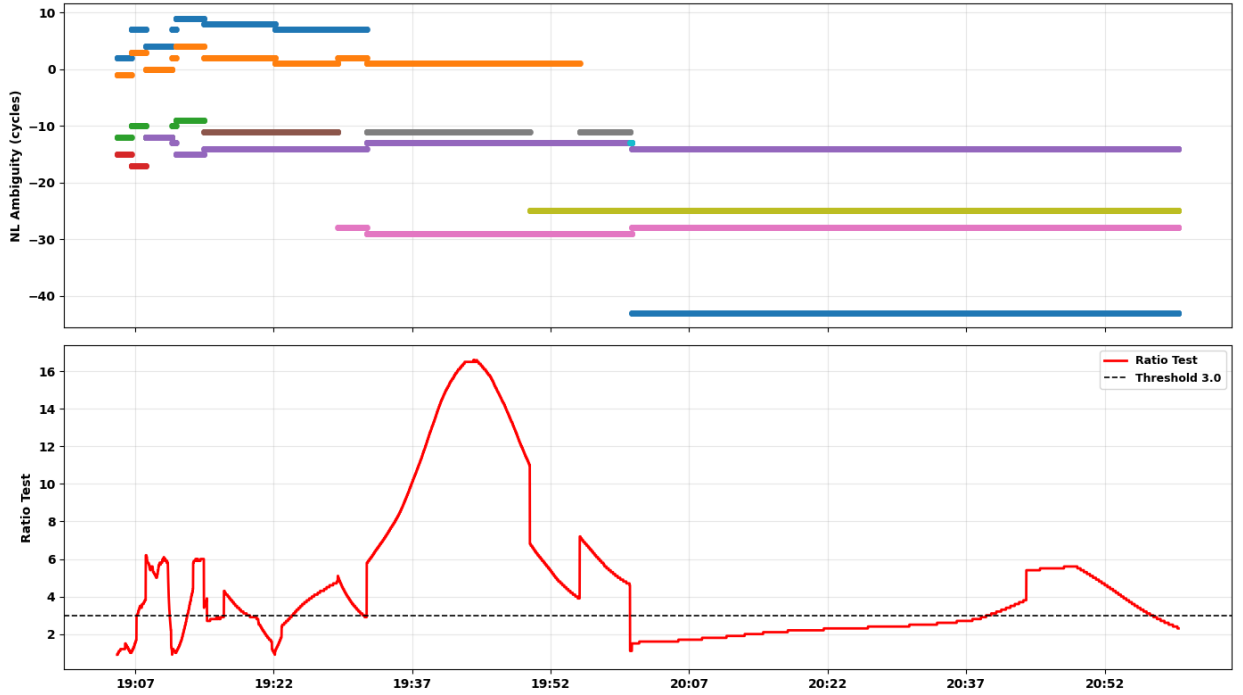


Figure 5.11: Time series of integer ambiguities and ratio test showing a distinct peak when fixing a subset of four satellites, indicating improved reliability in ambiguity resolution.

Integer Least-Squares

After successfully fixing a subset of ambiguities, the next step was to use these fixed values to compute a fixed position solution. The initial implementation followed the same procedure as in the raPPPid software (Glaner, 2022), where a Kalman filter is used to estimate the float solution, while a standalone least-squares adjustment is performed independently to obtain the fixed solution. A similar approach is also described in Huber, 2015.

However, during testing, it was observed that maintaining an independent fixed solution for the ambiguity resolution (AR) process can lead to significant errors in the resulting position estimates. A crucial requirement for successful PPP-AR performance is the high quality of the float ambiguity estimates. Although a subset of ambiguities may be successfully fixed, this does not necessarily guarantee an accurate final position solution. The positioning results and the corresponding ambiguity time series are illustrated in Figure 5.12 and Figure 5.13.

In Figure 5.13, it can be observed that the ambiguity resolution is achieved at approximately 19:30. After fixing, the ambiguities remain constant over time, as reflected in the time series. However, around 20:05, the positioning result begins to deviate from the zero-reference position. This behavior corresponds to the divergence problem discussed in Section 5.2.2 and is clearly visible in the float solution (blue line) in Figure 5.12, where noticeable drifts occur, particularly in the east and up components.

These results indicate that the standalone least-squares fixed solution strongly depends on the stability and accuracy of the float solution. When the float solution becomes unstable, such as under dynamic conditions, the fixed solution can inherit and amplify these errors, leading to significant positioning deviations.

Another limitation of the standalone least-squares approach is that it relies solely on the measurements from a single epoch, without considering the temporal information accumulated over

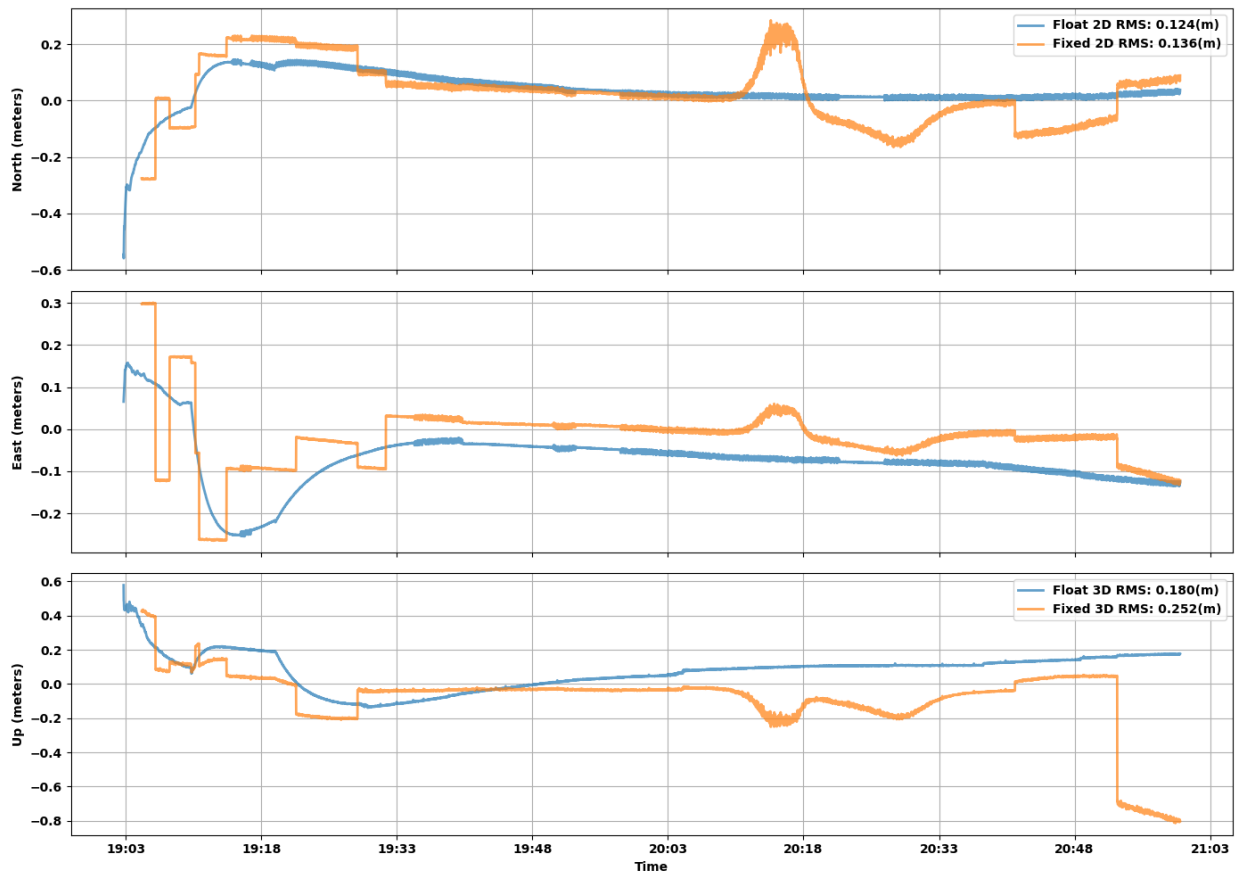


Figure 5.12: Positioning result using integer least-square adjustment

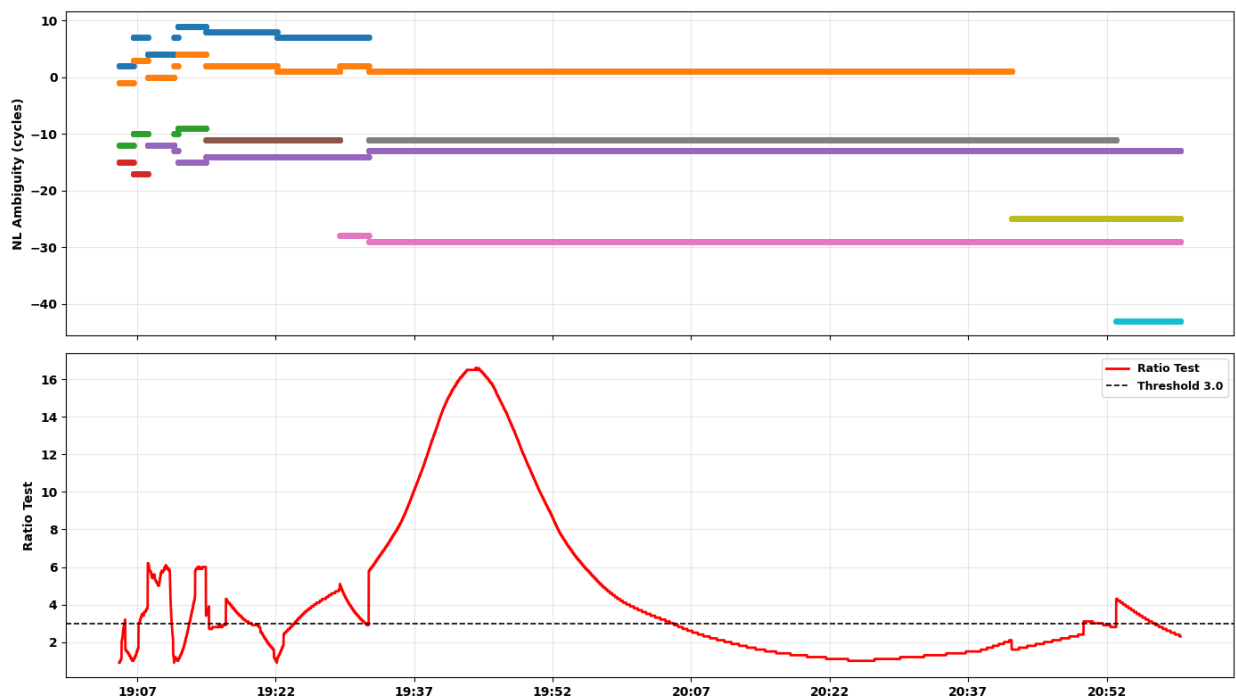


Figure 5.13: Time series of integer ambiguities and ratio test for the integer least-square adjustment

previous epochs. As a result, the temporal correlations and gradual improvements in parameter precision, naturally captured by the Kalman filter, are lost, making the ambiguity estimates less stable and more sensitive to measurement noise.

Furthermore, this independent least-squares method lacks a feedback mechanism to exploit the benefits of successful ambiguity resolution. In a tightly coupled framework, the residuals associated with fixed ambiguities can be redistributed among other model parameters (e.g., receiver position, clock, and tropospheric delay), thereby improving their accuracy. This feedback effect is absent in the standalone least-squares implementation, reducing the robustness and consistency of the overall solution.

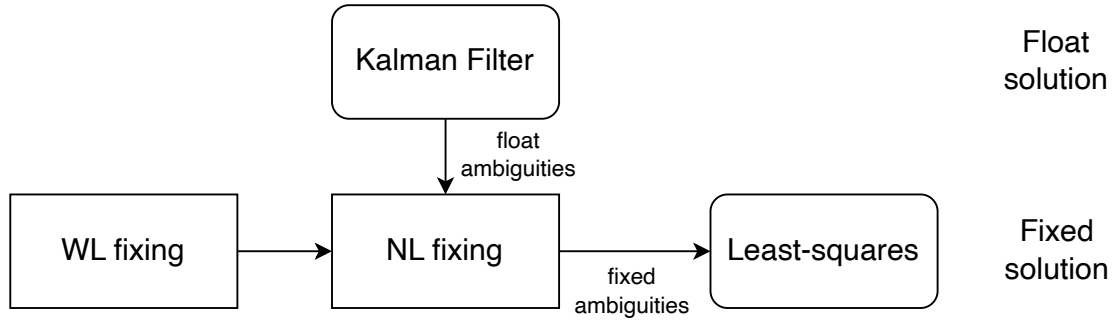


Figure 5.14: Standalone least-squares approach (raPPPid)

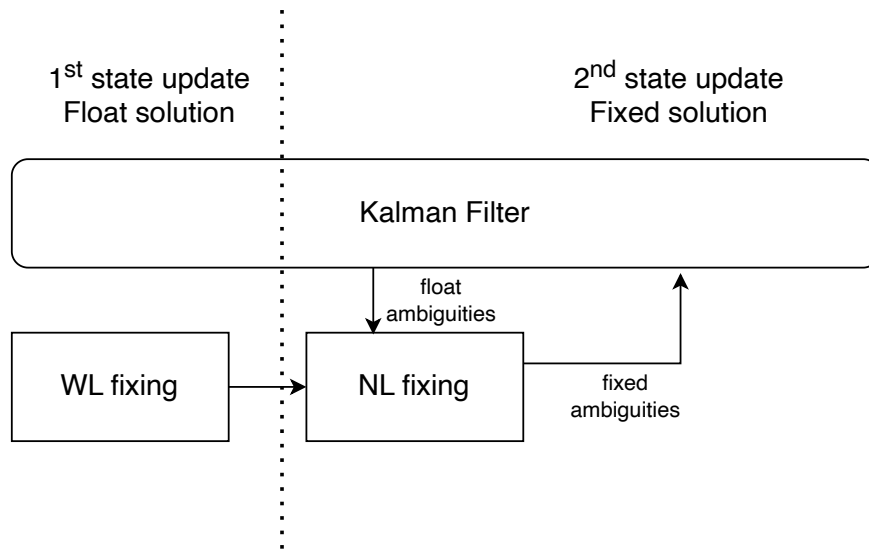


Figure 5.15: Proposed integrated Kalman filter approach

In Figure 5.14, the ambiguity resolution is performed independently from the Kalman filter using a separate least-squares adjustment. The float ambiguities estimated by the Kalman filter are passed to the LAMBDA method for integer fixing, and the resulting fixed ambiguities are used only to compute a secondary fixed solution. However, this solution is not fed back into the filter, meaning that the improved ambiguity precision does not contribute to refining other model parameters, such as the tropospheric delay or clock offset. Additionally, since the least-squares procedure uses only measurements from a single epoch, it does not benefit from temporal correlations or the

information accumulated across epochs, reducing stability in dynamic scenarios.

Figure 5.15 shows the proposed integrated Kalman filter approach implemented in this thesis. Once the wide-lane and narrow-lane ambiguities are fixed, the fixed integer values are reintroduced into the filter as pseudo-observations with tight constraints. This enables a second filter update, allowing the fixed ambiguities to influence and refine the entire state vector, including receiver position, clock, and tropospheric parameters. By incorporating the ambiguity constraints directly into the Kalman filter, this approach ensures better consistency, faster convergence, and higher positioning accuracy, particularly in dynamic environments.

Fixing remaining ambiguities

After successfully fixing the first subset of ambiguities using the ratio-based validation criteria, the next step is to extend the fixing process to the remaining unfixed ambiguities. A straightforward approach is to apply the same set of criteria to another subset of the unfixed ambiguities. The results of this attempt are illustrated in Figure 5.16.

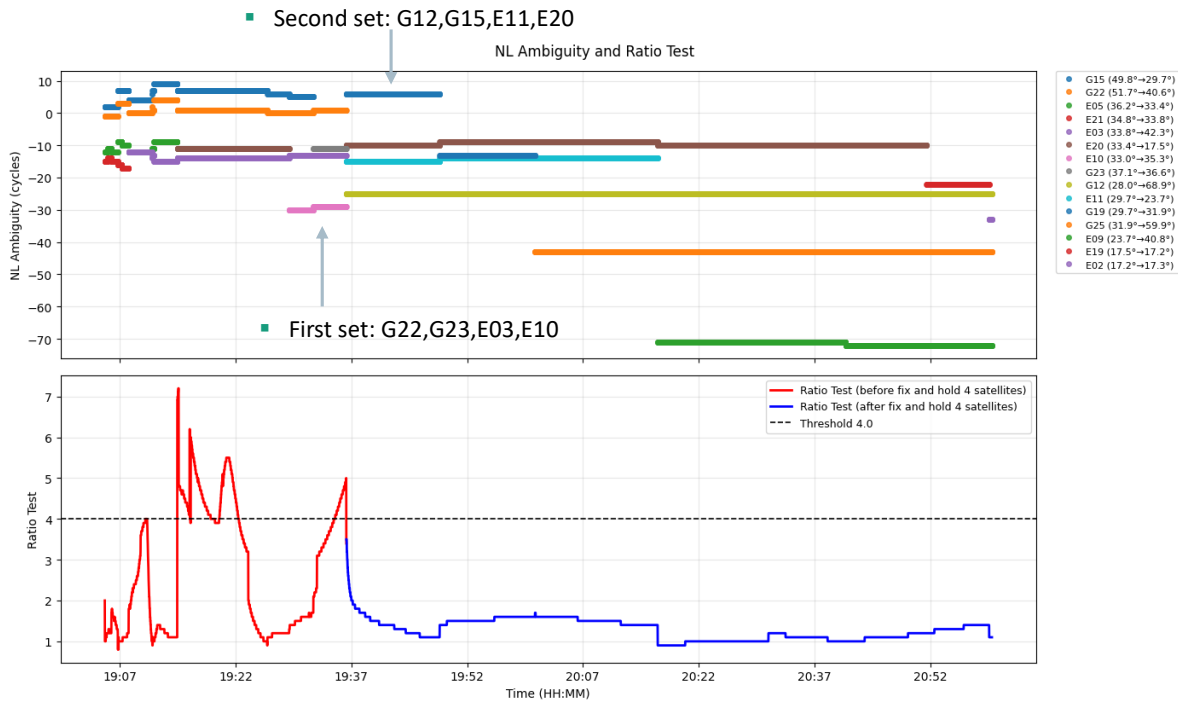


Figure 5.16: Time series of narrow-lane (NL) ambiguities and ratio test results. The red curve shows the ratio test values before fixing the first subset of four ambiguities, and the blue curve corresponds to subsequent attempts to fix additional satellites after the first subset was held fixed. The dashed line indicates the ratio test threshold.

In Figure 5.16, it can be observed that the first subset of ambiguities (G22, G23, E03, and E10) is successfully fixed after approximately 35 minutes. Subsequently, the LAMBDA method is applied to a new subset of ambiguities (G12, G15, E11, and E20), whose ratio test results are shown in blue. However, as illustrated in the figure, the blue curve does not exhibit any significant peak and remains consistently below the threshold, indicating that these ambiguities cannot be confidently fixed using the same ratio-based criterion.

Therefore, an alternative validation strategy is applied to the remaining unfixed ambiguities, as described in Section 4.2.5.

5.4 Positioning Results

After establishing the rationale behind the final ambiguity resolution implementation, it is essential to evaluate the positioning performance of the proposed approach against the standard float PPP solution. This comparison allows us to quantify the actual improvement achieved through ambiguity resolution.

In this section, the following results will be analyzed:

1. **Position deviations using Fix-and-Hold applied to the first subset of four satellites**, evaluated against the standard float solution.
2. **Positioning accuracy when fixing as many satellites as possible**, achieved by applying the deviation-based second-stage fixing criterion on top of the initially validated four-satellite subset.
3. **Influence of the tropospheric delay model**, assessed by comparing performance with and without simulated tropospheric delay (representing either a highly accurate model or ideal correction data).
4. **Influence of incorrect ambiguity fixing**, investigated by manually modifying selected fixed ambiguities to analyze the resulting degradation in positioning accuracy.

Float and fixed solution before ambiguity resolution

Before presenting the results of the float and fixed solutions in the following sections, it is important to clarify an implementation-related detail. According to the fixing procedure described in Section 4.2.5, the fixed solution should be identical to the float solution before the initial ambiguity-fixing criteria are met. In theory, this means that the blue (float) and orange (fixed) curves should fully overlap during the pre-fixing interval.

In practice, however, the two solutions do not overlap perfectly in the figures. The reason is related to the testing methodology: a single binary data file was recorded and subsequently reprocessed multiple times during the development phase, each time simulating a real-time processing scenario. Due to the nature of the Kalman filter used in the BNC implementation, which relies on stochastic processes and numerical operations, the solution is not strictly deterministic. This causes small run-to-run differences in the estimated states, even when using the same input data.

Despite these slight numerical variations, extensive testing has shown that the overall order of magnitude and the statistical performance of the solutions remain consistent across runs. As a consequence, minor discrepancies may appear between the float and fixed solutions before ambiguity fixing (where they should theoretically match), as well as small differences between the RMS values in the tables and the statistics displayed in the plot legends.

These differences do not affect the validity of the results but should be kept in mind when interpreting the figures.

Result of fixing four ambiguities

The first result presented concerns the case in which only the four ambiguities selected based on the elevation criterion. Specifically, the two highest non-reference satellites from each constellation are fixed. After validating these ambiguities using the criteria described in Section 4.2.5, this initial

subset of four ambiguities is resolved and subsequently held constant, unless their corresponding satellites fall below the elevation mask of 15° . The resulting positioning performance is shown in Fig. 5.17.

The fixed-and-hold ambiguities, together with the corresponding ratio-test time series, are depicted in Fig. 5.18. It can be observed that the four ambiguities remain constant over time as long as the associated satellites stay above the elevation threshold. Once the ratio test exceeds the defined threshold and all additional validation criteria are satisfied, the ambiguities are accepted and held fixed.

The numerical root mean square (RMS) errors for both the float and fixed solutions are summarized in Table 5.3. Both statistics are computed only after successful ambiguity resolution has been achieved.

Table 5.3: Root Mean Square (RMS) errors of fixing the first subset of four ambiguities.

	RMS_N	RMS_E	RMS_U	RMS_{2D}	RMS_{3D}
Float	2.50 cm	5.49 cm	12.63 cm	6.03 cm	13.99 cm
Fixed	3.39 cm	1.27 cm	6.16 cm	3.62 cm	7.14 cm

We observe that the ambiguity resolution leads to a substantial improvement in positioning accuracy. In the East component, the RMS error decreases from 5.49 cm to 1.27 cm (a reduction of approximately 77%), while the Up component improves from 12.63 cm to 6.16 cm (about 51% reduction). The overall 3D RMS decreases from 13.99 cm to 7.14 cm, corresponding to an improvement of roughly 49%.

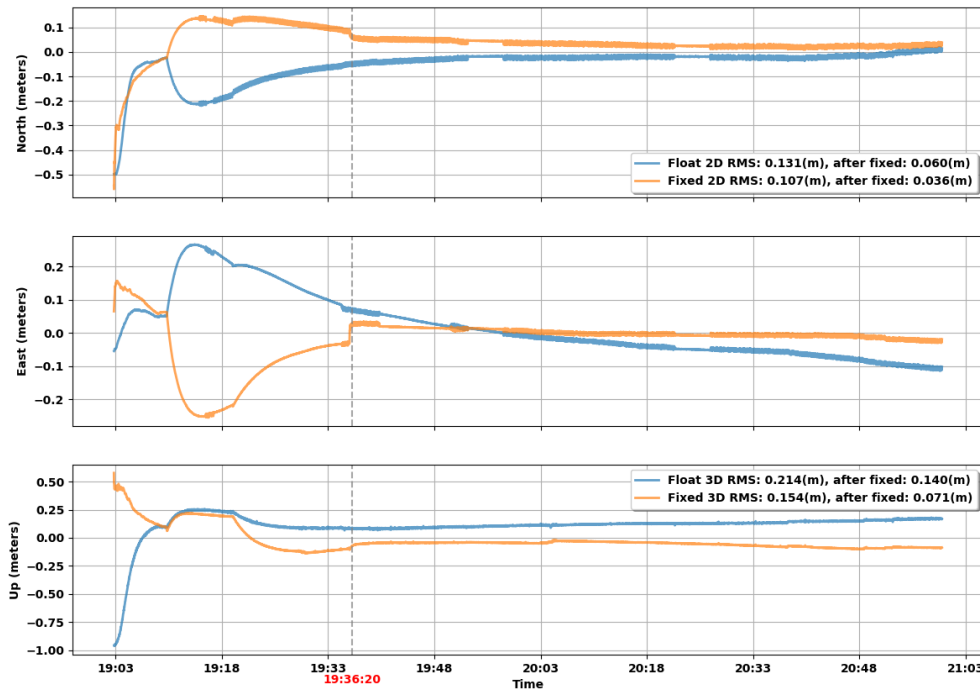


Figure 5.17: NEU position deviations after fixing the first subset of four ambiguities (G22, G23, E03, E10).

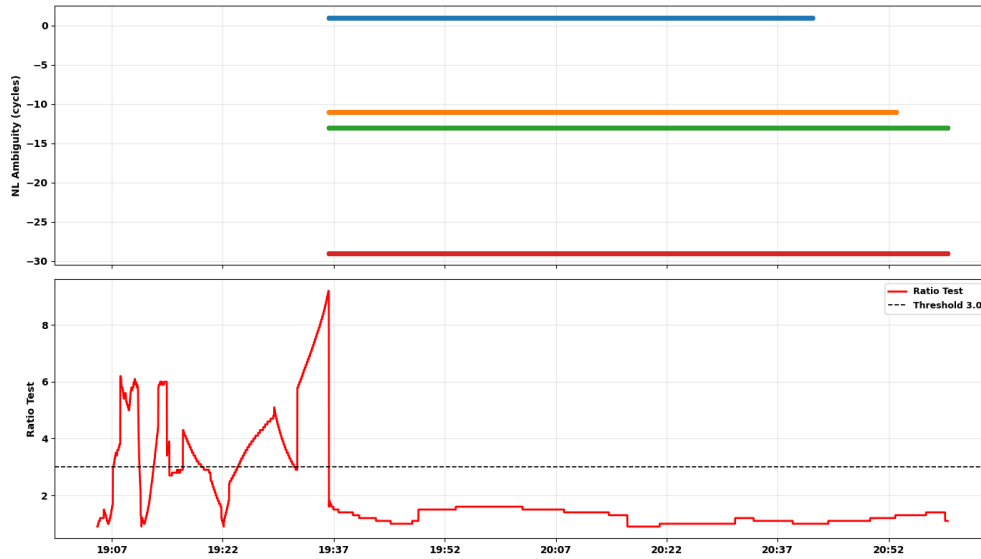


Figure 5.18: Ambiguity time series and ratio test for the first subset of fixed satellites. The four selected ambiguities remain constant once fixed and until the satellites fall below the elevation mask.

Result of fixing more satellites

After presenting the first result, the next step is to evaluate the positioning performance when extending the ambiguity resolution beyond the initial four-satellite subset. Once the first subset of ambiguities is successfully fixed using the criteria described in Section 4.2.5, the remaining unfixed ambiguities are processed using the second-stage deviation-based threshold criterion, which compares the float and integer narrow-lane ambiguities. This additional step allows further ambiguities to be fixed when they meet the consistency requirement, thereby increasing the number of fixed satellites beyond the initial subset.

The resulting positioning performance and ambiguity time series are shown in Figure 5.19 and Figure 5.20, respectively. The positioning performance after applying both the initial fixing of

Table 5.4: Root Mean Square (RMS) errors of fixing additional satellites using the deviation-based criterion.

	RMS_N	RMS_E	RMS_U	RMS_{2D}	RMS_{3D}
Float	2.50 cm	5.49 cm	12.63 cm	6.03 cm	13.99 cm
Fixed	2.00 cm	1.33 cm	9.24 cm	2.40 cm	9.55 cm

the four high-elevation satellites and the subsequent deviation-based fixing of the remaining satellites is summarized in Table 5.4. Compared to the float solution, the fully fixed solution exhibits substantial improvements across all components. In particular, the horizontal accuracy increases significantly: the RMS in the north direction improves from 2.50 cm to 2.00 cm, and in the east direction from 5.49 cm to 1.33 cm. This reduction leads to an overall improvement in the 2D RMS from 6.03 cm to 2.40 cm, corresponding to nearly a 60% enhancement. The vertical error (RMS_U) also decreases from 12.63 cm to 9.24 cm, although the improvement is less pronounced, which is typical given the poorer satellite geometry in the vertical direction.

When comparing this result to the first fixing strategy, where only the four highest-elevation satellites were fixed, the extended fixing approach yields further gains, especially in the horizontal plane. The 2D RMS is reduced from 3.62 cm (four-satellite fixing) to 2.40 cm, demonstrating a clear benefit of fixing additional satellites when the deviation-based criterion is satisfied. While

the 3D RMS (9.55 cm) is slightly larger than the corresponding value in the four-satellite case (7.14 cm), the overall accuracy remains substantially improved compared to the float solution, and the horizontal components in particular benefit from the increased number of fixed ambiguities.

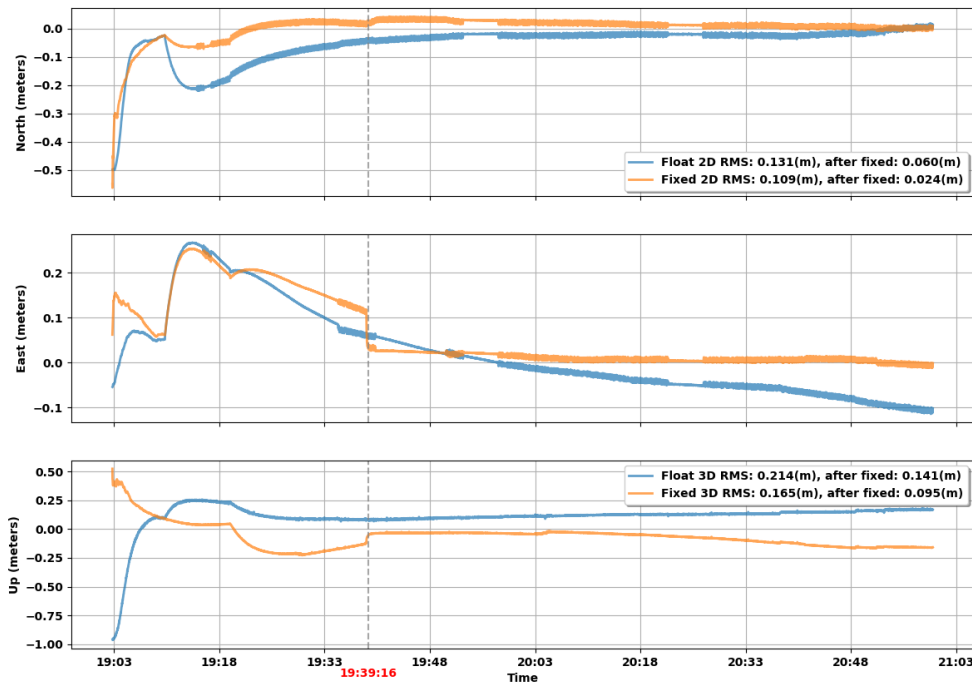


Figure 5.19: NEU position deviations after fixing the initial four ambiguities and subsequently fixing additional satellites using the deviation-based criterion.

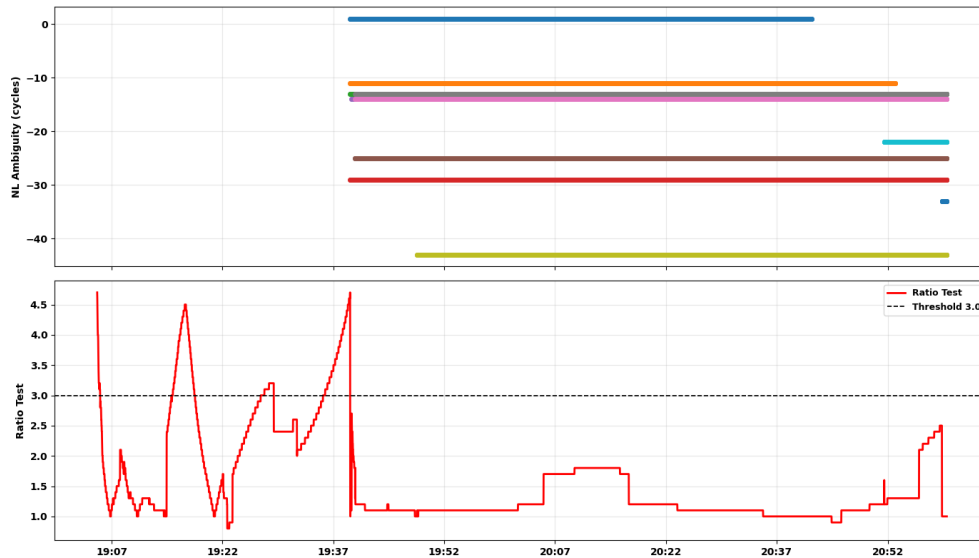


Figure 5.20: Ambiguity time series showing the initial fixed subset and additional ambiguities fixed using the second-stage deviation threshold.

Influence of the tropospheric delay

During the implementation of the ambiguity resolution approach, the influence of the tropospheric delay model was also investigated. To assess whether an ideally modeled or perfectly corrected troposphere could improve the ambiguity fixing performance, an additional simulation was conducted in which the tropospheric delay was entirely disabled. This allows us to evaluate two aspects: (1) whether removing the tropospheric uncertainty enables a larger number of ambiguities to be fixed, and (2) what level of positioning accuracy could theoretically be achieved if highly accurate tropospheric corrections were available.

To remain consistent with realistic processing conditions, the tropospheric delay parameter was still kept in the estimation, even though the delay itself was removed from the simulated signal. This ensures that the filter behavior remains comparable to nominal processing and, in addition, allows verification that the tropospheric term truly contains no atmospheric delay: the estimated tropospheric parameter converges to approximately -2.3 m, effectively compensating for the model correction applied in the PPP processing.

The resulting positioning performance in the NEU directions for this idealized scenario is shown in Figure 5.21.

Table 5.5: Root Mean Square (RMS) errors with tropospheric delay disabled.

	RMS_N	RMS_E	RMS_U	RMS_{2D}	RMS_{3D}
Float	1.62 cm	6.04 cm	8.03 cm	6.26 cm	10.18 cm
Fixed	2.13 cm	1.47 cm	1.93 cm	2.59 cm	3.23 cm

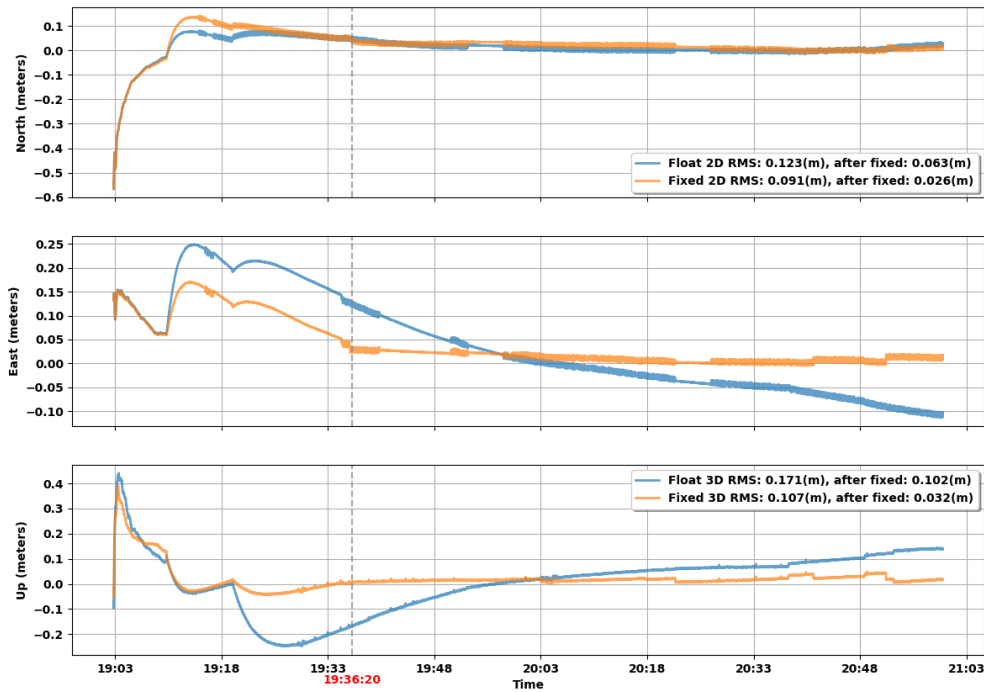


Figure 5.21: NEU position deviations for the scenario without tropospheric delay, using the full ambiguity-fixing strategy (initial subset + deviation-based fixing for remaining satellites).

The corresponding RMS errors for this idealized scenario are summarized in Table 5.5. Compared with the float solution, the fixed solution shows substantial improvement in all components. In

particular, the East and Up directions benefit the most, with RMS errors reduced from 6.04 cm to 1.47 cm in the East direction and from 8.03 cm to 1.93 cm in the Up direction. The overall 3D accuracy improves from 10.18 cm to 3.23 cm, representing a performance level that is significantly better than the previous simulation scenario.

Compared with the earlier results, where either only four satellites were fixed or as many satellites as possible were fixed under nominal conditions, the removal of the tropospheric delay leads to noticeably higher accuracy in all coordinate components, enabling very precise positioning. This clearly demonstrates that the tropospheric delay is a major limiting factor in PPP-AR performance. If perfect or near-perfect tropospheric information were available, centimeter-level accuracy across all directions would be achievable.

Influence of wrong fixing

While the previous sections demonstrated the benefits of correct ambiguity resolution, it is equally important to investigate the impact of *incorrect* fixing. PPP-AR is highly sensitive to ambiguity integrity: even a single wrongly fixed ambiguity can severely degrade the positioning solution, often leading to large drifts or abrupt jumps.

To illustrate this effect, a controlled experiment was performed in which one of the correctly resolved narrow-lane ambiguities was deliberately altered by adding an offset of one cycle. The modified ambiguity set was then applied within the Fix-and-Hold framework to observe how the filter reacts to a faulty integer constraint.

The resulting positioning errors clearly demonstrate the catastrophic effect of wrong fixing, highlighting the necessity of strict validation criteria and integrity monitoring in any reliable PPP-AR implementation.

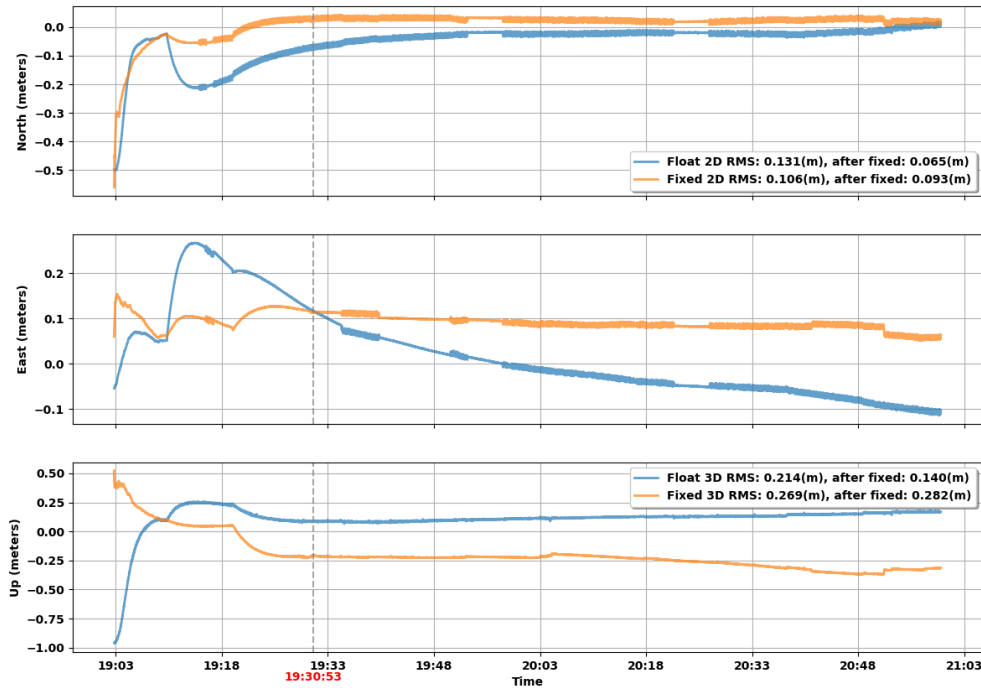


Figure 5.22: NEU position deviations showing the impact of intentionally incorrect ambiguity fixing, demonstrating how wrong fixes immediately degrade the positioning solution.

The positioning result is shown in Figure 5.22. As expected, the solution becomes unstable im-

mediately after the incorrect ambiguity is introduced. Even though only a single ambiguity was modified, the error rapidly propagates through the filter, leading to pronounced deviations in all three coordinate components. This experiment clearly illustrates the high sensitivity of PPP-AR to incorrect fixing and reinforces the importance of robust ambiguity validation strategies.

Discussion

There are several concerns that emerged during the development and validation process. Some of these arise from the inherent time constraints of a master's thesis, while others are related to limitations within the BNC software itself. For example, the same simulation scenario cannot be reproduced with perfectly identical results, and it is difficult to establish a true ground-truth statistics value for the ambiguities. The main limitations identified in this work are:

1. Only a single simulation scenario was deployed and tested. To fully assess the repeatability and generalization capability of the proposed algorithm, additional scenarios with different trajectories, environmental conditions, and correction quality should be evaluated in future work.
2. The number of ambiguities fixed in the first stage was restricted to four. This is not typical in PPP-AR processing, where a larger subset (or ideally the full set) of ambiguities is fixed simultaneously whenever possible. The underlying cause of this limitation remains unclear and should be further investigated. The author initially suspected a systematic modeling error, particularly related to the tropospheric delay, but dedicated testing ultimately ruled out this possibility. Identifying the true source of this behavior requires deeper analysis in future studies.

From the author's perspective, additional tests and simulation campaigns are necessary to better understand the behavior of the implemented algorithm and to validate its robustness. The following experiments are suggested for future work:

1. **Static scenario testing:** Design a static scenario to evaluate whether the majority of ambiguities can be fixed using the ratio test alone. This would help determine whether the limitation observed in the dynamic simulation is caused by motion-induced effects or by other underlying issues.
2. **Scenario simplification:** Reduce the complexity of the simulation environment (e.g., fewer satellites, reduced dynamics, simplified error sources) to systematically rule out potential sources of systematic errors in the current implementation.
3. **Scenario repetition and variation:** Generate multiple additional scenarios with similar characteristics to assess whether the observed limitation is consistent and reproducible. This would help identify whether the issue is scenario-dependent or reflects a more fundamental constraint in the current model or implementation.

Chapter 6 Conclusion and Future Work

6.1 Conclusion

This thesis demonstrates the feasibility of achieving real-time PPP-AR in a dynamic scenario using simulated Galileo HAS corrections. Through the implementation of wide-lane and narrow-lane ambiguity fixing, careful quality control, and a tightly integrated ESKF framework, the positioning accuracy was significantly improved compared to the float solution. The extensions developed for the BNC software, particularly the ambiguity-fixing module and enhanced configuration interface, proved effective for enabling reliable ambiguity resolution under realistic motion and signal conditions.

In addition, this study thoroughly examines the LAMBDA ambiguity resolution method. A complete validation strategy was implemented and discussed, together with its limitations and potential drawbacks. The interaction between ambiguity resolution and tropospheric estimation was also briefly explored, highlighting sensitivities within the PPP-AR process.

The implemented ESKF-based approach was compared with a standalone integer least-squares formulation, demonstrating substantial improvements in the simulation environment. The underlying reasons for these differences were analyzed in detail, emphasizing the importance of selecting an estimation strategy that integrates float and fixed ambiguities rather than treating them as separate processes.

In conclusion, this thesis presents several key aspects of implementing a real-time PPP-AR algorithm in a dynamic context, addressing potential challenges and offering practical solutions. The results and insights provided here contribute valuable guidance for future researchers working toward operational real-time PPP-AR with emerging high-accuracy services.

6.2 Future Work

While this thesis has achieved several important advancements and has demonstrated the feasibility of real-time PPP-AR in a dynamic scenario, a number of questions remain open for future research.

1. **Multi-frequency, multi-constellation processing:** This thesis utilizes only dual-frequency observations from GPS and Galileo. Incorporating additional frequencies and constellations could further improve the PPP-AR performance. With more observations available, the convergence time may be reduced, and the positioning robustness enhanced. Moreover, when more than two frequencies are used, the traditional ionosphere-free (IF) combination becomes less suitable due to multiple possible linear combinations. In such cases, adopting an uncombined model would be more appropriate, allowing all frequencies to be processed directly without forming predefined combinations.

2. **Repeatability:** A limitation of this thesis is the relatively small number of test runs, which is mainly due to the time constraints of a master's project. With more available time, a broader range of scenarios and repeated experiments could be conducted to strengthen the statistical significance of the results. Nevertheless, given the controlled nature of the simulation environment and the consistent configuration across tests, the author expects the demonstrated scenarios to be highly repeatable.
3. **Real-world testing of HAS SIS:** In this thesis, the Galileo HAS signal-in-space (SIS) corrections were simulated, as the real HAS phase bias corrections are still under development. Once these corrections become officially available, it will be important to investigate the real-world performance of HAS SIS for PPP-AR. Future testing should evaluate how the actual broadcast corrections including phase biases affect convergence time, positioning accuracy, and overall system robustness under realistic operational conditions.
4. **PPP-RTK:** In this thesis, the atmospheric delays were handled using a standard PPP approach: the dry component of the troposphere was corrected using a model, while the wet component was estimated in the filter. The ionospheric delay was removed through the ionosphere-free (IF) linear combination, which inherently introduces noise and contributes to the convergence time. In a PPP-RTK setting, external atmospheric corrections, such as ionospheric delay corrections and high-resolution tropospheric products, can be provided to the user. Incorporating these corrections would reduce the number of estimated parameters and therefore significantly shorten the convergence time. Evaluating the performance of PPP-RTK with such external atmospheric information would be a valuable extension of this work.
5. **Iterative Partial Ambiguity Fixing:** An iterative PAR strategy could be implemented to increase the number of ambiguities fixed in the first stage. Instead of selecting a fixed set of four satellites, the subset size could be expanded iteratively, adding additional high-quality ambiguities as long as the fixing criteria continue to be satisfied. This approach has the potential to increase the total number of reliably fixed ambiguities and thereby further improve positioning performance. For more details, see Li and Zhang, 2015.
6. **Investigation of Limited Ambiguity Fixing in the First Stage:** The observation that only four ambiguities could be fixed in the first stage (an atypical behavior for PPP-AR) warrants deeper investigation. Although tropospheric modeling errors were initially suspected, targeted tests ruled out this source. Future work should include additional simulation campaigns to identify the true cause, such as comparing static versus dynamic scenarios, simplifying the simulation environment to isolate potential systematic effects, and evaluating whether the limitation persists under different configurations. This would help clarify whether the issue stems from model deficiencies, filter behavior, or scenario-specific conditions.

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